12.0 RADIATION PROTECTION

12.1 ENSURING THAT OCCUPATIONAL RADIATION EXPOSURES ARE AS LOW AS REASONABLY ACHIEVABLE

12.1.1 POLICY CONSIDERATIONS

Administrative programs and procedures, in conjunction with facility design, ensure that the occupational radiation exposure to personnel will be kept as low as reasonably achievable (ALARA) as required by 10 CFR 20, Subpart B.

12.1.1.1 Design and Construction Policies

The ALARA philosophy was applied during the initial design of the plant and implemented via internal design reviews. Each design was given an ALARA review by engineers in the project organization of the architect/engineer. These reviewers included professional nuclear engineers and health physicists with a number of years of experience in ALARA nuclear power plant design and operation.

To comply with the ALARA policy, inspection and testing of plant shielding will be conducted to verify that the shielding performs its function of reducing radiation to design levels. During construction, a visual inspection was made to ensure that there were no major defects in the shield walls as they were poured. During initial power operations, radiation surveys will be conducted to ensure that there are no defects in the shielding that might seriously affect personnel exposures during normal operation and maintenance of the plant.

The ALARA reviews ensure that the design philosophies established in Regulatory Guide 8.8 are considered at the design stage. At the same time, these reviews ensure that a radiation exposure assessment similar to that recommended in Regulatory Guide 8.19 is performed. Design features are considered for potential exposure, and changes are recommended to reduce potentially high doses. Examples of changes effected by these ALARA reviews are:

- Additional shielding for level A filter valve galleries. Steel plates were provided to reduce direct radiation into corridor areas.

- Addition of a concrete slab over the pressurizer relief tank (PRT). The operating deck grating over the PRT was replaced with a concrete slab in order to minimize personnel exposure due to a sudden surge of reactor coolant to the PRT.

- Additional shielding for spent resin lines. Concrete floor slabs in pipe chases were locally thickened to reduce radiation levels from spent resin lines (during resin transfer) to less than 2.5 mrem/h in access corridors.

12.1.1.2 Operation Policies

It is the responsibility of the plant manager to implement a program to maintain occupational radiation exposures ALARA, consistent with recommendations of Section C.1 of Regulatory Guide 8.8, Revision 3, and Regulatory Guide 8.10, Revision 1.

To verify that the overall radiation protection program is functioning properly, the plant manager will review the written reports from audits of the ALARA program. Independent audits will be conducted by the quality assurance department, as delineated in administrative procedures.

The VEGP plant procedures manual is one of the major means of instituting the operational ALARA policy. It is available to each member of the plant staff, and each staff member will receive training covering the contents. This policy is further promoted in the health physics program, the training program, and plant procedures. (See section 12.5.) The "radiation protection plan," as defined in Section 5 of NUREG-0761, is a part of the plant procedures.

In addition to defining management's commitment to ALARA, the procedures manual designates the plant personnel who have the responsibility and authority to implement the ALARA program. The plant manager, as with all other aspects of plant operation, bears the final responsibility for implementing the ALARA program but will delegate this responsibility to the health physics manager. The authority to prevent unsafe practices and to direct steps to prevent any unnecessary radiation exposures also lies within the health physics manager's responsibilities. The health physics support supervisor will assist the health physics manager in ensuring that radiation exposures are maintained ALARA. The health physics support supervisor will report to the health physics manager who serves as the radiation protection manager referred to in Regulatory Guide 8.8. The plant health physicist or health physics support supervisor may also serve as the radiation protection manager if qualified to do so. The health physics foremen handle the day-to-day operation of the site radiation protection program and report to the health physics manager and health physics support supervisor. The foremen will implement the ALARA program and supervise the health physics technicians who perform the various surveys for radiation protection. For a more detailed discussion on the responsibility and authority of the key supervisory positions discussed above and the qualifications of the personnel who fill them, see chapter 13 and section 12.5.

It is the responsibility of the health physics manager to see that company employees and contractors are trained in radiation protection requirements to comply with 10 CFR 19, 10 CFR 20, 10 CFR 55, and the recommendations of Regulatory Guides 8.8, 8.10, and 8.13 and to verify that personnel follow the radiation protection procedures. To ensure compliance with this policy, the health physics manager, health physics support supervisor, and the health physics foremen are charged with the responsibility to promptly advise higher management of any unsafe practices which exceed their authority to correct. They have the authority to halt any operation which, in their judgment, is unsafe. It is also the prerogative of the working health physics technicians to halt any operation which, in their judgment, is unsafe pending review by the health physics foreman.

In addition to reviews by management, all employees are encouraged to express their concerns for ALARA through a formal suggestion program. The program provides the basis to evaluate proposed ALARA improvements. In addition, the plant modification program also contributes to the ALARA improvements.

Prior to unit startup, personnel assigned to that unit will be trained in radiation protection procedures and techniques. Personnel will be tested prior to startup of their respective units to verify that they understand how these procedures relate to the safe performance of their jobs. Plant personnel, whose assignments require it, will be trained in radiation protection procedures and techniques on a 3-year cycle and will be tested annually. This training program will ensure

compliance with 10 CFR 19.12. Contractors who work in the radiologically controlled area of the plant after initial fuel loading will receive radiation protection training commensurate with the safe performance of their jobs, and they will be tested on this phase of their training. Construction personnel will also be instructed in site emergency procedures.

Maintenance, refueling, and radwaste system operating procedures which involve significant radiation exposures will be reviewed to verify adherence to ALARA policy prior to their use. This review will be performed in accordance with the established ALARA program. ALARA will also be considered when the system or plant modifications are reviewed by the plant review board. A qualified health physics technician or health physics foremen will observe implementation of selected activities to identify situations in which exposures can be reduced. Those operations with higher exposure potentials will receive greater attention.

12.1.1.3 Conformance with Regulatory Guide 8.8

VEGP's design has been reviewed to ensure operator doses will be ALARA. Many of the recommendations of Revision 3 (June 1978) to Regulatory Guide 8.8, Information Relevant to Ensuring That Occupational Radiation Exposures at Nuclear Power Stations Will Be As Low As Reasonably Achievable, which was issued 4 years after the construction permit issue date, have been incorporated in the design.

12.1.1.4 Conformance with Regulatory Guide 8.10

VEGP's plant procedures will ensure conformance with the requirements of Revision 1R of Regulatory Guide 8.10, Operating Philosophy for Maintaining Occupational Radiation Exposures As Low As Reasonably Achievable.

12.1.2 DESIGN CONSIDERATIONS

This subsection discusses the methods and features by which the policy considerations of subsection 12.1.1 are applied. Provisions and designs for maintaining personnel exposures as low as reasonably achievable (ALARA) are presented.

The engineers and supervisors assigned to the VEGP project have, in general, performed similar design work on other nuclear power plants. Through this experience, they have developed knowledge of radiation protection aspects of design which have been applied to VEGP. Design engineers are also made aware of nuclear plant operating experiences through Nuclear Regulatory Commission (NRC) inspection and enforcement bulletins, generic letters, circulars, and information notices; NRC power reactor events; and internal procedures of the participating design organizations.

The VEGP is committed to the concept of ALARA. The project initiates formal reviews of the facility and equipment design to ensure that ALARA concerns are addressed. Designers employ the guidelines of NRC Regulatory Guide 8.8, Revision 3, and VEGP conforms to the intent of this regulatory guide. Design alterations are made when deficiencies are identified, as practicable. The ALARA reviews form the basis for identifying conformance and exceptions to the regulatory guide.

The basic philosophy of VEGP plant management, to ensure that occupational radiation exposures are ALARA, can be expressed as:

A. Design of structures, systems, and components to ensure increased reliability and maintainability, thereby effectively reducing the maintenance requirements on radioactive components.

B. Design of structures, systems, and components to reduce the radiation fields to ensure that operation, maintenance, and inspection activities are performed in the minimum radiation field feasible.

C. Design of structures, systems, and components to reduce the time spent in radiation fields during operation, maintenance, and inspection.

D. Design of structures, systems, and components to accommodate remote and semiremote operation, maintenance, and inspection procedures.

12.1.2.1 General Design Considerations for ALARA Exposures

General design considerations and methods employed to maintain in-plant radiation exposures ALARA consistent with the recommendations of NRC Regulatory Guide 8.8 have two objectives:

A. Minimizing the necessity for, and amount of, personnel time spent in radiation areas.

B. Minimizing radiation levels in routinely occupied plant areas in the vicinity of plant equipment expected to require personnel attention.

Equipment and facility designs are considered in maintaining exposures ALARA during plant operations, including:

- Normal operation.
- Maintenance and repairs.
- Refueling operations and fuel storage.
- Inservice inspection and calibrations.
- Radioactive waste handling and disposal.
- Other anticipated operational occurrences.

The actual design features employed are described in general in subsection 12.3.1.

Paragraphs 12.1.2.2 and 12.1.2.3 list general design considerations for maintaining exposures ALARA in accordance with Regulatory Guide 8.8.

12.1.2.2 Equipment General Design Considerations for ALARA

A. Equipment general design considerations to minimize the necessity for, and amount of, personnel time spent in a radiation area include:

1. Reliability, durability, construction, and design features of equipment, components, and materials to reduce or eliminate the need for repair or preventive maintenance.

2. Servicing convenience for anticipated maintenance or potential repair, including ease of disassembly and modularization of components for replacement or removal to a lower radiation area for repair.

3. Provisions, where practicable, to remotely or mechanically operate, repair, service, monitor, or inspect equipment.

4. Redundancy of equipment or components to reduce the need for immediate repair when radiation levels may be high and when no feasible method is available to reduce radiation levels.

B. Equipment general design considerations directed toward minimizing radiation levels proximate to equipment or components requiring personnel attention include:

1. Provisions for draining, flushing, or, if necessary, remote cleaning of equipment containing radioactive material.

2. Design of equipment, piping, connections, and valves to minimize the buildup of radioactive material and to facilitate flushing of crud traps.

3. Use of high quality valves, valve packings, and gaskets to minimize leakage and spillage of radioactive materials.

4. Provisions for minimizing the spread of contamination into equipment service areas, including direct drain connections.

5. Provisions for isolating equipment from radioactive process fluids.

6. Provision for a spent fuel pool cleanup system to maintain the radiation level of the fuel pool area within the Zone II limit. See table 12.3.1-1 for the description of radiation zones.

12.1.2.3 Facility Layout General Design Considerations for ALARA

A. Facility general design considerations to minimize the amount of personnel time spent in a radiation area include:

1. Locating equipment, instruments, and sampling stations which will require routine maintenance, calibration, operation, or inspection, for ease of access and minimum of required occupancy time in radiation areas.

2. Laying out plant areas to allow remote or mechanical operation, service, monitoring, or inspection of highly radioactive equipment.

3. Providing, where practicable, for transportation of equipment or components requiring service to a lower radiation area.

B. Facility general design considerations directed toward minimizing radiation levels in plant access areas and in the vicinity of equipment requiring personnel attention include:

1. Separating radiation sources and occupied areas where practicable (e.g., pipes or ducts containing potentially highly radioactive fluids do not pass through occupied areas). Redundant components requiring maintenance which are a source of radiation are located in separate compartments to allow maintenance of one component while the other component is in operation. Equipment such as demineralizers and filters are separated by

shielding from nonradioactive equipment to facilitate unrestricted maintenance on the latter. Areas between radiation sources and access and service areas are adequately shielded.

2. Providing labyrinth entrances to radioactive pump, equipment, and valve rooms. Adequate space is provided in labyrinth entrances for easy access. Highly radioactive passive components with minimal maintenance requirements are located in completely enclosed compartments and are provided with access via a shielded hatch or removable blocks.

3. Where appropriate, separating equipment or components in service areas with permanent shielding. Also provided are means and adequate space for utilizing movable shielding for sources within the service area when required.

4. Plant layout incorporates restrictions and control of access to the various radiation zones. In the case of an abnormal occurrence or accident, the zone restrictions may change due to increased dose rates. Special access controls would be implemented at these times to minimize personnel exposure.

   Access to a given radiation zone generally does not require passing through a higher radiation zone. For example, personnel are not normally required to pass through a Zone IV area to gain access to a Zone III area. Exception to this is taken within the containment. The plant design incorporates personnel escape routes which allow personnel to rapidly exit from an area in which the radiation exposure level has unexpectedly increased.

5. Locating equipment, instruments, and sampling sites in the lowest practicable radiation zone.

6. Providing means to control contamination, to facilitate decontamination of potentially contaminated areas, and for decontamination of service areas.

7. Providing backflushable and cartridge filter systems for specified high radioactive filtration requirements.

12.1.3 OPERATIONAL CONSIDERATIONS

In accordance with Southern Nuclear Operating Company (SNC) policy, and consistent with the recommendations of Regulatory Guides 8.8 and 8.10, the radiation exposure of plant personnel will be kept as low as reasonably achievable (ALARA) by means of the health physics program discussed in section 12.5. The radiation protection policies and practices contained therein are initiated through the training program discussed in section 13.2 and through the plant procedures manual discussed in subsection 12.1.1 and section 12.5.

The following will meet the requirements of NUREG-0737, Item III.D.1.1.

VEGP is committed to a program to reduce leakage from systems outside containment which could contain highly radioactive fluids during a serious transient or accident to an as low as practical level in accordance with the Technical Specifications.

This program includes inspection and leak rate tests with the exception of those systems that are known to contain highly radioactive fluids and/or when a leak is discovered in a high radiation area. In both of these exceptions, leak rate testing will not be performed due to the personnel hazard from a radiation exposure standpoint. The systems that are known to contain highly radioactive fluids will be pressurized and inspected for leaks. The procedures have been written for inspection and surveillances to require a work request upon identification of leakage. Leak rate measurements and/or inspections will be performed periodically at the intervals specified in the Technical Specifications. Quantitative leak testing will be performed on the gaseous system covered by this program.

12.1.3.1 Procedures

Procedures for radiation area jobs that routinely occur at an operating reactor will be written and approved for use at VEGP. For unusual or first-time operations which will involve significant radiation exposure, operating procedures or work control documents will normally be prepared by or with the assistance of the section doing the work. If the operation requires plant review board review (section 13.4), then it will be reviewed and approved by the board. If plant review board review is not required, the health physics superintendent or his designee will review the procedure for ALARA purposes according to the ALARA program.

Techniques which will be embodied in plant procedures, training, and/or work practices for ALARA purposes as appropriate are discussed below. Also discussed are the criteria and/or conditions for their use. These techniques will be written into plant procedures as appropriate and must be followed in accordance with SNC requirements for procedure compliance.

A large percentage of exposure at an operating reactor occurs during plant outages from maintenance and inspection activities and not from normal operating activities. This is to be expected since, during operation, instrumentation and valves can be operated from outside the shield walls, and operators only have to enter cubicles containing radioactive equipment for short periods of time to check equipment. Maintenance and inspection personnel, in order to perform their job, are often in close proximity to lines, valves, instruments, or other pieces of equipment which are radiation sources. For this reason, the ALARA-related procedures place emphasis on maintenance and non-routine operations.

12.1.3.2 General ALARA Techniques

Described below are several general ALARA techniques. These methods will be incorporated into preplanning of tasks and procedure development. Further information on ALARA techniques incorporated into procedures is given in section 12.5.

A. Permanent shielding is used, where possible, by having workers stay behind walls or in areas of low-level radiation when not actively working in the radiation area. On some jobs, temporary shielding may be used. Temporary shielding will be used only if the total exposure, which includes exposure received during installation and removal of the temporary shielding, will be effectively reduced.

B. Piping systems and other pieces of equipment which are subject to crud buildup have been equipped with connections which can be used for flushing the system to eliminate potential hot-spot buildup.

   Prior to performing maintenance work, consideration will be given to flushing and/or chemically decontaminating the involved system or piece of equipment in order to reduce crud levels and, hence, personnel exposure.

C. Work involving whole body exposure rates in excess of 100 mrem/h and exposure of greater than 1 person-rem per job or removable contamination levels in excess of 10^6 dpm/100 cm^2 will be preplanned. Other jobs which involve significant total man-rem exposure, but at lower dose rates, may also be preplanned. The purpose of preplanning is to carefully prepare for the job so that it can be performed in a proper, safe manner with a minimum of personnel exposure.

D. On complex jobs or jobs with exceptionally high radiation levels, dry-run training may be utilized, including mockup training as appropriate. The intent of dry-run or mockup training is to improve worker efficiency and to lower exposure times.

E. As much as practicable, jobs will be performed outside radiation areas. This includes items such as reading instruction manuals or maintenance procedures, adjusting tools or jigs, repairing valve internals, and prefabricating components.

F. For long-term repair jobs, consideration will be given to setting up a communications network to assist supervising personnel in checking on work progress from a lower radiation area.

G. Special tools or jigs will be used when their use permits the job to be performed more efficiently or prevents errors, thus reducing the time spent in a radiation area. Special tools may also be used if their use would increase the distance from the source to the worker, thereby reducing the exposure received. These special tools will be used only if the total exposure, including that received during installation and removal, is significantly reduced.

H. Entry and exit points will be established in low-level radiation areas to allow personnel access and exit in as low a level of radiation as practicable. This is done because personnel may spend a significant amount of time changing protective clothing and respiratory equipment in those access areas. These access points will be setup to limit the spread of contamination from the jobsite to as small an area as practicable.

I. Protective clothing and respiratory equipment will be selected to minimize the discomfort of workers so that efficiency will be increased and less time will be spent in radiation areas. Protective clothing prescribed by health physics will be commensurate with the hazards involved, and the requirements cannot be modified by other personnel.

J. Contamination containments (i.e., glove bags, polybottles, tents, etc.) will be used, where practicable, to allow personnel to work on highly contaminated equipment while minimizing the spread of contamination during the work.

K. Individuals will be instructed to remain in low-level radiation areas as much as possible, consistent with performing their assigned jobs. In addition, on certain jobs, surveys will be provided to clearly delineate the areas of high radiation levels and hot spots to prevent inadvertent entry into an area of much higher radiation level.

L. Personnel will be assigned self-reading dosimeters to allow determination of accumulated exposure at any time during the job. In addition, personnel performing work in containment will be assigned dosimetry capable of alarming excessive dose rate and dose. Additional discussion of personnel dosimetry is provided in paragraph 12.5.3.6.

M. On jobs where general area radiation levels are unusually high, remotely readable dosimetry or a timekeeper will keep track of the total exposure time. This will ensure that personnel do not stay in a radiation field long enough to exceed applicable dose limits.

N. On major maintenance jobs, especially those which involve high or complex radiation levels, the job preplanning will include estimates of the man-rem needed to complete the job. At the completion of the work, a debriefing session will be held with the personnel who performed the work (when practical) in an effort to determine how the work could have been completed more efficiently, resulting in less accumulated exposure. All radiation aspects (i.e., radiation, contamination, airborne radioactivity, and external or internal personnel contamination) will be compiled and filed for future reference to provide guidance during preplanning of future similar work situations. This will incorporate experience gained in performing these tasks into future work efforts.

12.2 RADIATION SOURCES

This section discusses and identifies the sources of radiation that form the basis for shield design calculations and the sources of airborne radioactivity used for the design of personnel protection measures and dose assessment.

12.2.1 CONTAINED SOURCES

The shielding design source terms are based upon the three plant conditions of normal full-power operation, shutdown, and design basis accident events.

12.2.1.1 Sources for Full-Power Operation

The primary sources of radioactivity during normal full-power operation are direct core radiation, coolant activation processes, leakage of fission products from pinhole defects in fuel rod cladding, and activation of reactor coolant corrosion products. The design basis for the shielding source terms for fission products in this section is cladding defects in fuel rods producing 1 percent of the core thermal power. The design basis for activation and corrosion product activities is derived from measurements at operating plants and is independent of fuel defect level. The design basis of 0.25 percent fuel cladding defects for shielding source terms is used to establish shielding provisions for the radwaste processing facility.

12.2.1.1.1 Reactor Core

The primary radiation from the reactor core during normal operation is neutrons and gamma rays. Figures 12.2.1-1 and 12.2.1-2 show neutron and gamma multigroup fluxes on the inside surface of the primary shield wall at the core centerline. Gamma dose rate incident on the primary shield wall at the core midplane is shown in figure 12.2.1-3. These figures are based on nuclear parameters discussed in chapter 4. Table 12.2.1-1 lists core gamma sources after shutdown for shielding requirements during shutdown and inservice inspection.

12.2.1.1.2 Reactor Coolant System (RCS)

Sources of radiation in the RCS are fission products released from fuel and activation, and corrosion products that are circulated in the reactor coolant. These sources and their bases are discussed in section 11.1.

The activation product, N-16, is the predominant contributor to the activity in the reactor coolant pumps, steam generators, and reactor coolant piping during operation. The N-16 activity in each of the components depends on the total transit time to the component and the average residence time in the component. Table 12.2.1-2 presents the RCS N-16 activity as a function of transport time in a reactor coolant loop. The N-16 activity for the pressurizer is tabulated in table 12.2.1-3.

Fission and corrosion product activities circulating in the RCS and out-of-core crud deposits comprise the remaining significant radiation sources during full-power operation. The fission and corrosion product activities circulating in the reactor coolant are given in section 11.1. The fission and corrosion product source strengths in the reactor coolant pressurizer liquid and

vapor phases are given in table 12.2.1-4. The isotopic composition and specific activity of typical out-of-core crud deposits are given in table 12.2.1-5. Typically, 1 mg of deposited crud material is found on 1 cm^2 of a relatively smooth surface. This may be as much as 50 times higher in crud trap areas. Crud trap areas are generally locations of high turbulence, areas of high momentum change, gravitational sedimentation areas, high affinity material areas, and possibly thin boundary layer regions.

The N-16 activity is not a factor in the radiation sources for systems and components located outside containment due to its short, 7.11-s, half-life and the greater than 1-min transport time before the letdown flow exits the containment.

12.2.1.1.3 Main Steam Supply System

For the purpose of evaluating the steam generator blowdown processing system, radioactivity in the main steam supply system is based on a steam generator tube leakage rate of 1 gal/min concurrent with 1 percent failed fuel. Continuous operation with primary-to-secondary leakage is assumed. The RCS radionuclide concentrations used are those in table 11.1-2. The resulting steam generator secondary side water and steam fission and corrosion product concentrations are discussed in subsection 10.4.8.

It is assumed that the maximum blowdown rate to the blowdown demineralizers is 1.2 percent of the main steam rate.

12.2.1.1.4 Auxiliary Systems

12.2.1.1.4.1 Chemical and Volume Control System (CVCS). Radiation sources in the CVCS consist of radionuclides carried in the reactor coolant. The design of the CVCS ensures that most of the N-16 has decayed before the letdown stream leaves the containment by placing a delay mechanism in the letdown flowpath. All CVCS system heat exchangers other than the regenerative heat exchanger and excess letdown heat exchanger are located in the auxiliary building.

The shielding design is based on the maximum activity in each component. These sources are listed in table 12.2.1-6.

A. CVCS Heat Exchangers

The regenerative and excess letdown heat exchangers are located in the containment building. They provide the initial cooling for the reactor coolant letdown. Their radiation sources include N-16.

The magnitude of the N-16 source strength is highly sensitive to the location of these heat exchangers with respect to the RCS loop piping. Therefore, the N-16 source strengths for these heat exchangers are based on the average value in the crossover leg between the steam generator and the reactor coolant pump. The shielding design takes into account the N-16 decay from the crossover leg to each heat exchanger.

The letdown heat exchanger provides second-stage cooling for the reactor coolant prior to entering the demineralizers. The seal water heat exchanger cools the water from several sources, including the reactor coolant discharged from the excess letdown heat exchangers. The tube side of the letdown reheat

heat exchanger heats letdown water before the water enters the boron thermal regeneration demineralizers during boron release.

The radiation source strengths for the boron thermal regeneration moderating, chiller, and shell side of the letdown reheat units account for radionuclides removed by the demineralizers upstream of the units. The source strengths for the tube side of the regenerative heat exchanger account for the removal of radionuclides by the CVCS demineralizers and the volume control tank.

B. CVCS Ion Exchangers

The mixed bed demineralizer is in continuous use and removes fission products in cation and anion form. It also is highly effective in removing corrosion products. The cation bed demineralizer is used intermittently to remove lithium for pH control. It also is highly effective in removing the monovalent cations, cesium and rubidium. The short-lived isotopes are assumed to build up to saturation activities on both beds. For the long-lived isotopes, the activity retained is assumed to be evenly distributed between the two demineralizers.

The boron thermal regeneration demineralizers are used to regulate the boron concentration in the reactor coolant water. They are used during load follow operations and in removing boron from the coolant as the nuclear fuel is depleted. These demineralizers collect radioactive anions, such as iodine and bromine, which were not retained by the mixed bed demineralizer. The radionuclides retained are assumed to be evenly distributed between two of the five 90-ft3 beds during load follow operation.

C. CVCS Filters

The design criterion for CVCS filter shielding is based primarily on operating experience.

The source strengths for the reactor coolant filter correspond to an exposure rate of 500 rem/h at contact; the source strengths for the remaining filters correspond to an exposure rate of 100 rem/h at contact. These dose rates are arrived at assuming the filters are homogeneous sources with the dimensions and composition given in table 12.2.1-6.

D. Tanks

1. Volume Control Tank

The radiation sources in the volume control tank are based on a nominal operating level in the tank of 200 ft^3 in the liquid phase and 200 ft^3 in the vapor phase and on the stripping fractions given in table 11.1-1, assuming no volume control tank purge.

2. Boric Acid Storage Tank

Removal from service of recycle evaporator has eliminated this source.

12.2.1.1.4.2 Steam Generator Blowdown System. The source terms for the steam generator blowdown system are provided in tables 12.2.1-7 through 12.2.1-13.

12.2.1.1.4.3 Boron Recycle System. The source terms for the boron recycle system are listed in table 12.2.1-15.

12.2.1.1.5 Nuclear Service Cooling Water System and Component Cooling Water System

The nuclear service cooling water and component cooling water systems are normally nonradioactive or, because of inleakage, of very low level activity. The process radiation monitor for these systems is described in section 11.5. For shielding and dose assessment purposes, the nuclear service cooling water and component cooling water systems do not yield substantive doses.

12.2.1.1.6 Spent Fuel Storage and Transfer

The predominant radioactivity sources in the spent fuel storage and transfer areas in the fuel handling building are the spent fuel assemblies. The source strengths employed to determine the minimum water depth above spent fuel and shielding walls around the spent fuel pool, as well as shielding of the spent fuel transfer tube, are given in table 12.2.1-18. For shielding design, the spent fuel pool is assumed to contain the design maximum number of fuel assemblies. Limitations are placed on fuel storage and transfer operations to limit occupational doses to personnel in the fuel handling building. These limitations are administratively controlled.

12.2.1.1.7 Spent Fuel Pool Cooling and Cleanup System

Sources in the spent fuel pool cooling and cleanup system are a result of transfer of radioactive isotopes from the reactor coolant into the spent fuel pool during refueling operations. The reactor coolant activities for fission, corrosion, and activation products are decayed for the amount of time required to remove the reactor vessel head following shutdown; are reduced by operation of the CVCS purification ion exchangers; and are diluted by the total volumes of the water in the reactor vessel, refueling pool, and spent fuel pool. This activity then undergoes subsequent decay and accumulation on the spent fuel pool cooling and cleanup system filters and demineralizer.

The design and normal fission and corrosion product activities in the spent fuel pools are given in table 12.2.1-19. The source terms for the spent fuel pool demineralizers and filters are provided in tables 12.2.1-20 through 12.2.1-23.

12.2.1.1.8 Main Steam Supply and Power Conversion Systems

Potential radioactivity in the main steam supply and power conversion systems is a result of steam generator tube leaks and fuel cladding defects as discussed in paragraph 12.2.1.1.3. This radioactivity is sufficiently low so that no radiation shielding for equipment in secondary systems, other than portions of the steam generator blowdown system (paragraph 12.2.1.5), is required to meet the radiation zone requirements.

12.2.1.1.9 Liquid Systems

Radioactive inputs to the radwaste system sources include fission and activation product radionuclides produced in the core and reactor coolant. The components of the radwaste systems contain varying degrees of activity.

The concentrations of radionuclides present in the process fluids at various locations in the radwaste systems, such as pipes, tanks, filters, ion exchangers, and evaporators, are discussed in section 11.1. Shielding for each component of the radwaste systems as based on maximum activity conditions is shown in tables 12.2.1-25 through 12.2.1-30. Radiation sources in the various pumps in this system are assumed to be identical to the liquid sources in the tank from which the pump takes suction.

12.2.1.1.10 Gaseous Radwaste System

Radiation sources for each component of the waste gas system are based on operation with the maximum activity conditions as given in sections 11.1 and 11.3.

Table 12.2.1-33 lists the distribution of the radioactive gas inventory associated with operation of the gaseous radwaste system. A volume control tank purge rate of 0.7 sf^3/min is assumed. The values represent the design activity distribution with 1-percent defective fuel and the purge system operating. They are used in defining the maximum activities in the waste gas system.

The volume control tank stripping fractions used in establishing the activity distributions are given in table 12.2.1-34. These stripping fractions are calculated using the equation given in table 11.1-1 and assuming a 0.7-sf^3/min volume control tank purge rate.

The Kr-85 inventory in the gaseous radwaste system for various full-power operating times is listed in table 12.2.1-35. These values assume no leakage from the system.

The radioactive gases removed from the RCS at the volume control tank are continuously recirculated through a gas decay tank and other gaseous radwaste system equipment, including the hydrogen recombiners and waste gas compressors. The gamma ray source strengths for this equipment are derived from cold shutdown procedures during which the radioactive gases are stripped from the RCS. Since the gases are continuously recirculated, the gamma ray source strengths for the hydrogen recombiner, waste gas compressor, and gas decay tank are identical. Table 12.2.1-36 lists the activities for the recirculation equipment.

12.2.1.1.11 Instrument Calibration Source

For the purpose of instrument calibration, a Cs-137 source is provided in the instrument calibration lab. This source is shielded to keep personnel exposures limits within ALARA limits.

12.2.1.2 Sources for Shutdown

In the reactor shutdown condition, the only additional significant sources requiring permanent shielding consideration are the spent reactor fuel, the residual heat removal system, and the incore detector system. Individual components may require shielding during shutdown due to deposited crud material. Estimates of accumulated crud are given in paragraph 12.2.1.1. The radiation sources in the RCS and other systems addressed in paragraph 12.2.1.1 are bounded by the sources given for full power operation with the exception of a short time period, i.e., less than 24 h, following shutdown, during which the fission product spiking phenomenon and crud

bursts can result in increased radiation sources. The spiking phenomenon involves the release of a portion of the accumulated water soluble salts from the interior cladding surface, e.g., iodine and cesium, and gases, e.g., xenon and krypton, of defective fuel rods during the shutdown and coolant depressurization.(1) Crud bursts are the resuspension or solubilization of a portion of the accumulated deposited corrosion products into the RCS during shutdown such as during oxygenization of the reactor coolant. However, special shielding considerations to accommodate these increases should be unnecessary due to several factors, including:

A. The spike or crud burst release is of short duration (generally less than 6 h).

B. The CVCS is generally in operation at full reactor coolant purification capability during the shutdown (130 gal/min).

C. The reactor coolant activity is limited by the Technical Specifications to approximately one-fourth of the design basis activity levels.

The maximum gamma ray source strengths in the residual heat removal system are given in table 12.2.1-37 for 4 and 8 h after reactor shutdown and the contributing nuclides are identified in tables 12.2.1-38 and 12.2.1-39. The system may be placed in operation at approximately 4 h following a shutdown at the maximum shutdown rate. The system removes decay heat from the reactor for the duration of the shutdown. The sources given are maximum values with credit for 4 and 8 h of fission and corrosion product decay and purification.

Core average gamma ray source strengths are given in table 12.2.1-1. These source strengths are used in the evaluation of radiation levels within and around the shutdown reactor. These sources are based on a 3-region core with the regions operated at 300, 600, and 900 effective full-power days, respectively. Spent fuel gamma ray source strengths are given in table 12.2.1-18. These source strengths are used in the evaluation of radiation levels for spent fuel handling, storage, and shipping. These sources are based on 1 core region operated for 900 effective full-power days. All of these sources may be put on a per unit volume of homogenized core basis by multiplying by the power density of 109.2 W/cm^3. Core average and spent fuel neutron source strengths are given in table 12.2.1-40.

The only source materials, byproduct material, or special nuclear material requiring shielding consideration is the neutron source material used in primary and secondary source rods. Each of the two primary source rods contains a californium-252 spontaneous fission neutron source which emits approximately 6×10^8 neutrons/s when initially placed in the reactor core. The gamma ray and neutron source strengths for the secondary source rods, irradiated for 400 days, are given in table 12.2.1-41. The source rods will be stored in the spent fuel pool after use.

The absorber material used in the control rods is hafnium or silver-indium-cadmium. The gamma ray source strengths associated with the absorber material are listed in table 12.2.1-42 for various times after shutdown. The values are based on an irradiation period of 4 years. There are no significant gamma ray sources associated with the BC absorber.

The absorber material used in the burnable poison rods is borosilicate glass. There are no significant gamma ray sources associated with this absorber.

The material used for the control rod cladding, primary source rod cladding, secondary source rod cladding, burnable poison rod cladding, and inner sheath is type 304 stainless steel with a maximum cobalt content of 0.12 weight percent. The gamma ray source strengths associated with the stainless steel are listed in table 12.2.1-43 for various times after shutdown. The values are based on an irradiation time of 15 years.

Irradiated incore detector and drive cable maximum gamma ray source strengths are given in table 12.2.1-44. These source strengths are used in determining shielding requirements when

detectors are being moved during or following a flux mapping of the reactor core. These source strengths are given for a detector irradiation period of 30 days and a drive cable irradiation period of 400 days and are given in terms of per cm^3 of detector and drive cable. Irradiated incore detector drive cable average gamma ray source strengths are given in table 12.2.1-45. These source strengths are used in determining shielding requirements when the detectors are not in use and for shipment when the detectors have failed. The values are given in terms of per cm^3 of drive cable after an irradiation period of 400 days. Irradiated incore flux thimble gamma ray source strengths are given in table 12.2.1-46. These source strengths are used in determining shielding requirements during refueling operations when the flux thimbles are withdrawn from the reactor core. The values are given in terms of per cm^3 stainless steel for an irradiation period of 15 years. The flux thimbles are made of type 316 stainless steel with a maximum cobalt impurity content of 0.12 weight percent.

12.2.1.3 Sources for Design Basis Accident

The radiation sources of importance for the design basis accident are the containment source and the residual heat removal and containment spray system sources.

The fission product radiation sources considered to be released from the fuel to the containment following a maximum credible accident are based on the assumptions given in TID-14844.[(2)] These assumptions are consistent with those provided in Regulatory Guide 1.4 and Section II.B.2 of NUREG-0737. The integrated gamma ray and beta particle source strengths for various time periods following the postulated accident are given in table 12.2.1-47.

The residual heat removal system and shielding are designed to allow limited access to the residual heat removal pumps following a design basis accident. The sources are based on the assumptions in TID-14844 with only the nongaseous activity being retained in the sump water, which flows in the residual heat removal loop. Noble gases formed by the decay of halogens in the sump water are assumed to be released to the containment and not retained in the water. Credit has been taken for dilution by the RCS volume plus the contents of the refueling water storage tank. Gamma ray source strengths for radiation sources circulating in the residual heat removal loop and associated equipment are given in table 12.2.1-48.

Isotopic fission product sources from the design basis accident, based on the assumptions in TID-14844, are given in table 12.2.1-49 and chapter 15.

12.2.1.4 Stored Radioactivity

The principal sources of activity not enclosed by the power block buildings are:

- The independent spent fuel storage installation.
- The refueling water storage tank.
- The reactor makeup water storage tank.
- The condensate storage tanks.
- The storage of used spent fuel racks.
- The radwaste processing facility.

- Designated radioactive material storage areas.
- The outage storage building.

The content of these tanks is processed by the spent fuel pool purification system, liquid waste processing system, or boron recycle system until the activity in the fluids is sufficiently low to allow the shielding afforded by the concrete tank walls to result in surface dose rate less than 0.25 mR/h.

Radionuclide inventories of the refueling water storage tank, reactor makeup water storage tank, and condensate storage tanks are presented in tables 12.2.1-50 through 12.2.1-52. There are no other significant amounts of radioactive fluids permanently stored outside the power block and radwaste buildings.

Spent fuel is stored in the spent fuel pool until it is placed in a spent fuel cask for temporary storage in the independent spent fuel storage installation pending transport offsite. Storage space is allocated in the radwaste processing facility for storage of spent filter cartridges and packaged spent resins. Radioactive wastes stored inside plant structures are shielded so that there is Zone I access outside the structures. If it becomes necessary to temporarily store radioactive wastes/ materials outside plant structures, radiation protection measures are to be taken by the radiation protection staff to ensure compliance with 10 CFR 20 and to be consistent with the recommendations of Nuclear Regulatory Commission Regulatory Guide 8.8.

Normally low level contaminated DAW is collected in bags and metal containers and is shipped to an offsite vendor for processing and disposal. In the event that an offsite disposal facility is not available then low level DAW can be stored onsite at various locations. DAW with higher levels of contamination or DAW that cannot be sent offsite due to its construction (noncompressible material, nonburnable material, metals, and so forth) are placed in metal containers and stored onsite at various locations in the owner controlled area. This DAW may also be shipped offsite in the future if an NRC approved facility becomes available.

The outage storage building will store materials with low levels of radioactive contamination that are not classified as radioactive waste. Storage will be controlled to comply with the requirements of 10 CFR 20.

The used spent fuel racks will be stored on site in an area other than those already designated or described. This storage location will comply with the requirements of 10 CFR 20.

12.2.1.5 Field Run Pipe Routing

The procedures for routing of radioactive piping are discussed in paragraph 12.3.1.1.2.

12.2.1.6 References

1. Lutz, R. J., and Chubb, W., "Iodine Spiking - Cause and Effect," ANS Transactions, Vol 28, p 649, June 1978.
2. DiNunno, J. J., et al., "Calculation of Distance Factors for Power and Test Reactor Sites," TID-14844, March 1962.

12.2.2 AIRBORNE RADIOACTIVE MATERIAL SOURCES

This section deals with the models, parameters, and sources required to evaluate airborne concentration of radionuclides during plant operations in various plant radiation areas where personnel occupancy is expected.

These sources include the concentrations of radionuclides in the primary system, secondary system, spent fuel pool, and refueling pool. Sources of airborne radioactive material in equipment cubicles, corridors, or operating areas normally occupied by operating personnel may be obtained from the reactor coolant activities given in section 11.1.

The assumptions and parameters required to evaluate the isotopic airborne concentrations in the various applicable regions are listed in table 12.2.2-1. The chemical and volume control system and the residual heat removal system are designed to provide the capability to purify reactor coolant through the purification demineralizer after reactor shutdown and cooldown. This mode of operation will ensure that the effect of activity spikes does not significantly contribute to the containment airborne activity during refueling operations.

Sources resulting from the removal of the reactor vessel head and the movement of spent fuel are dependent on a number of operating characteristics (e.g., coolant chemistry, fuel performance, etc.) and operating procedures followed during and after shutdown. The permissible coolant activity levels following depressurization are based on the noble gases evolved from the reactor coolant system water upon removal of the reactor vessel head. The endpoint limit for coolant cleanup and degasification is established based on maximum permissible concentration considerations and containment ventilation system capabilities of the plant. Operating plant experience has indicated that coolant Xe-133 concentrations of less than 0.05 μCi/g have posed no problem to operating personnel during vessel head removal.

The exposure rates at the surface of the reactor cavity and spent fuel pool water are dependent on the purification capabilities of the reactor cavity and spent fuel pool cleanup systems. A water total activity level of less than 0.005 μCi/g for the dominant gamma-emitting isotopes at the time of refueling has been shown in operating experience to maintain the dose rate at the water surface at less than 2.5 mrem/h.

The detailed listing of the expected airborne isotopic concentrations in all the various plant regions is presented in table 12.2.2-2. The final design of the plant ensures that all the expected airborne isotopic concentrations in all the applicable regions are well below the maximum permissible concentration for the critical organ for the appropriate isotope for occupational workers, adjusted on the basis of expected weekly occupancy in the regions.

12.2.2.1 Model for Calculating Airborne Concentrations

For those regions characterized by a constant leakrate of the radioactive source at constant source strength and a constant exhaust rate of the region, the peak or equilibrium airborne concentration of the radioisotope in the regions is calculated using the following equation:

$$C_i(t) = \frac{(LR)_i A_i (PF)_i (1 - e\lambda_{Ti} t)}{V_{\lambda_{Ti}}}$$

where:

$(LR)_i$ = leak or evaporation rate of the ith radioisotope in the applicable region (g/s).

A_i = activity concentration of the ith leaking or evaporating radioisotope (μCi/g).

$(PF)_i$ = partition factor or the fraction of the leaking activity that is airborne for the ith radioisotope.

λ_{Ti} = total removal rate constant for the ith radioisotope from the applicable region (s^{-1}).

λ_{Ti} = $\lambda_{di} + \lambda_e$, the removal rate constants in s^{-1} due to radioactive decay for the ith radioisotope and the exhaust from the applicable region, respectively.

t = time elapsed from the start of the leak and the time at which the concentration is evaluated (s).

V = free volume of the region in which the leak occurs (cm^3).

$C_i(t)$ = airborne concentration of the ith radioisotope at time t in the applicable region ($\mu Ci/cm^3$).

From the above equation, it is evident that the peak or equilibrium concentration, $C_{\Sigma gi}$, of the ith radioisotope in the applicable region will be given by the following expression:

$$C_{\Sigma gi} = (LR)_i A_i (PF) / V\lambda_{Ti}$$

With high exhaust rates, this peak concentration will be reached within a few hours.

12.2.2.2 Sources Resulting from Design Basis Accidents

The radiation sources from design basis accidents include the design basis inventory of radioactive isotopes in the reactor coolant, plus postulated fission product releases from the fuel. Accident parameters and sources are discussed and evaluated in chapter 15.

TABLE 12.2.1-1

CORE AVERAGE GAMMA RAY SOURCE STRENGTHS AT VARIOUS TIMES AFTER SHUTDOWN

Energy Group (MeV/gamma)	Source Strength at Time after Shutdown (MeV/W-s)				
	12 h	24 h	100 h	1 Week	1 Month
0.20 - -.40	1.8×10^{9}	1.5×10^{9}	8.2×10^{8}	5.8×10^{8}	1.4×10^{8}
0.40 - 0.90	1.1×10^{10}	9.4×10^{9}	6.2×10^{9}	5.3×10^{9}	3.3×10^{9}
0.90 - 1.35	1.9×10^{9}	1.2×10^{9}	5.8×10^{8}	4.2×10^{8}	1.2×10^{8}
1.35 - 1.80	3.8×10^{9}	3.3×10^{0}	2.7×10^{9}	2.3×10^{9}	6.6×10^{8}
1.80 - 2.20	2.6×10^{8}	1.8×10^{8}	1.2×10^{8}	9.2×10^{7}	3.7×10^{7}
2.20 - 2.60	2.5×10^{8}	1.9×10^{8}	1.6×10^{8}	1.4×10^{8}	4.0×10^{7}
2.60 - 3.00	6.5×10^{6}	3.3×10^{6}	2.7×10^{6}	2.4×10^{6}	6.9×10^{5}
3.00 - 4.00	4.5×10^{6}	1.3×10^{6}	1.1×10^{6}	9.4×10^{5}	2.7×10^{5}
	3 Month	6 Months	1 Year	5 Years	
0.20 - 0.40	4.6×10^{7}	2.6×10^{7}	1.5×10^{7}	1.6×10^{4}	
0.40 - 0.90	1.8×10^{9}	9.0×10^{8}	3.2×10^{8}	8.0×10^{7}	
0.90 - 1.35	3.0×10^{7}	1.8×10^{7}	1.3×10^{7}	4.8×10^{6}	
1.35 - 1.80	3.8×10^{7}	1.1×10^{7}	8.1×10^{6}	1.3×10^{6}	
1.80 - 2.20	1.6×10^{7}	1.2×10^{7}	7.5×10^{6}	2.2×10^{5}	
2.20 - 2.60	1.5×10^{6}	1.2×10^{4}	0	0	
2.60 - 3.00	2.7×10^{5}	2.0×10^{2}	0	0	
3.00 - 4.00	1.0×10^{4}	0	0	0	

TABLE 12.2.1-2

RADIATION SOURCES
REACTOR COOLANT NITROGEN-16 ACTIVITY

Position in Loop	Loop Transit Time(s)	Nitrogen-16 Activity (μCi/g)
Leaving core	0.0	189
Leaving reactor vessel	1.1	170
Entering steam generator	1.4	164
Leaving steam generator	5.4	112
Entering reactor coolant pump	6.0	106
Entering reactor vessel	6.8	98
Entering core	9.0	86
Leaving core	9.7	189

Nitrogen-16 Energy Emission

Energy (MeV/gamma)	Intensity (%)
1.75	0.13
2.74	9.76
6.13	60.0
7.12	5.0

TABLE 12.2.1-3

NITROGEN-16 RADIATION SOURCES - PRESSURIZER

Discrete Energy (MeV/gamma)	Specific Source Strength (MeV/g-s)
1.75	1.1×10^{3}
2.74	9.9×10^{3}
6.13	2.0×10^{6}
7.12	1.7×10^{5}

TABLE 12.2.1-4

RADIATION SOURCES - PRESSURIZER

Liquid Phase (1080 ft^3)

Enery Group (MeV/gamma)	Specific Source Strength (MeV/g-s)
0.20 - 0.40	$4.7 \times 10^{5(a)}$
0.40 - 0.90	6.7×10^{5}
0.90 - 1.35	2.9×10^{5}
1.35 - 1.80	1.7×10^{5}
1.80 - 2.20	1.7×10^{5}
2.20 - 2.60	1.8×10^{5}
2.60 - 3.00	2.2×10^{4}
3.00 - 4.00	9.0×10^{3}

Vapor Phase (720 ft^3)

Enery Group (MeV/gamma)	Specific Source Strength (MeV/cm^3-s)
0.20 - 0.40	$9.4 \times 10^{5(a)}$
0.40 - 0.90	8.2×10^{4}
0.90 - 1.35	2.1×10^{3}
1.35 - 1.80	3.4×10^{3}
1.80 - 2.20	4.9×10^{3}
2.20 - 2.60	1.0×10^{4}
2.60 - 3.00	5.3×10^{1}

a. Includes 80-keV Xe-133.

TABLE 12.2.1-5

ISOTOPIC COMPOSITION AND SPECIFIC ACTIVITY OF TYPICAL OUT-OF-CORE CRUD DEPOSITS[a]

Composition (Nuclide)	Activity (μCi/mg) of Deposited Crud for Effective Full Power Years of Plant Operation			
	1 Year	2 Years	5 Years	10 Years
Mn-54	1.0	1.1	1.3	1.4
Fe-59	0.5	0.5	0.5	0.5
Co-58	12.0	12.0	12.0	12.0
Co-60	1.5	2.3	4.0	6.0

a. In addition to corrosion products, about 1.0 μg of mixed actinides and fission products may be present for each 1 g of deposited crud.

TABLE 12.2.1-6 (SHEET 1 OF 15)

RADIATION SOURCES - CVCS
REGENERATIVE HEAT EXCHANGER SOURCE STRENGTHS

	Source Strength (MeV/g-s)	
Energy Group (MeV/gamma)	Tube Side	Shell Side
0.2 - 0.4	3.9×10^{5}[a]	4.7×10^{5}[a]
0.4 - 0.9	3.3×10^{5}	6.7×10^{5}
0.9 - 1.35	1.6×10^{5}	2.9×10^{5}
1.35 - 1.8	6.9×10^{4}	1.8×10^{5}
1.8 - 2.2	1.1×10^{5}	1.7×10^{5}
2.2 - 2.6	7.1×10^{4}	1.8×10^{5}
2.6 - 3.0	2.1×10^{4}	1.1×10^{5}
3.0 - 4.0	6.2×10^{3}	9.0×10^{3}
6.0 - 7.0	-	1.7×10^{7}
7.0 - 7.5	-	1.4×10^{6}

TABLE 12.2.1-6 (SHEET 2 OF 15)

RADIATION SOURCES - CVCS
REGENERATIVE HEAT EXCHANGER SPECIFIC ACTIVITY

	Activity (μCi/g)	
Nuclide	Tube Side	Shell Side
Kr-88	1.3	3.7
Kr-89	-	1.1×10^{-1}
Xe-133	2.6×10^{2}	2.7×10^{2}
Xe-135	5.3	7.3
I-132	-	2.8
I-133	-	4.2
I-135	-	2.3
Rb-88	4.8	4.8
Cs-134	2.3	2.3
Cs-136	2.9	2.9
Cs-138	9.6×10^{-1}	9.6×10^{-1}
N-16	-	1.1×10^{2}

TABLE 12.2.1-6 (SHEET 3 OF 15)

RADIATION SOURCES - CVCS
EXCESS LETDOWN HEAT EXCHANGER (TUBE SIDE)
SOURCE STRENGTHS

Energy Group (MeV/gamma)	Source Strength (MeV/g-s)
0.2 - 0.4	4.7×10^{5}[(a)]
0.4 - 0.9	6.7×10^{5}
0.9 - 1.35	2.9×10^{5}
1.35 - 1.8	1.8×10^{5}
1.8 - 2.2	1.7×10^{5}
2.2 - 2.6	1.8×10^{5}
2.6 - 3.0	1.1×10^{5}
3.0 - 4.0	9.0×10^{3}
6.0 - 7.0	1.7×10^{7}
7.0 - 7.5	1.4×10^{6}

TABLE 12.2.1-6 (SHEET 4 OF 15)

RADIATION SOURCES - CVCS
EXCESS LETDOWN HEAT EXCHANGER (TUBE SIDE)
SPECIFIC ACTIVITY

Nuclide	Activity (μCi/g)
Kr-88	3.7
Kr-89	1.1×10^{-1}
Xe-133	2.7×10^{2}
Xe-135	7.3
I-132	2.8
I-133	4.2
I-135	2.3
Rb-88	4.8
Cs-134	2.3
Cs-136	2.9
Cs-138	9.6×10^{-1}
N-16	1.1×10^{2}

TABLE 12.2.1-6 (SHEET 5 OF 15)

RADIATION SOURCES - CVCS
LETDOWN HEAT EXCHANGER AND SEAL WATER HEAT
EXCHANGER (TUBE SIDE) SOURCE STRENGTHS

Energy Group (MeV/gamma)	Source Strength (MeV/cm^3-s)
0.2 - 0.4	4.7×10^{5}[(a)]
0.4 - 0.0	6.7×10^{5}
0.9 - 1.35	2.9×10^{5}
1.35 - 1.8	1.7×10^{5}
1.8 - 2.2	1.7×10^{5}
2.2 - 2.6	1.8×10^{5}
2.6 - 3.0	2.2×10^{4}
3.0 - 4.0	9.0×10^{3}

TABLE 12.2.1-6 (SHEET 6 OF 15)

RADIATION SOURCES - CVCS
LETDOWN HEAT EXCHANGER AND SEAL WATER HEAT EXCHANGER (TUBE SIDE) SPECIFIC ACTIVITY

Nuclide	Activity (μCi/g)
Kr-88	3.7
Kr-89	1.1×10^{-1}
Xe-133	2.7×10^{2}
Xe-135	7.3
I-132	2.8
I-133	4.2
I-135	2.3
Rb-88	4.8
Cs-134	2.3
Cs-136	2.9
Cs-138	9.6×10^{-1}

TABLE 12.2.1-6 (SHEET 7 OF 15)

RADIATION SOURCES - CVCS
LETDOWN REHEAT HEAT EXCHANGER SOURCE STRENGTHS

	Source Strength (MeV/g-s)	
Energy Group (MeV/gamma)	Tube Side	Shell Side
0.2 - 0.4	4.7×10^{5}[(a)]	4.3×10^{5}[(a)]
0.4 - 0.9	6.7×10^{5}	3.7×10^{5}
0.9 - 1.35	2.9×10^{5}	1.8×10^{5}
1.35 - 1.8	1.7×10^{5}	1.1×10^{5}
1.8 - 2.2	1.7×10^{5}	1.7×10^{5}
2.2 - 2.6	1.8×10^{5}	1.8×10^{5}
2.6 - 3.0	2.2×10^{4}	2.2×10^{4}
3.0 - 4.0	9.0×10^{3}	8.2×10^{3}

TABLE 12.2.1-6 (SHEET 8 OF 15)

RADIATION SOURCES - CVCS
LETDOWN REHEAT HEAT EXCHANGER SPECIFIC ACTIVITY

	Activity (μCi/g)	
Nuclide	Tube Side	Shell Side
Kr-88	3.7	3.7
Kr-89	1.1×10^{-1}	1.1×10^{-1}
Xe-133	2.7×10^{2}	2.7×10^{2}
Xe-135	7.3	7.3
I-132	2.8	-
I-133	4.2	-
I-135	2.3	-
Rb-88	4.8	4.8
Cs-134	2.3	2.3
Cs-136	2.9	2.9
Cs-138	9.6×10^{-1}	9.6×10^{-1}

TABLE 12.2.1-6 (SHEET 9 OF 15)

RADIATION SOURCES - CVCS
LETDOWN CHILLER HEAT EXCHANGER (TUBE SIDE) AND
MODERATING HEAT EXCHANGER SOURCE STRENGTHS

Energy Group (MeV/gamma)	Source Strength (MeV/cm^3-s)
0.2 - 0.4	4.3 x 10$^{5(a)}$
0.4 - 0.9	3.7 x 10^5
0.9 - 1.35	1.8 x 10^5
1.35 - 1.8	1.1 x 10^5
1.8 - 2.2	1.7 x 10^5
2.2 - 2.6	1.8 x 10^5
2.6 - 3.0	2.2 x 10^4
3.0 - 4.0	8.2 x 10^3

LETDOWN CHILLER HEAT EXCHANGER (TUBE SIDE) AND
MODERATING HEAT EXCHANGER SPECIFIC ACTIVITY

Nuclide	Activity (µCi/g)
Kr-88	3.7
Kr-89	1.1 x 10^{-1}
Xe-133	2.7 x 10^2
Xe-135	7.3
Rb-88	4.8
Cs-134	2.3
Cs-136	2.9
Cs-138	9.6 x 10^{-1}

TABLE 12.2.1-6 (SHEET 10 OF 15)

RADIATION SOURCES - CVCS
MIXED BED DEMINERALIZER SOURCE STRENGTHS
FOR 30 FT^3 OF RESIN

Energy Group (MeV/gamma)	Source Strength (MeV/cm^3-s)
0.2 - 0.4	1.8×10^8
0.4 - 0.9	7.7×10^8
0.9 - 1.35	7.7×10^7
1.35 - 1.8	2.6×10^7
1.8 - 2.2	1.4×10^6
2.2 - 2.6	8.4×10^5
2.6 - 3.0	1.9×10^5
3.0 - 4.0	5.7×10^4

MIXED BED DEMINERALIZER SPECIFIC ACTIVITY

Nuclide	Activity ($\mu Ci/cm^3$)
Br-84	6.5×10^{-1}
I-131	1.4×10^4
I-132	1.7×10^2
I-135	3.9×10^2
Rb-88	3.3×10^1
Cs-134	8.5×10^3
Cs-136	7.8×10^2
Ba-137m	6.7×10^3
Cs-138	1.3×10^1
Co-60	1.5×10^2
La-140	3.4×10^1

TABLE 12.2.1-6 (SHEET 11 OF 15)

RADIATION SOURCES - CVCS
CATION BED DEMINERALIZER SOURCE STRENGTHS
FOR 20 FT^3 OF RESIN

Energy Group (MeV/gamma)	Source Strength (MeV/cm^3-s)
0.2 - 0.4	1.1×10^7
0.4 - 0.9	9.8×10^8
0.9 - 1.35	6.5×10^7
1.35 - 1.8	2.3×10^7
1.8 - 2.2	8.3×10^5
2.2 - 2.6	2.8×10^5
2.6 - 3.0	2.8×10^5
3.0 - 4.0	6.6×10^4

CATION BED DEMINERALIZER SPECIFIC ACTIVITY

Nuclide	Activity ($\mu Ci/cm^3$)
Rb-88	5.0×10^1
Cs-134	1.3×10^4
Cs-136	1.2×10^3
Cs-138	1.9×10^1
Ba-137m	1.0×10^4

TABLE 12.2.1-6 (SHEET 12 OF 15)

RADIATION SOURCES - CVCS
BORON THERMAL REGENERATION DEMINERALIZER SOURCE STRENGTHS
FOR 150 FT^3 OF RESIN

Energy Group (MeV/gamma)	Source Strength (MeV/cm^3-s)
0.2 - 0.4	2.7×10^6
0.4 - 0.9	1.8×10^6
0.9 - 1.35	4.4×10^5
1.35 - 1.8	2.4×10^5
1.8 - 2.2	2.1×10^4
2.2 - 2.6	1.4×10^4

BORON THERMAL REGENERATION DEMINERALIZER
SPECIFIC ACTIVITY

Nuclide	Activity ($\mu Ci/cm^3$)
I-131	2.3×10^2
I-132	4.4
I-133	3.7×10^1
I-135	1.0×10^1

TABLE 12.2.1-6 (SHEET 13 OF 15)

RADIATION SOURCES - CVCS
CHEMICAL AND VOLUME CONTROL SYSTEM FILTER
SOURCE STRENGTHS ASSUMED TO ESTABLISH
SHIELDING REQUIREMENTS

Reactor Coolant Filter

Energy Group (MeV/gamma)	Source Strength (MeV/cm^3-s)
0.4 - 0.9	5.7×10^7
0.9 - 1.35	1.5×10^7

Seal Water Injection Filter

Energy Group (MeV/gamma)	Source Strength (MeV/cm^3-s)
0.4 - 0.9	4.8×10^7
0.9 - 1.35	1.2×10^7

Seal Water Return Filter

Energy Group (MeV/gamma)	Source Strength (MeV/cm^3-s)
0.4 - 0.9	1.1×10^7
0.9 - 1.35	3.0×10^6

DIMENSIONS AND COMPOSITION OF CVCS FILTERS
ASSUMED TO ESTABLISH SHIELDING REQUIREMENTS

Filter	Source Dimensions (in.)	Source Composition volume %
Reactor coolant and seal water return	Radius - 3.375 Length - 19	Air – 62 Water - 38
Seal water injection	Radius - 1.3 Length - 20	Air - 90 Stainless steel - 10

TABLE 12.2.1-6 (SHEET 14 OF 15)

RADIATION SOURCES - CVCS
VOLUME CONTROL TANK LIQUID PHASE SOURCE STRENGTHS
FOR A 200-FT^3 LIQUID PHASE

Energy Group (MeV/gamma)	Source Strength (MeV/cm^3-s)
0.2 - 0.4	3.9×10^{5}[(a)]
0.4 - 0.9	3.3×10^{5}
0.9 - 1.35	1.6×10^{5}
1.35 - 1.8	6.9×10^{4}
1.8 - 2.2	1.1×10^{5}
2.2 - 2.6	7.1×10^{4}
2.6 - 3.0	2.1×10^{4}
3.0 - 4.0	6.2×10^{3}

TABLE 12.2.1-6 (SHEET 15 OF 15)

RADIATION SOURCES - CVCS
VOLUME CONTROL TANK VAPOR PHASE SOURCE STRENGTHS
FOR A 200-FT^3 VAPOR PHASE

Energy Group (MeV/gamma)	Source Strength (MeV/cm^3-s)
0.2 - 0.4	5.6×10^{6}[a]
0.4 - 0.9	4.8×10^{5}
0.9 - 1.35	1.1×10^{5}
1.35 - 1.8	3.3×10^{5}
1.8 - 2.2	5.9×10^{5}
2.2 - 2.6	1.2×10^{6}
2.6 - 3.0	6.8×10^{3}
3.0 - 4.0	3.3×10^{3}

a. Includes 80-keV Xe-133.

TABLE 12.2.1-7

STEAM GENERATOR BLOWDOWN DEMINERALIZER SOURCE STRENGTHS FOR 75 FT^3 OF RESIN

Energy Group (MeV/gamma)	Source Strength (MeV/cm^3-s)
0.2 - 0.4	1.3×10^6
0.4 - 0.9	1.3×10^8
0.9 - 1.35	8.0×10^6
1.35 - 1.8	3.1×10^6
1.8 - 2.2	8.5×10^3
2.2 - 2.6	2.3×10^3

TABLE 12.2.1-8

STEAM GENERATOR BLOWDOWN DEMINERALIZER
SPECIFIC ACTIVITY

Nuclide	Activity ($\mu Ci/cm^3$)
I-131	7.8×10^{1}
I-132	3.5
I-135	1.1
Cs-134	1.8×10^{3}
Cs-136	1.3×10^{2}
Ba-137m	1.3×10^{3}
La-140	2.0×10^{-1}

TABLE 12.2.1-9

STEAM GENERATOR BLOWDOWN SPENT RESIN STORAGE TANK
SOURCE STRENGTHS

Energy Group (MeV/gamma)	Source Strength Resin	Liquid (MeV/g-s)
0.2 - 0.4	1.3×10^{6}	1.3×10^{2}
0.4 - 0.9	1.3×10^{8}	1.3×10^{4}
0.9 - 1.35	8.0×10^{6}	8.0×10^{2}
1.35 - 1.8	3.1×10^{6}	3.1×10^{2}
1.8 - 2.2	8.5×10^{3}	8.5×10^{-1}
2.2 - 2.6	2.3×10^{3}	2.3×10^{-1}

TABLE 12.2.1-10

STEAM GENERATOR BLOWDOWN SPENT RESIN STORAGE TANK
SPECIFIC ACTIVITY

	Activity (μCi/g)	
Nuclide	Resin	Liquid
I-131	7.8×10^{1}	7.8×10^{-3}
I-132	3.5	3.5×10^{-4}
I-135	1.1×10^{-1}	1.1×10^{-5}
Cs-134	1.8×10^{3}	1.8×10^{-1}
Cs-136	1.3×10^{2}	1.3×10^{-2}
Ba-137m	1.3×10^{3}	1.3×10^{-1}
La-140	2.0×10^{-1}	2.0×10^{-5}

TABLE 12.2.1-11

STEAM GENERATOR BLOWDOWN PREFILTER SOURCE STRENGTHS

Energy Group (MeV/gamma)	Source Strength (MeV/cm^3-s)
0.2 - 0.4	1.4×10^2
0.4 - 0.9	1.9×10^3
0.9 - 1.35	8.7×10^2
1.35 - 1.8	1.3×10^2
1.8 - 2.2	2.2×10^1
2.2 - 2.6	8.6

TABLE 12.2.1-12

STEAM GENERATOR BLOWDOWN OUTLET AND SPENT RESIN SLUICE FILTER SOURCE STRENGTHS

Energy Group (MeV/gamma)	Source Strength (MeV/cm^3-s)
0.2 - 0.4	1.3×10^6
0.4 - 0.9	1.3×10^8
0.9 - 1.35	8.0×10^6
1.35 - 1.8	3.1×10^6
1.8 - 2.2	8.5×10^3
2.2 - 2.6	2.3×10^3

TABLE 12.2.1-13

DIMENSIONS AND COMPOSITION OF STEAM GENERATOR BLOWDOWN PURIFICATION SYSTEM FILTERS

Filter	Source Dimensions (in.)	Source Composition (volume %)
Blowdown prefilter[a]	Radius - 5.0	Air - 66
	Length - 36	Water - 32
		Stainless steel - 2
Blowdown outlet and spent resin sluice	Radius - 3.375	Air - 62
	Length - 19	Water - 38

a. These are the parameters assumed to establish shielding requirements.

TABLE 12.2.1-14

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TABLE 12.2.1-15

RECYCLE HOLDUP TANK SPECIFIC ACTIVITIES

Nuclide[a]	Liquid Phase 112,000 gal (μCi/g)	Vapor Phase 500 ft^3 ($\mu Ci/cm^3$)
Kr-83m	-	1.1
Kr-85m	-	4.7
Kr-85	-	1.7×10^{1}
Kr-87	1.3	2.9
Kr-88	3.7	8.5
Kr-89	1.1×10^{-1}	2.5×10^{-1}
Xe-131m	-	5.1
Xe-133m	-	4.0×10^{1}
Xe-133	2.7×10^{2}	6.2×10^{2}
Xe-135m	-	1.1
Xe-135	7.3	1.7×10^{1}
Xe-137	-	4.0×10^{-1}
Xe-138	6.4×10^{-1}	1.5
I-132	2.8×10^{-1}	-
I-135	2.3×10^{-1}	-
Rb-88	4.8×10^{-1}	-
Cs-134	-	-
Cs-136	2.9×10^{-1}	-
Cs-138	9.6×10^{-2}	-

a. The nuclides listed by activity are the significant contributors to the source strengths and represent greater than 1 percent of gamma source strengths.

TABLE 12.2.1-16

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TABLE 12.2.1-17

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TABLE 12.2.1-18

SPENT FUEL GAMMA RAY SOURCE STRENGTHS AT VARIOUS TIMES AFTER SHUTDOWN

Energy Group (MeV/gamma)	Source Strength at Time after Shutdown (MeV/W-s)				
	12 h	24 h	100 h	1 Week	1 Month
0.20 - 0.40	1.8×10^{9}	1.6×10^{9}	8.4×10^{5}	6.0×10^{5}	1.6×10^{8}
0.40 - 0.90	1.1×10^{10}	9.7×10^{9}	6.5×10^{9}	5.6×10^{9}	3.6×10^{9}
0.90 - 1.35	2.0×10^{9}	1.4×10^{9}	7.0×10^{8}	5.3×10^{8}	1.6×10^{8}
1.35 - 1.80	3.6×10^{9}	3.2×10^{9}	2.6×10^{9}	2.2×10^{9}	6.4×10^{8}
1.80 - 2.20	3.2×10^{8}	2.5×10^{8}	1.7×10^{8}	1.4×10^{8}	5.7×10^{7}
2.20 - 2.60	2.4×10^{8}	1.8×10^{8}	1.5×10^{8}	1.3×10^{8}	3.8×10^{7}
2.60 - 3.00	5.9×10^{6}	3.2×10^{6}	2.6×10^{8}	2.3×10^{6}	6.6×10^{5}
3.00 - 4.00	4.2×10^{6}	1.2×10^{6}	1.0×10^{6}	9.0×10^{5}	2.6×10^{5}

	3 Months	6 Months	1 Year	5 Years
0.20 - 0.40	5.3×10^{7}	3.1×10^{7}	1.8×10^{7}	2.2×10^{6}
0.40 - 0.90	2.0×10^{9}	1.1×10^{9}	4.8×10^{8}	1.3×10^{8}
0.90 - 1.35	4.6×10^{7}	3.0×10^{7}	2.3×10^{7}	8.7×10^{6}
1.35 - 1.80	4.6×10^{7}	1.8×10^{7}	1.3×10^{7}	2.4×10^{6}
1.80 - 2.20	2.0×10^{7}	1.4×10^{7}	8.8×10^{6}	2.5×10^{5}
2.20 - 2.60	1.5×10^{4}	1.1×10^{4}	0	0
2.60 - 3.00	2.6×10^{4}	2.0×10^{2}	0	0
3.00 - 4.00	1.0×10^{4}	0	0	0

TABLE 12.2.1-19 (SHEET 1 OF 2)

SPENT FUEL POOL SPECIFIC ACTIVITIES
FOR DETERMINING SHIELD WALL THICKNESSES[a]

Nuclide	Concentration for Maximum Failed Fuel (μCi/g)	Concentration for Expected Failed Fuel (μCi/g)
I-130	4.4901-11	4.9425-12
I-131	3.9336-05	3.9248-06
I-133	4.2069-07	4.0335-08
Rb-86	4.3508-07	1.7611-09
Cs-134	5.8617-05	6.4578-07
Cs-136	5.1219-05	2.4711-07
Cs-137	3.8453-05	4.8762-07
Cr-51	2.4155-04	2.4155-04
Mn-54	5.0467-05	5.0467-05
Fe-55	5.1036-08	4.0749-08
Fe-59	2.4597-05	2.4597-05
Co-58	4.9747-04	4.9747-04
Co-60	7.5971-05	7.5971-05
Sr-89	1.0019-07	8.2401-09
Sr-90	3.0765-09	2.5874-10
Y-90	1.4140-10	5.1408-12
Y-91	1.3462-08	1.5375-09
Zr-95	2.4832-05	2.4832-05

TABLE 12.2.1-19 (SHEET 2 OF 2)

Nuclide	Concentration for Maximum Failed Fuel (μCi/g)	Concentration for Expected Failed Fuel (μCi/g)
Nb-95	1.4518-08	1.1500-09
Mo-99	3.3173-06	3.6894-07
Ru-103	2.4432-05	2.4432-05
Ru-106	3.5438-09	2.5288-10
Te-125m	6.6049-09	6.8091-10
Te-127m	7.1135-08	6.8825-09
Te-129m	4.2167-07	3.1090-08
Te-131m	1.3749-08	1.3754-09
Te-132	1.6719-06	1.6204-07
Ba-140	7.3745-08	3.8577-09
La-140	1.9926-09	2.2546-10
Ce-141	1.4593-06	1.4593-06
Ce-143	4.2130-10	3.2189-11
Ce-144	7.5669-07	7.5669-07
Pr-143	1.1310-08	9.0264-10
Np239	--	3.9583-09
Ag-110m	3.5226-08	--

a. These activities are used to verify shield wall thicknesses. For dose assessment (section 12.4), activities in the pool are assumed to be limited administratively so that pool surface dose rates are less than 2.5 mrem/h.

TABLE 12.2.1-20

SPENT FUEL POOL DEMINERALIZER SOURCE STRENGTHS
FOR 30 FT^3 OF RESIN

Energy Group (MeV/gamma)	Source Strength (MeV/cm^3-s)
0.2 - 0.4	2.5×10^4
0.4 - 0.9	1.6×10^6
0.9 - 1.35	7.5×10^5
1.35 - 1.8	1.5×10^4

TABLE 12.2.1-21

SPENT FUEL POOL DEMINERALIZER SPECIFIC ACTIVITY

Nuclide	Activity ($\mu Ci/cm^3$)
Cr-51	23.0
Mn-54	4.5
Fe-59	2.3
Co-58	45.0
Co-60	7.0
Zr-95	2.3
Ru-103	2.3

TABLE 12.2.1-22

SPENT FUEL POOL FILTER SOURCE STRENGTHS

Energy Group (MeV/gamma)	Source Strength (MeV/cm^3-s)
0.4 - 0.9	1.1×10^7
0.9 - 1.35	3.0×10^6

TABLE 12.2.1-23

SPENT FUEL POOL FILTER DIMENSIONS AND COMPOSITION
ASSUMED TO ESTABLISH SHIELDING REQUIREMENTS

Filter	Source Dimensions (in.)	Source Composition (volume %)
Spent fuel pool	Radius - 3.375	Air - 62
	Length - 19	Water - 38

TABLE 12.2.1-24

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TABLE 12.2.1-25

WASTE MONITOR TANK DEMINERALIZER
SPECIFIC ACTIVITY

Nuclide	Activity ($\mu Ci/cm^3$)
I-131	1.2×10^2
I-132	2.0×10^1
I-133	1.0×10^2
I-135	3.0×10^1
Cs-134	1.0×10^2
Cs-136	1.2×10^2

TABLE 12.2.1-26

REACTOR COOLANT DRAIN TANK LIQUID PHASE SPECIFIC ACTIVITY

Nuclide	Activity (μCi/g)
Kr-88	3.7
Kr-89	1.1×10^{-1}
Xe-133	2.7×10^{2}
Xe-135	7.3
I-132	2.8
I-138	4.2
I-135	2.3
Rb-88	4.8
Cs-134	2.3
Cs-136	2.9
Cs-138	9.6×10^{-1}

TABLE 12.2.1-27

REACTOR COOLANT DRAIN TANK VAPOR PHASE SPECIFIC ACTIVITY

Nuclide	Activity ($\mu Ci/cm^3$)
Kr-85m	8.7×10^{-1}
Kr-85	1.6×10^{2}
Kr-87	1.6×10^{-1}
Kr-88	9.9×10^{-1}
Xe-131m	2.1×10^{1}
Xe-133m	6.6×10^{1}
Xe-133	1.8×10^{3}
Xe-135	6.0

TABLE 12.2.1-28

FLOOR DRAIN TANK, WASTE MONITOR TANKS, AND
WASTE HOLDUP TANK SPECIFIC ACTIVITY

Nuclide	Activity (μCi/g)
I-131	2.8
I-132	2.8
I-133	4.2
I-135	2.3
Cs-134	2.3
Cs-136	2.9

TABLE 12.2.1-29

CHEMICAL DRAIN TANK SPECIFIC ACTIVITY

Nuclide	Activity (μCi/g)
Co-60	1.9×10^{-3}
Cs-134	1.1
Ba-137m	7.0×10^{-1}

TABLE 12.2.1-30

SPENT RESIN STORAGE TANK SPECIFIC ACTIVITY

	Activity (μCi/g)	
Nuclide	Resin	Liquid
Br-84	6.5×10^{-1}	6.5×10^{-5}
I-131	1.4×10^{4}	1.4
I-132	1.7×10^{2}	1.7×10^{-2}
I-135	3.9×10^{2}	3.9×10^{-2}
Rb-88	3.3×10^{1}	3.3×10^{-3}
Cs-134	8.5×10^{3}	8.5×10^{-1}
Cs-136	7.8×10^{2}	7.8×10^{-2}
Ba-137m	6.7×10^{3}	6.7×10^{-1}
Cs-138	1.3×10^{1}	1.3×10^{-3}
Co-60	1.5×10^{2}	1.5×10^{-2}
La-140	3.4×10^{1}	3.4×10^{-3}

TABLE 12.2.1-31

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TABLE 12.2.1-32

DELETED

TABLE 12.2.1-33

RADIOACTIVE GAS DISTRIBUTION - GASEOUS RADWASTE SYSTEM OPERATING

Nuclide	Reactor Coolant Activity (Ci/g)	Reactor Coolant Inventory (Ci)	Volume Control Tank Inventory (Ci)	Gas Decay Tank Inventory (Ci)	Total Ci
Kr-83m	4.5×10^{-2}	1.0×10^{2}	1.5×10^{1}	4.5	1.2×10^{2}
Kr-85	5.9×10^{-2}	1.4×10^{2}	4.0	(a)	(a)
Kr-85m	1.9	4.4×10^{2}	8.1×10^{1}	5.8×10^{1}	5.8×10^{2}
Kr-87	1.3	2.9×10^{2}	2.7×10^{1}	5.4	3.2×10^{2}
Kr-88	3.6	8.2×10^{2}	1.2×10^{2}	5.5×10^{1}	1.0×10^{3}
Kr-89	1.1×10^{-1}	2.4×10^{1}	1.4×10^{1}	1.1×10^{-3}	2.4×10^{1}
Xe-131m	2.5×10^{-1}	5.7×10^{1}	1.4×10^{1}	6.2×10^{2}	6.9×10^{2}
Xe-133	6.2×10^{-1}	1.4×10^{1}	3.3×10^{3}	6.6×10^{4}	8.3×10^{2}
Xe-133m	7.4	1.7×10^{3}	3.9×10^{2}	3.3×10^{3}	5.4×10^{3}
Xe-135	6.1	1.4×10^{3}	3.3×10^{2}	4.8×10^{2}	2.2×10^{3}
Xe-135m	4.8×10^{-1}	1.1×10^{2}	4.6×10^{1}	1.9	1.6×10^{2}
Xe-137	1.7×10^{-1}	4.0×10^{1}	2.7×10^{-1}	2.7×10^{-3}	4.0×10^{1}
Xe-138	6.4×10^{-1}	1.5×10^{2}	3.4	1.3×10^{-1}	1.5×10^{2}

a. The Kr-85 activity does not reach equilibrium. (See table 12.2.1-34.)

TABLE 12.2.1-34

VOLUME CONTROL TANK NOBLE GAS STRIPPING FRACTIONS[a]

Nuclide	Stripping Fraction
Kr-83m	0.78
Kr-85m	0.65
Kr-85	0.43
Kr-87	0.83
Kr-88	0.72
Kr-89	0.99
Xe-131m	0.34
Xe-133m	0.37
Xe-133	0.35
Xe-135m	0.93
Xe-135	0.47
Xe-137	0.98
Xe-138	0.94

a. Assuming 0.7-sf^3/min volume control tank purge rate.

TABLE 12.2.1-35

WASTE GAS PROCESSING SYSTEM Kr-85 INVENTORY

Time (effective full-power years)	GRS Kr-85 Inventory (Ci)
1	2.7×10^3
2	5.2×10^3
4	1.0×10^4
6	1.3×10^4
10	2.0×10^4
20	3.1×10^4
30	3.7×10^4
40[a]	3.9×10^4

[a] The renewed operating licenses authorized a 20-year period of extended operation for both VEGP units, resulting in a total plant operating life of 60 years. Since the inventory in the Waste Gas Decay Tanks (WGDTs) has been routinely released during the first 20 years of operation and is expected to continue to be routinely released during future operation, the inventory of the WGDTs accumulated during the first 20 years of operation will be released prior to entering the period of extended operation. Therefore, the stated design capacity of the GWPS remains sufficient, and the analysis of the maximum fission product inventory in the GWPS over a 40-year plant life remains bounding for a 60-year plant life.

TABLE 12.2.1-36

HYDROGEN RECOMBINER, WASTE GAS COMPRESSOR,
AND GAS DECAY TANK SPECIFIC ACTIVITY

Nuclide	Activity ($\mu Ci/cm^3$)
Kr-85m	9.5
Kr-85	8.0×10^1
Kr-87	1.3
Kr-88	1.2×10^1
Xe-131m	2.0×10^1
Xe-133m	1.5×10^2
Xe-133	2.3×10^3
Xe-135	4.9×10^1

TABLE 12.2.1-37

RADIATION SOURCES
RESIDUAL HEAT REMOVAL SYSTEM

Energy Group (MeV/gamma)	Specific Source Strength (MeV/g-s) 4 h after Shutdown	8 h after Shutdown
0.20 - 0.40	4.0×10^{5}[a]	3.5×10^{5}[a]
0.40 - 0.90	2.9×10^{5}	1.7×10^{5}
0.90 - 1.35	1.3×10^{5}	7.3×10^{4}
1.35 - 1.80	3.9×10^{4}	1.5×10^{4}
1.80 - 2.20	4.0×10^{4}	1.1×10^{4}
2.20 - 2.60	4.2×10^{4}	1.1×10^{4}
2.60 - 3.00	3.2×10^{3}	8.5×10^{2}
3.00 - 4.00	1.2×10^{3}	3.3×10^{2}

a. Includes 80 keV Xe-133.

TABLE 12.2.1-38

RESIDUAL HEAT REMOVAL LOOP SPECIFIC ACTIVITY - 4 HOURS AFTER SHUTDOWN

Nuclide	Activity (μCi/g)
Kr-88	1.0
Xe-133	2.6×10^{2}
Xe-135	5.0
I-132	7.2×10^{-1}
I-133	2.4
I-135	9.7×10^{-1}
Rb-88	1.1
Cs-134	1.5
Cs-136	1.9

TABLE 12.2.1-39

RESIDUAL HEAT REMOVAL LOOP SPECIFIC ACTIVITY - 8 HOURS AFTER SHUTDOWN

Nuclide	Activity (μCi/g)
Kr-88	2.7×10^{-1}
Xe-133	2.5×10^{2}
I-132	3.0×10^{-1}
I-133	1.4
I-135	4.1×10^{-1}
Cs-134	9.7×10^{-1}
Cs-136	1.2
Rb-88	3.0×10^{-1}

TABLE 12.2.1-40

CORE AVERAGE AND SPENT FUEL NEUTRON SOURCE STRENGTHS AT VARIOUS TIMES AFTER SHUTDOWN[(a)]

Time after Shutdown	Core Average (neut/W-s)	Spent Fuel (neut/W-s)
12 h	9.6	24
24 h	9.6	24
100 h	9.5	24
1 week	9.5	23
1 month	9.1	23
3 months	8.3	21
6 months	7.4	19
1 year	6.3	16
5 years	4.8	12

a. Eighty-three to 93 percent of the neutron source strength is due to the spontaneous fission of curium-242 and curium-244. The curium spontaneous fission neutron spectrum is quite similar to that of californium-252. The californium-252 spontaneous fission neutron spectrum may be expressed by the Watt formula as follows:

$$\chi(E) = 0.37 \exp(-0.88E) \sinh\left(\sqrt{2.0\,E}\right)$$

where E is the neutron energy and x(E) is normalized so that

$$\int_0^\infty \chi(E)\, dE = 1$$

TABLE 12.2.1-41

IRRADIATED Sb-Be SECONDARY SOURCE ROD SOURCE STRENGTHS

	Gamma Ray Source Strength at Time after Shutdown (MeV/cm -s)					
Energy Group (MeV/gamma)	1 Day	1 Week	1 Month	6 Months	1 Year	5 Years
0.20 - 0.40	3.0×10^{10}	2.9×10^{10}	2.5×10^{10}	1.1×10^{10}	3.7×10^{9}	2.2×10^{7}
0.40 - 0.90	1.1×10^{13}	7.0×10^{12}	4.6×10^{12}	8.1×10^{11}	9.7×10^{10}	1.8×10^{8}
0.90 - 1.35	6.7×10^{11}	4.8×10^{11}	3.4×10^{11}	6.0×10^{10}	7.0×10^{9}	-
1.35 - 1.80	7.6×10^{12}	7.1×10^{12}	5.5×10^{11}	9.7×10^{11}	1.2×10^{11}	-
1.80 - 2.20	9.8×10^{11}	9.1×10^{11}	7.0×10^{11}	1.2×10^{10}	1.5×10^{10}	-

Neutron Source Strength at Time after Shutdown (neut/cm -s)

1 Day	1 Week	1 Month	6 Months	1 Year	5 Years
4.5×10^{8}	4.2×10^{8}	3.2×10^{8}	5.8×10^{7}	6.8×10^{6}	-

a. The secondary source cross-sectional area is 0.582 cm^2 per rod. The Sb-Be material density is 3.38 g/cm^3.

b. The secondary source rod cross-sectional area is 0.582 cm^2 per rod. The average neutron energy is 30 keV. The Sb-Be material density is 3.38 g/cm^3.

TABLE 12.2.1-42 (SHEET 1 of 2)

CONTROL ROD SOURCE STRENGTHS

Hafnium Control Rod[(a)]

400-Day Irradiation

Source Strength at Time after Shutdown (MeV/cm^3 -s)

Energy Group (MeV)	1 Day	1 Week	1 Month	6 Months	1 Year
0.20 - 0.40	2.2×10^{10}	2.0×10^{10}	1.4×10^{10}	1.3×10^{9}	1.0×10^{8}
0.40 - 0.90	1.9×10^{11}	1.7×10^{11}	1.2×10^{11}	1.0×10^{10}	5.0×10^{8}
0.90 - 1.35	2.6×10^{10}	2.5×10^{10}	2.2×10^{9}	8.9×10^{9}	2.9×10^{9}

15-Year Irradiation

Source Strength at Time after Shutdown (MeV/cm^3 -s)

Energy Group (MeV)	1 Day	1 Week	1 Month	6 Months	1 Year
0.20 - 0.40	4.7×10^{10}	4.3×10^{10}	2.9×10^{10}	3.8×10^{9}	6.2×10^{8}
0.40 - 0.90	3.5×10^{11}	3.2×10^{11}	2.2×10^{11}	1.9×10^{10}	9.3×10^{8}
0.90 - 1.35	2.8×10^{11}	2.7×10^{11}	2.4×10^{11}	9.7×10^{10}	3.1×10^{10}

TABLE 12.2.1-42 (SHEET 2 of 2)

Irradiated Ag-In-Cd Control Rod [b,c]

Energy Group (MeV/gamma)	Source Strength at Time after Shutdown (MeV/cm^3 -s)					
	1 Day	1 Week	1 Month	6 Months	1 Year	5 Years
0.20 - 0.40	2.3×10^8	2.3×10^8	2.2×10^8	1.4×10^8	8.5×10^7	1.5×10^6
0.40 - 0.90	1.1×10^{12}	1.1×10^{12}	1.0×10^{12}	6.6×10^{11}	4.0×10^{11}	7.1×10^9
0.90 - 1.35	2.0×10^{11}	1.9×10^{11}	1.8×10^{11}	1.2×10^{11}	7.2×10^{10}	1.3×10^9
1.35 - 1.80	3.7×10^{11}	3.7×10^{11}	3.4×10^{11}	2.3×10^{11}	1.4×10^{11}	2.5×10^9

a. Source strengths are expressed per cm^3 of absorber. Density of the hafnium absorber is 13.31 g/cm^3.

b. The absorber cross-sectional area is 0.589 cm^2/rod.

c. The absorber material density is 10.17 g/cm^3.

TABLE 12.2.1-43

IRRADIATED TYPE 304 STAINLESS STEEL (0.12 WEIGHT PERCENT COBALT)
CLADDING SOURCE STRENGTHS[a]

Source Strength at Time after Shutdown (Mev/cm -s)

Energy Group (MeV/gamma)	1 Day	1 Week	1 Month	6 Months	1 Year	5 Years
0.20 - 0.40	7.1×10^{9}	6.1×10^{9}	3.4×10^{9}	8.3×10^{7}	9.9×10^{5}	-
0.40 - 0.90	3.1×10^{10}	2.9×10^{10}	2.6×10^{10}	1.2×10^{10}	6.4×10^{9}	2.3×10^{8}
0.90 - 1.35	2.4×10^{11}	2.3×10^{11}	2.3×10^{11}	2.1×10^{11}	2.0×10^{11}	1.2×10^{10}
1.35 - 1.80	1.9×10^{8}	1.8×10^{8}	1.4×10^{8}	3.3×10^{7}	5.4×10^{6}	-

a. The various cladding cross-sectional areas per rod are:

Ag-In-Cd control rod	$0.136\ cm^2$
Cf-252 primary source rod	$0.136\ cm^2$
Sb-Be secondary source rod	$0.136\ cm^2$
Burnable poison rod (including inner sheath)	$0.159\ cm^2$
Hafnium control rod	$0.136\ cm^2$

TABLE 12.2.1-44

IRRADIATED INCORE DETECTOR AND DRIVE CABLE MAXIMUM WITHDRAWAL SOURCE STRENGTHS[(a)]

Energy Group (MeV/gamma)	Incore Detector (MeV/cm^3-s)	Drive Cable (MeV/cm^3-s)
0.20 - 0.40	3.8×10^{10}	6.0×10^{8}
0.40 - 0.90	1.6×10^{11}	5.1×10^{10}
0.90 - 1.35	1.1×10^{11}	1.6×10^{10}
1.35 - 1.80	1.1×10^{11}	3.1×10^{8}
1.80 - 2.20	2.9×10^{10}	3.8×10^{10}
2.20 - 2.60	3.1×10^{10}	1.3×10^{9}
2.60 - 3.00	1.6×10^{10}	1.3×10^{9}
3.00 - 4.00	2.1×10^{10}	3.1×10^{8}
4.00 - 5.00	1.5×10^{10}	-
5.00 - 6.00	1.4×10^{9}	-

a. The effective diameter and length of the incore detector are 0.48 and 5.33 cm, respectively. The effective cross-sectional area of the drive cable is 0.095 cm^2.

TABLE 12.2.1-45

IRRADIATED INCORE DETECTOR DRIVE CABLE SOURCE STRENGTHS[a]

Energy Group (MeV/gamma)	Source Strength at Time after Shutdown (MeV/cm -s)						
	8 h	1 Day	1 Week	1 Month	6 Months	1 Year	5 Years
0.20 - 0.40	5.8×10^{8}	5.7×10^{8}	4.9×10^{8}	2.8×10^{8}	9.8×10^{6}	2.8×10^{5}	-
0.40 - 0.90	1.7×10^{10}	1.2×10^{10}	1.2×10^{10}	1.1×10^{10}	7.8×10^{9}	5.0×10^{9}	2.0×10^{8}
0.90 - 1.35	1.6×10^{10}	1.6×10^{10}	1.5×10^{10}	1.2×10^{10}	5.6×10^{9}	4.2×10^{9}	2.5×10^{9}
1.35 - 1.80	2.1×10^{7}	1.1×10^{7}	1.1×10^{7}	8.5×10^{6}	2.0×10^{6}	3.3×10^{5}	-
1.80 - 2.20	4.5×10^{9}	6.1×10^{7}	-	-	-	-	-
2.20 - 2.60	1.5×10^{8}	2.0×10^{6}	-	-	-	-	-
2.60 - 3.00	1.6×10^{8}	2.1×10^{6}	-	-	-	-	-
3.00 - 4.00	3.6×10^{7}	5.0×10^{5}	-	-	-	-	-

a. The drive cable effective cross-sectional area is 0.095 cm^2.

TABLE 12.2.1-46

IRRADIATED TYPE 316 STAINLESS STEEL (0.12 WEIGHT PERCENT COBALT)
FLUX THIMBLE SOURCE STRENGTHS[a]

Energy Group (MeV/gamma)	Source Strength at Time after Shutdown (MeV/cm -s)						
	12 h	1 Day	1 Week	1 Month	6 Months	1 Year	5 Years
0.20 - 0.40	6.5×10^{9}	6.4×10^{9}	5.5×10^{9}	3.1×10^{9}	7.5×10^{7}	9.0×10^{5}	-
0.40 - 0.90	5.7×10^{10}	3.7×10^{10}	3.3×10^{10}	2.8×10^{10}	1.2×10^{10}	6.2×10^{9}	2.2×10^{8}
0.90 - 1.35	2.4×10^{11}	2.4×10^{11}	2.4×10^{11}	2.3×10^{11}	2.2×10^{11}	2.0×10^{11}	1.2×10^{11}
1.35 - 1.80	2.9×10^{8}	2.2×10^{8}	2.1×10^{8}	1.7×10^{7}	3.9×10^{7}	6.4×10^{6}	-
1.80 - 2.20	2.1×10^{10}	8.2×10^{8}	-	-	-	-	-
2.20 - 2.60	6.8×10^{8}	2.7×10^{7}	-	-	-	-	-
2.60 - 3.00	7.2×10^{8}	2.9×10^{7}	-	-	-	-	-
3.00 - 4.00	1.7×10^{8}	6.7×10^{2}	-	-	-	-	-

a. The flux thimble cross-sectional area is 0.270 cm^2.

TABLE 12.2.1-47 (SHEET 1 OF 2)

INTEGRATED GAMMA RAY AND BETA SOURCE STRENGTHS AT VARIOUS TIMES FOLLOWING A DESIGN BASIS ACCIDENT (TID-14844 RELEASE FRACTIONS)

Energy Group (MeV/gamma)	Source Strength at Time after Release (MeV/W)				
	0.5 h	1 h	2 h	8 h	1 Day
0.20 - 0.40	1.2×10^{12}	2.2×10^{12}	3.9×10^{12}	1.3×10^{13}	3.2×10^{13}
0.40 - 0.90	8.2×10^{12}	1.5×10^{13}	2.4×10^{13}	5.3×10^{13}	8.6×10^{13}
0.90 - 1.35	3.5×10^{12}	6.2×10^{12}	1.1×10^{13}	2.5×10^{13}	3.8×10^{13}
1.35 - 1.80	3.5×10^{12}	6.2×10^{12}	1.0×10^{13}	2.1×10^{13}	3.0×10^{13}
1.80 - 2.20	1.9×10^{12}	3.3×10^{12}	5.4×10^{12}	1.1×10^{13}	1.3×10^{13}
2.20 - 2.60	2.2×10^{12}	3.9×10^{12}	6.6×10^{12}	1.3×10^{13}	1.5×10^{13}
2.60 - 3.00	3.2×10^{11}	5.4×10^{11}	8.4×10^{11}	1.3×10^{12}	1.5×10^{12}
3.00 - 4.00	2.3×10^{11}	3.4×10^{11}	4.7×10^{11}	6.8×10^{11}	7.4×10^{11}
4.00 - 5.00	9.4×10^{10}	1.0×10^{11}	1.1×10^{11}	1.4×10^{11}	1.5×10^{11}
5.00 - 6.00	1.0×10^{9}	1.0×10^{9}	1.0×10^{9}	1.0×10^{9}	1.0×10^{9}
Beta	1.2×10^{13}	2.2×10^{13}	3.6×10^{13}	8.6×10^{11}	1.5×10^{14}

TABLE 12.2.1-47 (SHEET 2 OF 2)

Energy Group (MeV/gamma)	Source Strength at Time after Release (MeV/W)			
	1 Week	1 Month	6 Months	1 Year
0.20 - 0.40	1.2×10^{14}	2.3×10^{14}	2.5×10^{14}	2.6×10^{14}
0.40 - 0.90	1.7×10^{14}	2.7×10^{14}	5.1×10^{14}	5.9×10^{14}
0.90 - 1.35	4.9×10^{13}	5.3×10^{13}	5.2×10^{13}	6.0×10^{13}
1.35 - 1.80	4.8×10^{13}	7.3×10^{13}	8.4×10^{13}	8.6×10^{13}
1.80 - 2.20	1.4×10^{13}	1.5×10^{13}	1.8×10^{13}	1.9×10^{13}
2.20 - 2.60	1.6×10^{13}	1.8×10^{13}	1.9×10^{13}	1.9×10^{13}
2.60 - 3.00	1.7×10^{12}	1.7×10^{12}	1.7×10^{12}	1.7×10^{12}
3.00 - 4.00	7.4×10^{11}	7.5×10^{11}	7.6×10^{11}	7.6×10^{11}
4.00 - 5.00	1.5×10^{11}	1.5×10^{11}	1.5×10^{11}	1.5×10^{11}
5.00 - 6.00	1.0×10^{9}	1.0×10^{9}	1.0×10^{9}	1.0×10^{9}
Beta	3.8×10^{14}	6.5×10^{14}	1.0×10^{15}	1.3×10^{15}

TABLE 12.2.1-48

SOURCE STRENGTH IN THE RESIDUAL
HEAT REMOVAL LOOP AT VARIOUS TIMES FOLLOWING AN EQUIVALENT
FULL-CORE MELTDOWN ACCIDENT

Energy Group (MeV/gamma)	Source Strength at Time after Release (MeV/W) 0 h	0.5 h	1 h	2 h	8 hr
0.20 - 0.40	2.8×10^{-1}	1.9×10^{-1}	1.7×10^{-1}	1.5×10^{-1}	1.3×10^{-1}
0.40 - 0.90	3.2	2.5	2.0	1.4	5.6×10^{-1}
0.90 - 1.35	1.6	9.9×10^{-1}	8.6×10^{-1}	6.8×10^{-1}	3.1×10^{-1}
1.35 - 1.80	1.2	6.7×10^{-1}	5.7×10^{-1}	4.5×10^{-1}	2.0×10^{-1}
1.80 - 2.20	1.1	6.9×10^{-2}	5.4×10^{-2}	4.0×10^{-2}	1.4×10^{-2}
2.20 - 2.60	2.4×10^{-1}	4.7×10^{-2}	3.8×10^{-2}	2.9×10^{-2}	1.3×10^{-2}
2.60 - 3.00	3.0×10^{-1}	5.0×10^{-3}	2.4×10^{-3}	6.3×10^{-4}	-
3.00 - 4.00	1.6×10^{-1}	3.3×10^{-2}	1.6×10^{-2}	4.5×10^{-3}	-
4.00 - 5.00	2.8×10^{-1}	4.5×10^{-4}	2.3×10^{-4}	-	-
5.00 - 6.00	1.4×10^{-3}	-	-	-	-

	1 Day	1 Week	1 Month	6 Months	1 Year
0.20 - 0.40	1.2×10^{-1}	6.7×10^{-2}	9.7×10^{-3}	1.9×10^{-4}	1.1×10^{-4}
0.40 - 0.90	2.9×10^{-1}	5.5×10^{-2}	2.6×10^{-2}	6.4×10^{-3}	2.3×10^{-3}
0.90 - 1.35	7.3×10^{-2}	3.2×10^{-3}	8.3×10^{-4}	1.3×10^{-4}	9.6×10^{-5}
1.35 - 1.80	5.4×10^{-2}	1.7×10^{-2}	4.7×10^{-3}	8.2×10^{-5}	5.8×10^{-5}
1.80 - 2.20	3.2×10^{-3}	6.6×10^{-4}	2.6×10^{-4}	8.5×10^{-5}	5.4×10^{-5}
2.20 - 2.60	3.4×10^{-3}	9.8×10^{-4}	2.8×10^{-4}	-	-

TABLE 12.2.1-49 (SHEET 1 OF 3)

POST-ACCIDENT SOURCE TERMS
SPECIFIC ACTIVITIES FOR TID-14844 RELEASE FRACTIONS

Nuclide	Containment Atmosphere (Ci/cm^3)	Reactor Coolant (Ci/cm^3)	Containment Sump Ci/cm^3)
Sr-89 N/A	2.49×10^{-3}	5.16×10^{-4}	
Sr-90 N/A	2.24×10^{-4}	4.65×10^{-5}	
Y-90	N/A	2.36×10^{-4}	4.91×10^{-5}
Y-91	N/A	3.38×10^{-3}	7.01×10^{-4}
Zr-95 N/A	4.91×10^{-3}	1.02×10^{-3}	
Zr-97 N/A	4.91×10^{-3}	1.02×10^{-3}	
Nb-95 N/A	4.91×10^{-3}	1.02×10^{-3}	
Nb-95m	N/A	6.14×10^{-5}	1.27×10^{-5}
Nb-97 N/A	5.22×10^{-3}	1.08×10^{-3}	
Mo-99 N/A	5.53×10^{-3}	1.15×10^{-3}	
Tc-99m	N/A	4.91×10^{-3}	1.02×10^{-3}
Ru-103	N/A	5.22×10^{-3}	1.08×10^{-3}
Ru-106	N/A	1.81×10^{-3}	3.76×10^{-4}
Rh-103m	N/A	5.22×10^{-3}	1.08×10^{-3}
Rh-105	N/A	3.38×10^{-3}	7.01×10^{-4}
Rh-106	N/A	2.12×10^{-3}	4.40×10^{-4}
Ag-110m	N/A	1.97×10^{-5}	4.08×10^{-6}
Ag-111	N/A	1.75×10^{-4}	3.63×10^{-5}
Sb-125	N/A	3.01×10^{-5}	6.25×10^{-6}
Sb-127	N/A	3.38×10^{-4}	7.01×10^{-5}
Te-127	N/A	3.38×10^{-4}	7.01×10^{-5}

TABLE 12.2.1-49 (SHEET 2 OF 3)

Nuclide	Containment Atmosphere (Ci/cm^3)	Reactor Coolant (Ci/cm^3)	Containment Sump Ci/cm^3)
Te-129	N/A	1.01×10^{-3}	2.10×10^{-4}
Te-129m	N/A	2.73×10^{-4}	5.67×10^{-5}
Te-132	N/A	4.30×10^{-3}	8.92×10^{-4}
Cs-134	N/A	7.06×10^{-4}	1.47×10^{-4}
Cs-136	N/A	1.97×10^{-4}	4.08×10^{-5}
Cs-137	N/A	3.38×10^{-4}	7.01×10^{-5}
Ba-137m	N/A	3.01×10^{-4}	6.25×10^{-5}
Ba-140	N/A	5.22×10^{-3}	1.08×10^{-3}
La-140 N/A	5.53×10^{-3}	1.15×10^{-3}	
Ce-141	N/A	4.91×10^{-3}	1.02×10^{-3}
Ce-143	N/A	4.30×10^{-3}	8.92×10^{-4}
Ce-144	N/A	3.69×10^{-3}	7.65×10^{-4}
Pr-143 N/A	4.30×10^{-3}	8.92×10^{-4}	
Pr-144 N/A	3.69×10^{-3}	7.65×10^{-4}	
Nd-147	N/A	1.93×10^{-3}	4.02×10^{-4}
Nd-154	N/A	1.29×10^{-4}	2.68×10^{-5}
Pm-147	N/A	3.38×10^{-4}	7.01×10^{-5}
Pm-148	N/A	6.14×10^{-4}	1.27×10^{-4}
Pm-148m	N/A	2.73×10^{-4}	5.67×10^{-5}
Pm-149	N/A	1.93×10^{-3}	4.02×10^{-4}
Pm-154	N/A	1.63×10^{-4}	3.38×10^{-5}
Sm-153	N/A	1.93×10^{-3}	4.02×10^{-4}

TABLE 12.2.1-49 (SHEET 3 OF 3)

Nuclide	Containment Atmosphere (Ci/cm^3)	Reactor Coolant (Ci/cm^3)	Containment Sump Ci/cm^3)
Eu-156	N/A	1.01×10^{-3}	2.10×10^{-4}
Kr-83m	1.54×10^{-4}	3.69×10^{-2}	N/A
Kr-85m	3.47×10^{-4}	8.29×10^{-2}	N/A
Kr-85 8.48×10^{-6}	2.03×10^{-3}		N/A
Kr-87 6.29×10^{-4}	1.50×10^{-1}		N/A
Kr-88 8.99×10^{-4}	2.15×10^{-1}		N/A
Kr-89 1.12×10^{-3}	2.67×10^{-1}		N/A
Xe-131m	8.99×10^{-6}	2.15×10^{-3}	N/A
Xe-133m	3.72×10^{-4}	8.91×10^{-2}	N/A
Xe-133	2.44×10^{-3}	5.83×10^{-1}	N/A
Xe-135m	5.14×10^{-4}	1.23×10^{-1}	N/A
Xe-135	5.39×10^{-4}	1.29×10^{-1}	N/A
Xe-138	2.05×10^{-3}	4.91×10^{-1}	N/A
Br-82 N/A	4.61×10^{-4}	9.56×10^{-5}	
Br-83 N/A	1.84×10^{-2}	3.82×10^{-3}	
Br-84 N/A	3.07×10^{-2}	6.37×10^{-3}	
I-130 5.65×10^{-6}	2.70×10^{-3}	5.61×10^{-4}	
I-131 3.15×10^{-4}	1.50×10^{-1}	3.12×10^{-2}	
I-132 4.62×10^{-4}	2.21×10^{-1}	4.59×10^{-2}	
I-133 6.42×10^{-4}	3.07×10^{-1}	6.37×10^{-2}	
I-134 7.06×10^{-4}	3.38×10^{-1}	7.01×10^{-2}	
I-135 6.04×10^{-4}	2.89×10^{-1}	5.99×10^{-2}	

TABLE 12.2.1-50

REFUELING WATER STORAGE TANK
SPECIFIC ACTIVITIES

Nuclide	Specific Activity (μCi/g)	Nuclide	Specific Activity (μCi/g)	Nuclide	Specific Activity (μCi/g)
Br-83	-	Te-129m	1.1×10^{-3}	Pr-143	3×10^{-5}
Kr-83m	-	Te-129	7.5×10^{-4}	Pr-144	7.7×10^{-8}
Kr-85m	2.9×10^{-9}	Te-132	3×10^{-3}	Cr-51	1.3×10^{-5}
Kr-85	4×10^{-2}	I-129	4.1×10^{-8}	Mn-54	5.4×10^{-6}
Kr-88	-	I-130	8.4×10^{-9}	Fe-59	1.4×10^{-6}
Rb-88	-	I-131	9×10^{-2}	Co-58	3×10^{-5}
Sr-89	2.7×10^{-4}	I-132	3.1×10^{-3}	Co-60	2.6×10^{-5}
Sr-90	8.6×10^{-5}	I-133	2.3×10^{-4}	H-3	1.8
Sr-91	-	Xe-131m	9.6×10^{-2}	Ag-110m	1.5×10^{-4}
Sr-92	-	Xe-133m	8×10^{-2}	Ag-110	1.5×10^{-4}
Y-90	8.5×10^{-5}	Xe-133	6.0	Rh-106	1.8×10^{-5}
Y-91m	2.4×10^{-8}	Xe-135m	1.9×10^{-3}	Te-127m	2.1×10^{-4}
Y-91	4.2×10^{-5}	Xe-135	2.2×10^{-3}	Te-127	2.1×10^{-4}
Y-92	-	Cs-134	4.8×10^{-1}	Rh-103m	1.4×10^{-6}
Y-93	-	Cs-135	3.4×10^{-9}	Te-125m	1.8×10^{-5}
Zr-95	1.5×10^{-6}	Cs-136	1.3×10^{-1}	Fe-55	4.8×10^{-4}
Nb-95m	2.2×10^{-7}	Cs-137	1.1		
Nb-95	3.8×10^{-5}	Ba-137m	1.1		
Mo-99	5.5×10^{-3}	Ba-140	1.8×10^{-4}		
Tc-99m	7×10^{-3}	La-140	2×10^{-4}		
Tc-99	5×10^{-8}	Ce-141	8.1×10^{-8}		
Ru-103	1.4×10^{-6}	Ce-143	3.7×10^{-7}		
Ru-106	1.8×10^{-5}	Ce-144	7.7×10^{-8}		

TABLE 12.2.1-51

REACTOR MAKEUP WATER STORAGE TANK
SPECIFIC ACTIVITIES

Nuclide	Specific Activity (μCi/g)	Nuclide	Specific Activity (μCi/g)	Nuclide	Specific Activity (μCi/g)
Kr-83m	4.6×10^{-3}	Rb-86	1.1×10^{-5}	Zr-95	6.5×10^{-8}
Kr-85m	2×10^{-3}	Rb-88	2.4×10^{-3}	Nb-95	6.5×10^{-8}
Kr-95	7.3×10^{-3}	Rb-89	1.1×10^{-4}	Mo-99	7.5×10^{-5}
Kr-87	1.3×10^{-3}	Cs-134	1.2×10^{-3}	Tc-99m	6.9×10^{-5}
Kr-88	3.6×10^{-3}	Cs-136	1.5×10^{-3}	Ru-103	5.7×10^{-8}
Kr-89	1.1×10^{-4}	Cs-137	7.5×10^{-4}	Ru-106	1.4×10^{-8}
Xe-131m	2.2×10^{-3}	Ba-137m	1.4×10^{-4}	Rh-103m	5.7×10^{-8}
Xe-133m	1.7×10^{-2}	Cs-138	9.6×10^{-5}	Rh-106	1.4×10^{-8}
Xe-133	2.7×10^{-1}	H-3	3.5	Ag-110m	1.4×10^{-7}
Xe-135m	4.8×10^{-4}	Cr-51	5.5×10^{-7}	Te-125m	2.8×10^{-8}
Xe-135	7.2×10^{-3}	Mn-54	4.4×10^{-8}	Te-127m	2.9×10^{-7}
Xe-137	1.7×10^{-4}	Mn-56	2×10^{-6}	Te-127	1.2×10^{-6}
Xe-138	6.4×10^{-4}	Fe-55	2×10^{-7}	Te-129m	1.9×10^{-6}
Br-83	9.5×10^{-6}	Fe-59	5.2×10^{-8}	Te-129	1.8×10^{-6}
Br-84	4.7×10^{-7}	Co-58	1.5×10^{-6}	Te-131m	2.6×10^{-6}
Br-85	6×10^{-7}	Co-60	1.9×10^{-7}	Te-131	1.2×10^{-6}
I-127	-	Sr-89	4.3×10^{-7}	Te-132	2.9×10^{-5}
I-129	-	Sr-90	1.2×10^{-8}	Te-134	3×10^{-6}
I-130	2.1×10^{-5}	Sr-91	6.2×10^{-7}	Ba-140	4.2×10^{-7}
I-131	2.8×10^{-3}	Sr-92	1.3×10^{-7}	La-140	1.4×10^{-7}
I-132	2.8×10^{-3}	Y-90	3.4×10^{-9}	Ce-141	6.3×10^{-8}
I-133	4.2×10^{-3}	Y-91m	3.3×10^{-7}	Ce-143	5×10^{-8}
I-134	5.7×10^{-4}	Y-91	5.7×10^{-8}	Ce-144	3.9×10^{-8}
I-135	2.3×10^{-3}	Y-92	1.2×10^{-7}	Pr-143	6.3×10^{-8}
		Y-93	3.8×10^{-8}	Pr-144	3.9×10^{-8}

TABLE 12.2.1-52

CONDENSATE STORAGE TANK
SPECIFIC ACTIVITIES

Nuclide	Specific Activity (μCi/g)	Nuclide	Specific Activity (μCi/g)
Br-83	2.5×10^{-7}	Y-90	5.3×10^{-11}
Br-84	3.5×10^{-8}	Y-91m	2.9×10^{-9}
Br-85	4.3×10^{-10}	Y-91	6.1×10^{-10}
I-129	4.2×10^{-13}	Y-92	4×10^{-10}
I-130	1.3×10^{-7}	Y-93	2.3×10^{-10}
I-131	2.7×10^{-5}	Zr-95	6.6×10^{-10}
I-132	9.3×10^{-6}	Nb-95	6.6×10^{-10}
I-133	3.1×10^{-5}	Mo-99	6.9×10^{-7}
I-134	7×10^{-7}	Tc-99m	6.5×10^{-7}
I-135	1.1×10^{-5}	Ru-103	5.8×10^{-10}
Rb-86	1.95×10^{-7}	Ru-106	1.4×10^{-10}
Rb-88	1.05×10^{-6}	Rh-103m	5.8×10^{-10}
Rb-89	4×10^{-8}	Rh-106	1.4×10^{-10}
Cs-134	2.05×10^{-5}	Ag-110m	1.4×10^{-9}
Cs-136	2.6×10^{-5}	Te-125m	2.8×10^{-10}
Cs-137	1.35×10^{-5}	Te-127m	3.0×10^{-9}
Ba-137m	1.25×10^{-5}	Te-127	8.1×10^{-9}
Cs-138	2.75×10^{-7}	Te-129m	1.9×10^{-8}
H-3	1.8	Te-129	1.3×10^{-8}
Cr-51	5.6×10^{-9}	Te-131m	2.2×10^{-8}
Mn-54	4.5×10^{-10}	Te-131	4.4×10^{-9}
Mn-56	5.7×10^{-9}	Te-132	2.8×10^{-7}
Fe-55	2×10^{-9}	Te-134	2.9×10^{-9}
Fe-59	5.3×10^{-10}	Ba-140	4.2×10^{-9}
Co-58	1.5×10^{-8}	La-140	1.8×10^{-9}
Co-60	1.9×10^{-9}	Ce-141	6.4×10^{-10}
Sr-89	7.8×10^{-9}	Ce-143	4.3×10^{-10}
Sr-90	2.2×10^{-10}	Ce-144	3.9×10^{-10}
Sr-91	5×10^{-9}	Pr-143	6.4×10^{-10}
Sr-92	4.3×10^{-10}	Pr-144	3.9×10^{-10}

TABLE 12.2.2-1 (SHEET 1 OF 2)

PARAMETERS AND ASSUMPTIONS FOR CALCULATING
AIRBORNE RADIOACTIVE CONCENTRATIONS

Leak Rates (lb/day)	
Equivalent reactor coolant leak into containment during power for noble gases	5100
Equivalent reactor coolant leak into containment for halogens	5.1
Equivalent reactor coolant leak into auxiliary building	160
Equivalent reactor coolant leak into letdown heat exchanger valve gallery	7.4
Equivalent steam generator steam leak into turbine building	40,800

Evaporation Rates (g/min)	
From refueling pool into containment atmosphere	2960
From spent fuel pool into fuel building atmosphere	2670

Partition Factors	Noble Gases	Halogens	Particulates	Tritium
Auxiliary building	1	0.0075	0.0001	0.1
Fuel handling building	Negligible	Negligible	Negligible	1

TABLE 12.2.2-1 (SHEET 2 OF 2)

Ventilation Rates (ft^3/min)	
Containment during power	5000
Containment during refueling	15,000
Fuel handling building, in the vicinity of each spent fuel pool, during refueling	10,890
Auxiliary building	72,000
Auxiliary building letdown heat exchanger valve gallery	90
Turbine building	1.1×10^6
Volumes of the Regions (ft^3)	
Containment	2.75×10^6
Fuel handling building	5.0×10^5
Auxiliary building	1.9×10^6
Auxiliary building letdown heat exchanger valve gallery	2730
Turbine building	5.3×10^6
Miscellaneous Information	
Failed fuel percentage for fission products	0.12
Reactor coolant specific activities	Table 11.1-7
Steam generator steam activities	Table 11.1-7
Plant capacity factor (percent)	80

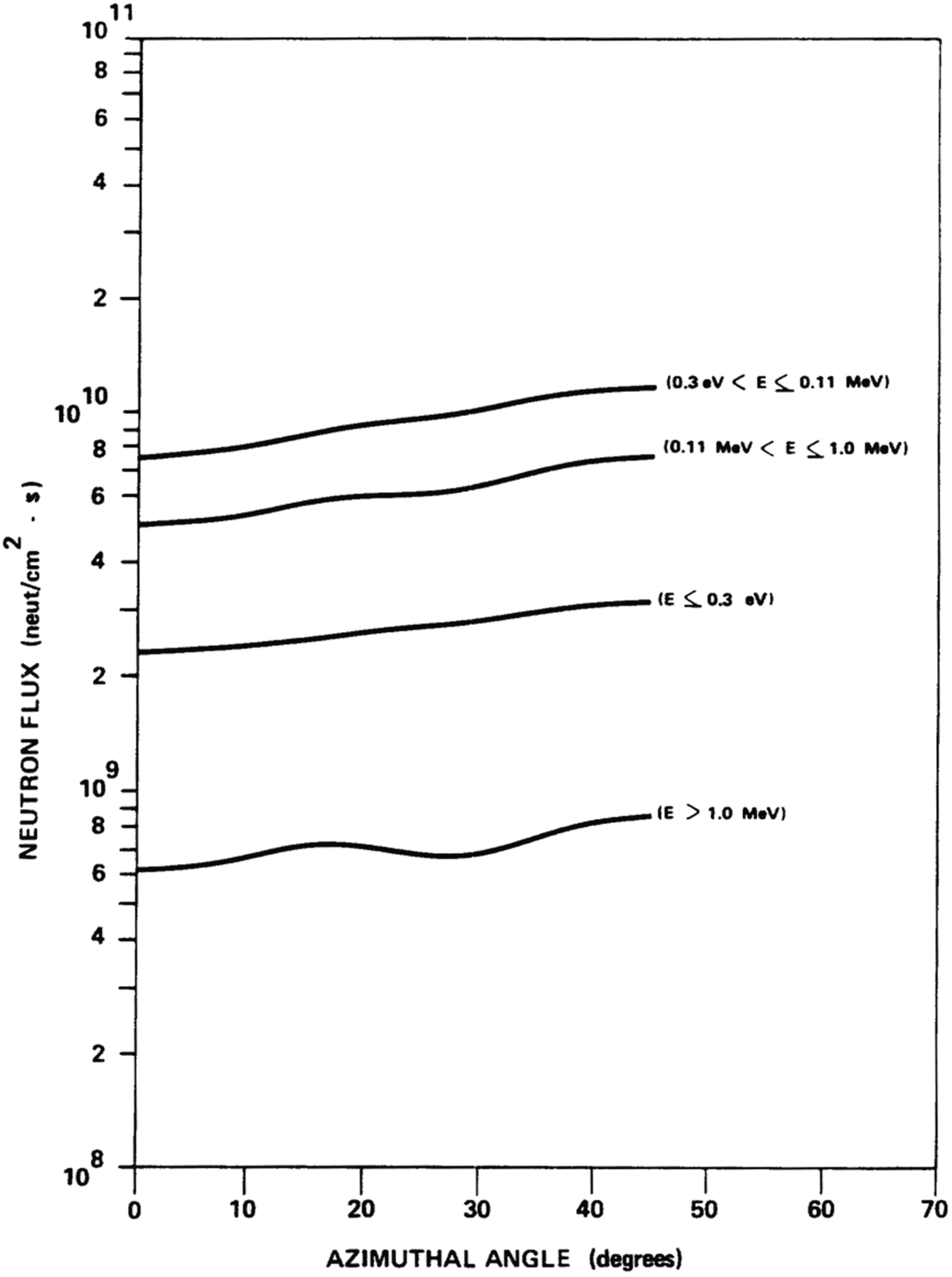
TABLE 12.2.2-2 (SHEET 1 OF 2)

AIRBORNE RADIOACTIVITY CONCENTRATIONS
($\mu Ci/cm^3$)

Nuclide	Containment (100%Power)	Containment (Refueling)	Fuel Handling Building (Refueling)	Turbine Building
H-3	1.50E-6	2.50E-6	2.50E-6	4.54E-10
N-16	-			-
Ar-41	9.97E-7			-
Mn-54	2.95E-10			-
Fe-59	1.01E-10			-
Co-58	1.01E-9			-
Co-60	4.56E-10			-
Br-83	1.67E-11			5.24E-16
Br-84	2.62E-12			7.74E-17
Br-85	2.98E-14			3.50E-19
Kr-83m	5.33E-8			2.46E-15
Kr-85m	4.46E-7			1.20E-14
Kr-85	5.40E-8			5.90E-16
Kr-87	1.16E-7			6.76E-15
Kr-88	6.60E-7			2.24E-14
Kr-89	5.25E-10			2.64E-16
Rb-86	-			3.36E-18
Rb-88	-			2.88E-16
Sr-89	2.28E-11			-
Sr-90	3.99E-12			-
I-130	1.70E-11			5.96E-16
I-131	3.05E-9			9.98E-14
I-132	3.26E-10			3.96E-14
I-133	3.45E-9			1.09E-13
I-134	7.22E-11			2.26E-15
I-135	1.20E-9			3.72E-14
Xe-131m	1.65E-7			1.86E-15
Xe-133m	8.63E-7			1.09E-14
Xe-133	4.39E-5			4.98E-14
Xe-135m	6.06E-9			1.33E-15
Xe-135	1.85E-6			3.42E-14
Xe-137	1.12E-9			5.22E-16
Xe-138	1.98E-8			4.24E-15
Cs-134	2.95E-10			7.72E-16
Cs-136	-			4.32E-16
Cs-137	5.10E-10			6.36E-16
Ba-137m	-			1.89E-16

TABLE 12.2.2-2 (SHEET 2 OF 2)

	Auxiliary Building	
Nuclide	Corridor	Letdown Heat Exchanger Valve Gallery
H-3	2.48E-9	9.19E-8
N-16	-	-
Mn-54	-	-
Fe-59	-	-
Co-58	-	-
Co-60	-	-
Br-83	8.93E-13	3.25E-11
Br-84	3.59E-13	1.25E-11
Br-85	9.13E-15	2.92E-13
Kr-83m	4.49E-10	1.62E-8
Kr-85m	2.23E-9	8.18E-8
Kr-85	1.19E-10	4.41E-9
Kr-87	1.24E-9	4.47E-8
Kr-88	4.26E-9	1.55E-7
Kr-89	2.13E-11	6.83E-10
Rb-86	2.20E-16	8.17E-15
Rb-88	2.87E-13	9.76E-12
Sr-89	-	-
Sr-90	-	-
I-130	4.17E-13	1.54E-11
I-131	5.19E-11	1.93E-9
I-132	1.81E-11	6.58E-10
I-133	7.32E-11	2.71E-9
I-134	7.36E-12	2.61E-10
I-135	3.73E-11	1.37E-9
Xe-131m	3.71E-10	1.38E-8
Xe-133m	2.12E-9	7.85E-8
Xe-133	1.01E-7	3.76E-6
Xe-135m	1.61E-10	5.42E-9
Xe-135	6.73E-9	2.48E-7
Xe-137	4.42E-11	1.42E-9
Xe-138	5.40E-10	1.82E-8
Cs-134	6.44E-14	2.39E-12
Cs-136	3.47E-14	1.29E-12
Cs-137	4.71E-14	1.75E-12
Ba-137m	1.09E-16	1.82E-13



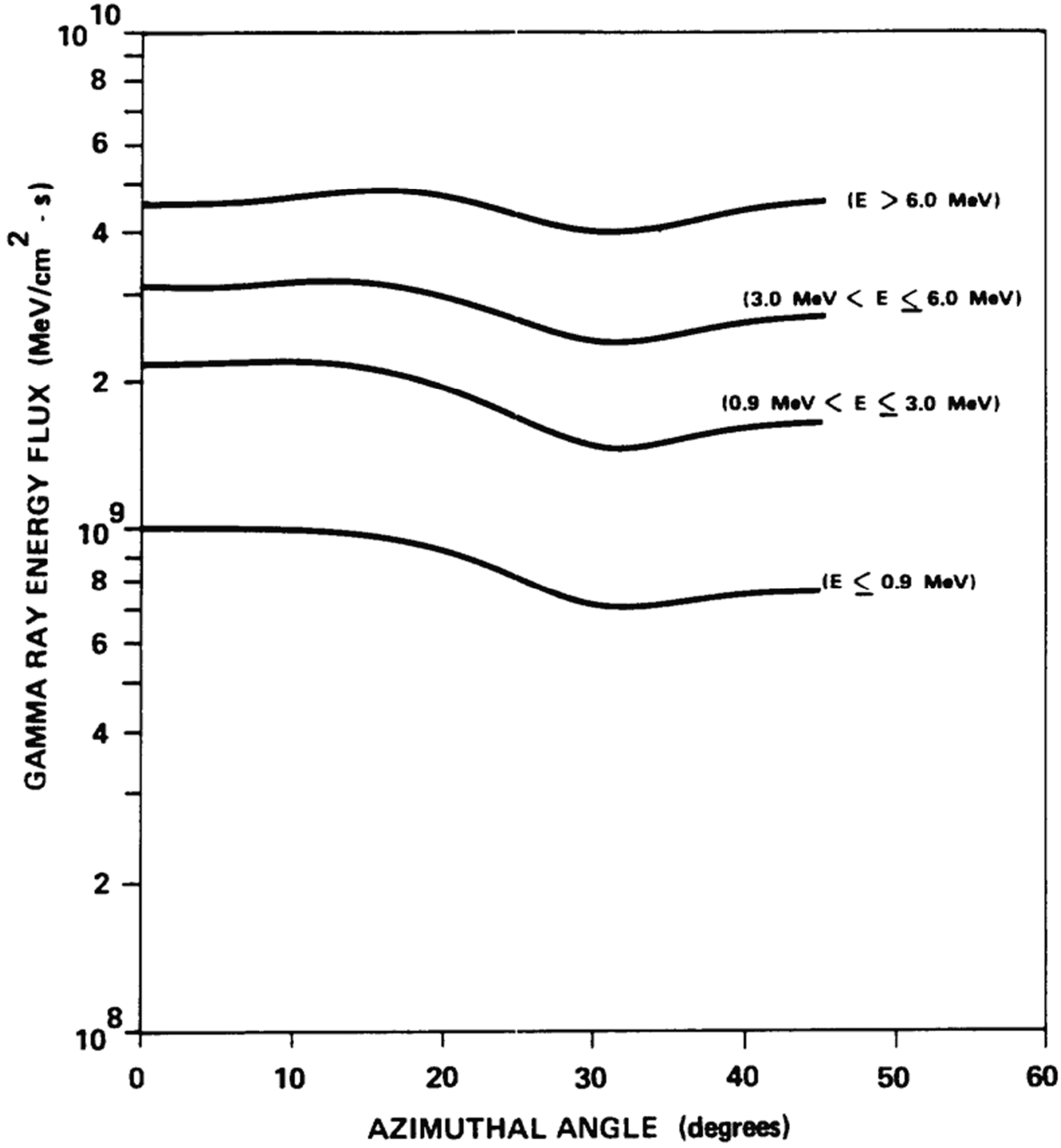
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ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

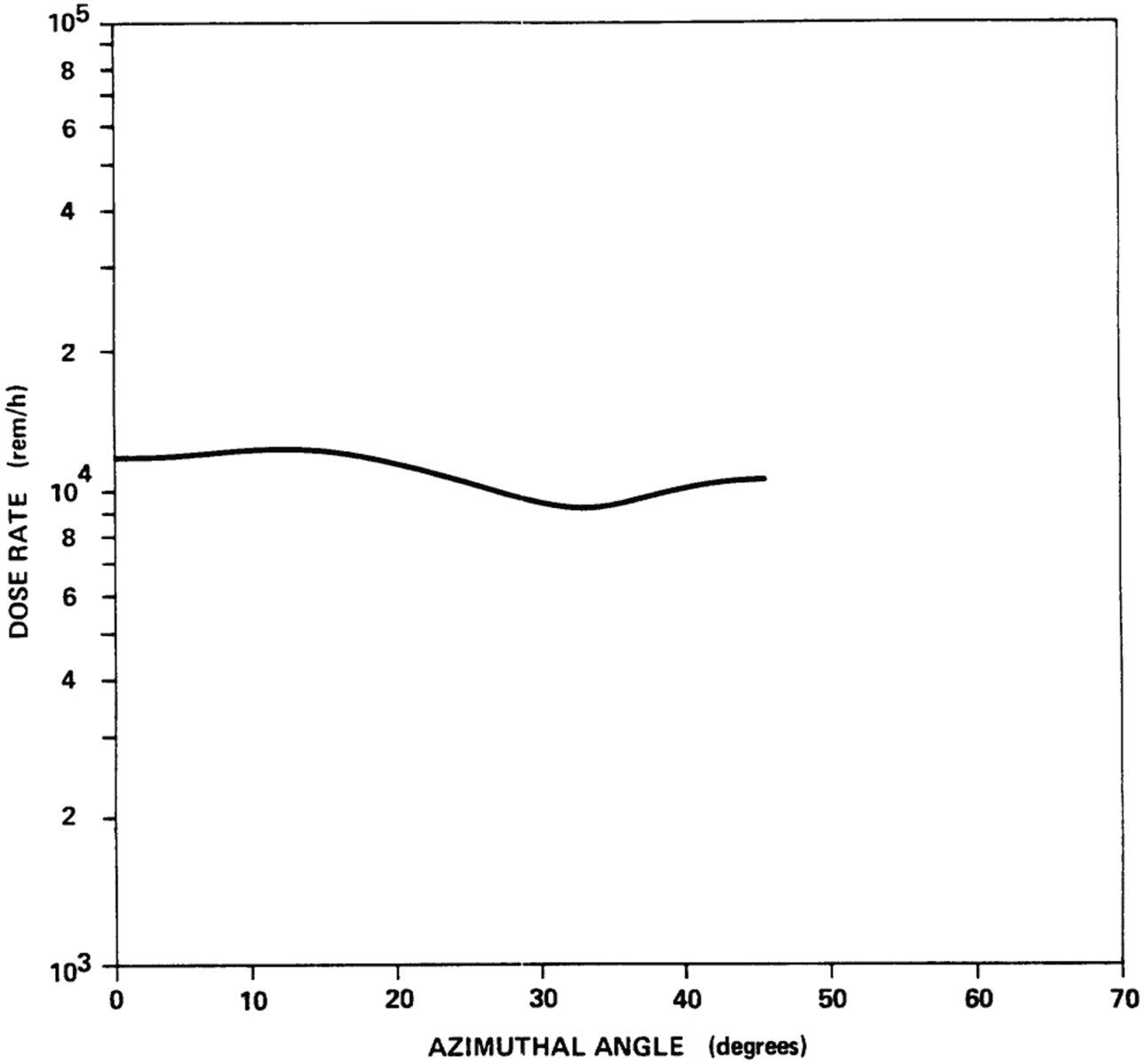
AZIMUTHAL DISTRIBUTION OF NEUTRON FLUX INCIDENT ON THE PRIMARY SHIELD AT THE REACTOR CORE MIDPLANE

FIGURE 12.2.1–1



REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	AZIMUTHAL DISTRIBUTION OF GAMMA RAY ENERGY FLUX INCIDENT ON THE PRIMARY SHIELD AT THE REACTOR CORE MIDPLANE
	FIGURE 12.2.1–2



REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

AZIMUTHAL DISTRIBUTION OF GAMMA RAY DOSE RATE INCIDENT ON THE PRIMARY SHIELD AT THE REACTOR CORE MIDPLANE

FIGURE 12.2.1–3

12.3 RADIATION PROTECTION DESIGN FEATURES

12.3.1 FACILITY DESIGN FEATURES

Specific design features for maintaining personnel exposure as low as reasonably achievable (ALARA) are discussed in this subsection. The design feature recommendations given in Regulatory Guide 8.8, Paragraph C.3, are utilized to minimize exposures to personnel.

12.3.1.1 Plant Design Description for ALARA

The equipment and plant design features employed to maintain radiation exposures ALARA are based upon the design considerations of subsection 12.1.2 and are outlined in this subsection for several general classes of equipment (paragraph 12.3.1.1.1) and several typical plant layout situations (paragraph 12.3.1.1.2).

12.3.1.1.1 Common Equipment and Component Designs for ALARA

This paragraph describes the design features utilized for several general classes of equipment or components. These classes of equipment are common to many of the plant systems; thus, the features employed for each system to maintain minimum exposures are similar and are discussed by equipment class in the following paragraphs. The locations of equipment and components are shown on drawings 1X4DE312, 1X4DE313, 1X4DE314, 1X4DE315, 1X4DE317, 1X4DE318, 1X4DE320, 1X4DE321, 1X4DE322, 1X4DE323, 1X4DE324, and 1X4DE325.

12.3.1.1.1.1 Balance of Plant Equipment.

A. Filters

Filters that accumulate radioactivity are supplied with the means either to backflush the filter remotely or to perform cartridge replacement with semiremote tools. For cartridge filters, adequate space is provided to allow removing, cask loading, and transporting the cartridge to the solid radwaste area.

Backflushable filters are designed so that filter internals may be remotely removed and placed in a shielded cask for offsite shipping and disposal in the unlikely event that a filter loses its backflush capability.

Liquid systems containing radioactive cartridge filters are provided with a remote filter handling system for the removal of spent radioactive filter cartridges from their housings and for their transfer to the drumming station for packaging and shipment from the site for burial. The process is accomplished in such a manner that exposure to personnel and the possibility of inadvertent radioactive release to the environment is minimized. Each filter is contained in a shielded compartment and provided with vent and drain valving, and individual compartments are provided with compartment drainage capabilities. The filter handling system has also been designed to be simple with a minimum of components susceptible to malfunction.

B. Demineralizers

Demineralizers for highly radioactive systems are designed so that spent resins can be remotely and hydraulically transferred to spent resin tanks so that fresh resin can be loaded into the demineralizer remotely. Underdrains are designed for full system pressure drop. The demineralizers and piping are designed with provisions for being flushed with demineralized water. Strainers are installed in the vent lines to prevent entry of spent resin into the exhaust duct.

C. Evaporators (Abandoned in place)

D. Pumps

Wherever practicable, pumps are purchased with mechanical seals to reduce seal servicing time. Pumps and associated piping are arranged to provide adequate space for access to the pumps for servicing. Small pumps are installed in a manner which allows easy removal if necessary. All pumps-in radioactive waste systems are provided with flanged connections for ease of removal. Pump casings are provided with drain connections for draining the pump for maintenance.

E. Tanks

Whenever practicable, tanks are provided with sloped bottoms and bottom outlet connections. Overflow lines are directed to the waste collection system to control any contamination within plant structures. For tanks outside structures, dikes are used to contain overflows.

F. Heat Exchangers

Heat exchangers are provided with corrosion-resistant tubes of stainless steel or other suitable materials to minimize leakage. Impact baffles are provided and tube side and shell side velocities are limited to minimize erosive effects. Wherever possible, the radioactive fluid passes through the tube side of the heat exchanger.

G. Instruments

Instrument devices are located in low radiation zones and away from radiation sources whenever practical. Primary instrument devices, which for functional reasons are located in high radiation zones, are designed for easy removal to a lower radiation zone for calibration. Transmitters and readout devices are located in low radiation zones, such as corridors and the control room, for servicing.

Some instruments in high radiation zones, such as thermocouples, are provided in duplicate to reduce access and service time required. In containment, instruments are located outside the secondary shield (area of lowest radiation at power and shutdown) whenever practical.

Integral radiation check sources for response verification for airborne radiation monitors and safety-related area radiation monitors are provided.

Chemical seals are provided on the instrument sensing lines on process piping, which may contain highly radioactive solids, to reduce the servicing time required to keep the lines free of solids. Instrument and sensing line connections are located slightly above the pipe midplane wherever practical to minimize radioactive crud or gas buildup.

Process control panels are shown in the equipment location drawings.

H. Valves

To minimize personnel exposures from valve operations, motor-operated, air-operated, or other remotely actuated valves are used where justified by the activity levels and frequency of use. Valves are located in valve galleries so that they are shielded separately from the major components. Long runs of exposed piping are minimized in valve galleries. In areas where manual valves are used on frequently operated process lines, either valve stem extenders or shielding is provided such that personnel need not enter a high radiation area for valve operation.

When equipment in high radiation areas is operated infrequently, only those manual valves associated with safe operation of the equipment are provided with remote-manual operators or reach rods. All other valve operations are performed with equipment in the shutdown mode.

For valves located in radiation areas, provisions are made to drain adjacent radioactive components when maintenance is required. To the extent practicable, valves are not located at piping low points. Valves in containment expected to exhibit stem leakage are provided with leakoff connections piped to the reactor coolant drain tank (RCDT or a reduced packing configuration with the valve stem leakoff line capped). Valves for clean, nonradioactive systems are separated from radioactive sources and are located in readily accessible areas. For most larger valves (2 1/2 in. and larger) in lines carrying radioactive fluids, a double set of packing with a lantern ring is provided. A stuffing box with a leakoff connection that may be piped to a drain header is also provided. Metal diaphragm or bellows seat valves are used on those systems where essentially no leakage can be tolerated.

Manually operated valves in the filter and demineralizer valve compartments required for normal operation and shutdown are equipped with reach rods extending through or over the valve gallery wall. Personnel do not enter the valve gallery during spent resin or cartridge transfer operations. The valve gallery shield walls are designed to minimize personnel exposure during maintenance of components within or adjacent to the gallery and to protect personnel who remotely operate the valves.

Relief valves are located in an associated equipment compartment or valve gallery. Check valves are located in the equipment compartment or associated valve gallery unless they are the locking type requiring manipulation during normal operation. In this case, check valves are treated as normal manual valves.

I. Piping

The piping in pipe chases is designed for the lifetime of the unit[a]. Wherever radioactive piping is routed through areas where routine maintenance is required, pipe chases are provided to reduce the radiation contribution from these pipes to levels appropriate for the inspection or maintenance requirements. Butt welds

[a] The operating licenses for both VEGP units have been renewed, and the original licensed operating terms have been extended by 20 years. In accordance with 10 CFR Part 54, appropriate aging management programs and activities have been initiated to manage the detrimental effects of aging to maintain functionality during the period of extended operation (see chapter 19).

are used to the fullest extent possible in radwaste piping utilized for transport of spent resins or slurries. Piping containing radioactive material is routed to minimize radiation exposure to the unit personnel.

J. Floor Drains

Floor drains and properly sloped floors are provided for each room or cubicle containing serviceable components containing radioactive liquids. When practicable, shielded pipe chases are used for radioactive pipes. If a radioactive drain line must pass through a plant area requiring personnel access, shielding is provided as necessary-to ensure radiation levels consistent with the required access.

K. Lighting

Wherever practicable, multiple electric lights are provided for each cell or room containing highly radioactive components so that the burnout of a single lamp will not require entry and immediate replacement of the defective lamp since sufficient illumination will still be available. Normally, incandescent lights are provided which require less time for servicing and, hence, the personnel exposure is reduced. The fluorescent lights which are used in some areas do not require frequent service due to the increased life of the tubes. Burned out lamps can be replaced when the system is secured and flushed out.

L. Heating, Ventilation and Air-Conditioning

The heating, ventilation, and air-conditioning (HVAC) system design facilitates replacement of the filter elements.

M. Hydrogen Recombiners

Provision is made for permanent shielding of the hydrogen recombiner control console from components that are highly radioactive.

N. Sample Stations

Sample stations for routine sampling of process fluids are located as shown in the equipment location and general arrangement drawings presented in section 1.2. Proper shielding and ventilation are provided at the local sample stations to maintain radiation zoning in proximate areas and minimize personnel exposure during sampling. The counting room and laboratory facilities are described in section 12.5. The use of concrete containing fly ash is minimized in the counting room and laboratory areas.

O. Clean Services

Whenever possible, clean services and equipment such as compressed air piping, clean water piping, ventilation ducts, and cable trays are not routed through radioactive pipeways.

12.3.1.1.1.2 Nuclear Steam Supply System (NSSS) Equipment.

A. Reactor Vessel

The reactor vessel nozzle welds are designed to accommodate remote inspection with ultrasonic sensors. The nozzle area is tapered along the reinforced areas to ensure a smooth transition, and pipe branch locations are

selected to ensure no interference from one branch to the next. All weld-to-pipe interfaces require a smooth, high quality finish.

B. Reactor Coolant Pumps

The reactor coolant pump design includes the use of an assembled cartridge seal for the No. 3 pump seal which reduces the time required for replacement. The cartridge seal is also expected to have a useful life double that of the old design. The reactor coolant pump design also includes a spool piece to facilitate separation and replacement of the motor from the pump.

C. Reactor Vessel Insulation

Insulation in the area of the reactor vessel nozzle welds is fabricated in sections with a thin reflective metallic sheet covering and quick disconnect clasps to facilitate removal of the insulation for inspection of the welds.

D. Steam Generators

The Model F steam generators incorporate several design features to facilitate maintenance and inspection in reduced radiation fields. The tube ends are designed to be flush with the tube sheet in the steam generator channel head to eliminate a potential crud trap. The steam generator manways (entrance to channel head) are sized to facilitate entrance and exit with protective clothing. Handholes to the secondary side are positioned to facilitate maintenance operations. Changes to increase steam generator reliability will also reduce occupational radiation exposure. Such changes include improved steam generator tube support plates (stainless steel and quatrefoil flow holes) and the use of all-volatile treatment (AVT) chemistry and the installation of a full flow condensate polishing demineralizer system on the secondary side.

E. Reactor Coolant Pipe Connections

To minimize crud buildup in branch lines, piping connections to the reactor coolant loops are located on or above the horizontal centerline of the pipe wherever possible.

F. Heat Exchangers

The heat exchangers are designed so that the shell-to-tube sheet joint need not be broken for inspection.

The shell and tube assembly can be lifted intact above the channel head to expose the tube ends for inspection and testing for leaks.

G. Valves

Valves are of the bolted body-to-bonnet forging type. This permits the use of ultrasonic testing in place of radiography for inspection and facilitates assembly and disassembly. This reduces inspection and maintenance time. Additionally, manual valves under 2 in. in diameter are designed for zero stem leakage.

H. Pumps

Pumps are designed with flanged connections to facilitate removal for maintenance. Depending on expected conditions, either canned pumps or pumps with high quality mechanical seals are used to reduce leakage and maintenance requirements.

I. Demineralizers

Demineralizer resin screens are constructed for substantially higher than normal pressure differential to accommodate higher than design flows without breakdown.

J. Filters

Filters are designed to be removable from the top with lifting bails in the middle of the head. The filter assemblies have bolt lead-ins for tool entry, and the filters are contained in a disposable cage assembly. These features facilitate remote removal, disposal, and assembly.

K. Materials

Equipment specifications for components exposed to high temperature reactor coolant contain specific limitations on the cobalt impurity content of the base metal as given in table 12.3.1-2, thereby controlling the potential for production of radioactive cobalt-60 from the base metal impurity cobalt-59. The estimated surface area of material in contact with the reactor coolant is given in table 12.3.1-3. The use of hard facing material with cobalt content such as stellite is limited to applications where its use is necessary for reliability considerations. Table 12.3.1-4 shows the estimated total surface area of stellite in the reactor coolant system (RCS). Nickel-based alloys in the RCS (cobalt-58 is produced from activation of the base metal nickel-58) are similarly used only where component reliability may be compromised by the use of other materials. The major use of nickel-based alloys in the RCS is the Inconel steam generator tubes. The surface area in contact with the RCS is given in table 12.3.1-3. From tables 12.3.1-3 and 12.3.1-4, it can be seen that the Inconel surface is the predominant area in contact with the RCS and that the stellite area is minimal.

L. Improved Head Closure System

The head closure system includes quick disconnect/connect stud tensioners which have quick-acting, hydraulically operated stud gripper devices as opposed to conventional tensioners which must be threaded onto the top of the studs. Also provided are six air motor-driven stud removal tools which can rapidly remove or insert the studs unlike the much slower manual stud removal and insertion used in the previous design. The stud tensioners are designed to operate simultaneously as are the stud removal tools.

12.3.1.1.2 Common Facility and Layout Designs for ALARA

This paragraph describes the design features utilized for standard type plant process and layout situations. These features are employed in conjunction with the general equipment described in paragraph 12.3.1.1.1 and include the features discussed in the following paragraphs.

A. Valve Galleries

Valve galleries are provided with shielded entrances for personnel protection. Floor drains are provided to control radioactive leakage. To facilitate decontamination in valve galleries, concrete surfaces are covered with a smooth surface coating which will allow easy decontamination.

B. Piping

Pipes carrying radioactive materials are routed through controlled access areas properly zoned for that level of activity. Each piping run is individually analyzed to determine the potential radioactivity level and surface dose rate. Where it is necessary that radioactive piping be routed through corridors or other low radiation zone areas, shielded pipeways are provided. Whenever practicable, valves and instruments are not placed in radioactive pipeways. Equipment compartments are used as pipeways only for those pipes associated with equipment in the compartment.

When possible and practical, radioactive and nonradioactive piping are separated to minimize personnel exposure. Should maintenance be required, provision is made to isolate and drain radioactive piping and associated equipment. Potentially radioactive piping is located in appropriately zoned and restricted areas. Process piping is monitored to ensure that access is controlled to limit exposure (section 12.5).

Piping is designed to minimize low points and dead legs. Drains are provided on piping where low points and dead legs cannot be eliminated. In radioactive systems, the use of nonremovable backing rings in the piping joints is prohibited. Whenever possible, branch lines having little or no flow during normal operation are connected above the horizontal midplane of the main pipe.

Piping which carries resin slurries or evaporator bottoms is run vertically as much as possible. Horizontal runs carrying spent resin are sloped toward the spent resin tanks. Large radius bends are utilized instead of elbows. Where sloped lines or large radius bends are impractical, adequate flush and drain capability is provided to prevent flow blockage and minimize crud traps.

C. Penetrations

To minimize radiation streaming through penetrations, as many penetrations as practicable are located with an offset between the source and the accessible areas. If offsets are not practicable, penetrations are located as far as possible above the floor elevation to reduce the exposure to personnel. If these two methods are not used, alternate means are employed, such as baffle shield walls or grouting the area around the penetration. As a result of radiation streaming into the electrical penetration areas, the radiation doses to personnel and equipment in these areas is higher than the balance of the control building. Post-accident radiation zones for the control building are shown on drawings 1X6DD100, 1X6DD101, AX6DD100, AX6DD101, AX6DD102, AX6DD103, AX6DD104, AX6DD105, AX6DD106, AX6DD107, AX6DD108, AX6DD109, AX6DD110, AX6DD111, AX6DD112, AX6DD113, AX6DD114, AX6DD115, AX6DD116, AX6DD117, AX6DD118, AX6DD119, and AX6DD120.

D. Contamination Control

Access control and traffic patterns are considered in the basic plant layout to minimize the spread of contamination. Equipment vents and drains from highly radioactive systems are piped directly to the collection system instead of allowing any contaminated fluid to flow across to the floor drain. All-welded piping systems are employed on contaminated systems to the maximum extent practicable to reduce system leakage and crud buildup at joints. The valves in some radioactive systems are provided with leakoff connections piped directly to the collection system.

Decontamination of potentially contaminated areas and equipment within the plant is facilitated by the application of decontaminable paints and suitable smooth-surface coatings to the concrete floors and walls. Floor drains with properly sloping floors are provided in all potentially contaminated areas of the plant. In addition, radioactive and potentially radioactive drains are separated from nonradioactive drains.

In controlled access areas where contamination is expected, radiation monitoring equipment is provided (subsection 12.3.4). Those systems that become highly radioactive, such as the spent resin lines in the radwaste system, are provided with flush and drain connections.

The role of the ventilation systems in minimizing the spread of airborne contamination is discussed in subsection 12.3.3.

E. Equipment Layout

In those systems where process equipment is a major radiation source (such as fuel pool cleanup, coolant and boric acid recycle, chemical waste, and miscellaneous waste) pumps, valves, and instruments are separated from the process component. This allows servicing and maintenance of these items in reduced radiation zones. In general, control panels are located in low radiation zones.

Major components such as tanks, demineralizers, and filters in radioactive systems are isolated in individual shielded compartments insofar as practical. Labyrinth entranceway shields or shielding doors are provided for each compartment from which radiation could stream or scatter to access areas and exceed the radiation zone dose limits for those areas. For potentially high radiation components (such as ion exchangers or heat exchangers and tanks in the primary coolant system) completely enclosed shielded compartments with hatch openings or removable concrete block walls are used. Provision is made on some major plant components for removal of these components to lower radiation zones for maintenance.

Equipment in nonradioactive systems which requires lubrication is located in lower radiation areas. Wherever practicable, lubrication of equipment in radiation areas is achieved with the use of tube-type extensions to reduce exposure during maintenance.

Exposure from routine in-plant inspection is controlled by locating, whenever possible, inspection points in properly shielded low-background radiation areas. Radioactive and nonradioactive systems are separated as far as practicable to limit radiation exposure from routine inspection of nonradioactive systems. For radioactive systems, emphasis is placed on adequate space and ease of motion in a properly shielded inspection area. Where longer times for routine inspection are required and permanent shielding is not feasible, sufficient space for portable shielding is provided. For example, a remotely operated device is provided for inservice inspections of the reactor vessel. When this is not practicable, written procedures are used which reduce the total time personnel are exposed to the radiation field. Also, access to high radiation areas is under the direct supervision of the unit health physics personnel.

F. Field Run Piping

Radioactive process piping is routed dimensionally on orthographic drawings by the plant architect/engineer. Fabrication isometrics of radioactive process piping are reviewed by the plant architect/engineer to ensure that adequate shielding is provided.

12.3.1.2 Radiation Zoning and Access Control

Access to areas inside the plant structures and plant yard area is regulated and controlled by radiation zoning and access control (section 12.5). Each radiation zone defines the radiation level range to which the aggregate of all contributing sources must be attenuated by shielding. During plant operation, personnel normally gain access to radiation controlled areas through designated access control points determined by health physics.

All plant areas are categorized into radiation zones according to expected radiation levels and anticipated personnel occupancy with consideration given toward maintaining personnel exposures ALARA and within the standards of 10 CFR 20. Each room, corridor, and pipeway of every plant building is evaluated for potential radiation sources during normal, shutdown, spent resin transfer, and emergency operations; for maintenance occupancy requirements; for general access requirements; and for material exposure limits to determine appropriate zoning. The radiation zone categories employed and their descriptions are given in table 12.3.1-1. The zoning for each plant area under normal conditions is shown on drawings 1X6DD001, 1X6DD002, 1X6DD003, 1X6DD004, 2X6DD001, 2X6DD002, 2X6DD003, AX6DD001, AX6DD002, AX6DD003, AX6DD004, AX6DD005, AX6DD006, AX6DD007, AX6DD008, AX6DD009, AX6DD010, AX6DD011, AX6DD012, AX6DD013, AX6DD014, AX6DD015, AX6DD016, AX6DD017, AX6DD018, AX6DD019, AX6DD020, AX6DD021, AX6DD022, AX6DD023, AX6DD024, AX6DD025, AX6DD026, AX6DD027, AX6DD028, AX6DD029, AX6DD030, AX6DD031, and AX6DD032. The zoning for each plant under accident conditions is shown on drawings 1X6DD100, 1X6DD101, AX6DD100, AX6DD101, AX6DD102, AX6DD103, AX6DD104, AX6DD105, AX6DD106, AX6DD107, AX6DD108, AX6DD109, AX6DD110, AX6DD111, AX6DD112, AX6DD113, AX6DD114, AX6DD115, AX6DD116, AX6DD117, AX6DD118, AX6DD119, and AX6DD120. Radiation zones shown in the figures are based upon conservative design data. Actual in-plant zones and control of personnel access will be based upon surveys conducted by health physics, as described in section 12.5.

In accordance with Section II.B.2 of NUREG-O737, a radiation and shielding design review was performed to identify vital areas and equipment. Areas which may require occupancy to permit an operator to aid in the long-term recovery from an accident are considered as vital. Vital areas include the control room, technical support center, safety-related motor control centers and switchgear in the control building, auxiliary building, diesel generator building, auxiliary feedwater pumphouse, radiochemistry laboratory, and the remote shutdown panels. Projected dose rates for these vital areas at various times after an accident are given in table 12.3.1-5. VEGP is designed to ensure the capability to achieve cold shutdown without subjecting personnel to excessive radiation exposure. This capability is further described in section 7.4. Radiation protection design features and access controls are described in sections 12.3 and 12.5. In the event that entry is desired into areas where excessive radiation exposures may occur, due consideration is given to the dose rates defined on drawings 1X6DD100, 1X6DD101, AX6DD100, AX6DD101, AX6DD102, AX6DD103, AX6DD104, AX6DD105, AX6DD106, AX6DD107, AX6DD108, AX6DD109, AX6DD110, AX6DD111, AX6DD112, AX6DD113, AX6DD114, AX6DD115, AX6DD116, AX6DD117, AX6DD118, AX6DD119, AX6DD120, and table 12.3.1-5, and appropriate time limits for presence in the area are imposed.

Ingress or egress of plant operating personnel to radiation controlled areas is controlled by the plant health physics staff to ensure that radiation levels and exposures are within the limits prescribed in 10 CFR 20. Any area having a radiation level that could result in an individual receiving a dose equivalent in excess of 5 mrem in 1 hour at 30 cm (12 in.) from the radiation source or from any surface which the radiation penetrates will be posted "Caution, Radiation Area." Any area having a radiation level that could result in an individual receiving a dose equivalent in excess of 100 mrem in 1 hour at 30 cm (12 in.) from the radiation source or from any surface which the radiation penetrates will be barricaded and posted "Caution, High Radiation Area" or "Danger, High Radiation Area." High radiation areas with radiation levels greater than 1000 mrem/h at 30 cm (12 in.) from the radiation source or from any surface which the radiation penetrates are provided with locked or alarmed barriers. For individual high radiation areas accessible to personnel with radiation levels of greater than 1000 mrem/h at 30 cm (12 in.) that are located within large areas where no enclosure exists for purposes of locking, and where no enclosure can be reasonably constructed around the individual area, that individual area shall be barricaded, conspicuously posted, and a flashing light shall be activated as a warning device. During periods when access to a high radiation area is required, positive control is exercised over each individual entry. Any area having a radiation level that could result in an individual receiving an absorbed dose in excess of 500 rads in 1 hour at 1 meter from the radiation source or from any surface which the radiation penetrates will be posted "Grave Danger, Very High Radiation Area." Measures taken to control access to very high radiation areas will meet the intent of Draft Regulatory Guide 8.38. To the extent practicable, the measured radiation level and the location of the source is posted at the entry to any radiation area or high radiation area.

Posting of radiation signs, control of personnel access, and use of alarms and locks are in compliance with requirements of 10 CFR 20.1601 and 1902. The flow of personnel is shown on drawings 1X6DD200, 1X6DD201, 1X6DD202, 1X6DD203, 2X6DD200, 2X6DD201, 2X6DD202, AX6DD200, AX6DD201, AX6DD202, AX6DD203, AX6DD204, AX6DD205, AX6DD206, AX6DD207, AX6DD208, AX6DD209, AX6DD210, AX6DD211, AX6DD212, AX6DD213, AX6DD214, AX6DD215, AX6DD216, AX6DD217, AX6DD218, AX6DD219, AX6DD220, AX6DD221, AX6DD222, AX6DD223, AX6DD224, AX6DD225, and AX6DD226.

Entry into high radiation areas is controlled by the Technical Specifications.

12.3.2 SHIELDING

The bases for the nuclear radiation shielding and the shielding configurations are discussed in this subsection.

12.3.2.1 Design Objectives

The objective of the plant radiation shielding is to reduce personnel and population exposures, in conjunction with a program of controlled personnel access to and occupancy of radiation areas, to levels that are within the dose standards of 10 CFR 50 and are as low as reasonably achievable (ALARA) within the dose standards of 10 CFR 20. Shielding and equipment layout and design are considered in ensuring that exposures are kept ALARA during anticipated personnel activities in areas of the plant containing radioactive materials, utilizing the design recommendations given in Regulatory Guide 8.8, paragraph C.3, where practicable.

Four plant conditions are considered in the nuclear radiation shielding design:

A. Normal, full-power operation.

B. Shutdown operation.

C. Spent resin transfer (radwaste areas and radwaste portion of the auxiliary building only).

D. Emergency operations (for required access to safety-related equipment).

The shielding design objectives for the plant during normal operation (including anticipated operational occurrences), for shutdown operations, and for emergency operations are listed below:

A. To ensure that radiation exposure to plant operating personnel, contractors, administrators, visitors, and approximate site boundary occupants are ALARA and within the limits of 10 CFR 20.

B. To ensure sufficient personnel access and occupancy time to allow normal anticipated maintenance, inspection, and safety-related operations required for each plant equipment and instrumentation area.

C. To reduce potential equipment neutron activation and to mitigate the possibility of radiation damage to materials.

D. To provide sufficient shielding for the control room so that for design basis accidents (DBAs) the direct dose plus the inhalation dose (calculated in chapter 15) will not exceed the limits of 10 CFR 50, Appendix A, General Design Criterion 19.

12.3.2.2 General Shielding Design

Shielding is provided to attenuate direct radiation through walls and penetrations and scattered radiation to less than the upper limit of the radiation zone for each area shown on drawings 1X6DD001, 1X6DD002, 1X6DD003, 1X6DD004, 2X6DD001, 2X6DD002, 2X6DD003, AX6DD001, AX6DD002, AX6DD003, AX6DD004, AX6DD005, AX6DD006, AX6DD007, AX6DD008, AX6DD009, AX6DD010, AX6DD011, AX6DD012, AX6DD013, AX6DD014, AX6DD015, AX6DD016, AX6DD017, AX6DD018, AX6DD019, AX6DD020, AX6DD021, AX6DD022, AX6DD023, AX6DD024, AX6DD025, AX6DD026, AX6DD027, AX6DD028, AX6DD029, AX6DD030, AX6DD031, and AX6DD032. General locations of the plant areas and equipment discussed in this subsection are shown in the general arrangement drawings of section 1.2. Design criteria for penetrations are consistent with the recommendations of Regulatory Guide 8.8 and are discussed in paragraph 12.3.1.1.2.

The material used for most of the plant shielding is ordinary concrete with a bulk density of approximately 137 lb/ft^3. Whenever poured-in-place concrete has been replaced by concrete blocks, design ensures protection on an equivalent shielding basis as determined by the density of the concrete block selected. Water is used as the primary shield material for areas above the spent fuel storage area.

12.3.2.2.1 Containment Shielding Design

During reactor operation, the containment protects personnel occupying adjacent plant structures and yard areas from radiation originating in the reactor vessel and primary loop components. The concrete containment wall and the reactor vessel and steam generator compartment shield walls reduce radiation levels outside the containment to less than 0.25 mrem/h from sources inside containment. The containment wall is a reinforced,

prestressed concrete structure completely surrounding the nuclear steam supply system. The wall and dome are of a minimum 3 ft 9 in. thickness.

For DBAs, the containment shield and the control room shielding reduce the plant radiation intensities from fission products inside the containment to acceptable emergency levels, as defined by 10 CFR 50, Appendix A, General Design Criterion 19, for the control room. (See paragraph 12.3.2.2.7.)

Where personnel locks and equipment hatches or penetrations pass through the containment wall, additional shielding is provided to attenuate radiation to the required level defined by the outside radiation zone during normal operation and shutdown and to acceptable emergency levels as defined by 10 CFR 50, Appendix A, General Design Criterion 19, during DBAs.

12.3.2.2.2 Containment Interior Shielding Design

During reactor operation, many areas inside the containment are Zone V and normally inaccessible. However, shielding is provided to reduce dose rates to approximately 100 mrem/h or less in areas of the containment that potentially require access at power. These are the Zone IV or lower areas shown on drawings 1X6DD001, 1X6DD002, 1X6DD003, 1X6DD004, 2X6DD001, 2X6DD002, 2X6DD003, AX6DD001, AX6DD002, AX6DD003, AX6DD004, AX6DD005, AX6DD006, AX6DD007, AX6DD008, AX6DD009, AX6DD010, AX6DD011, AX6DD012, AX6DD013, AX6DD014, AX6DD015, AX6DD016, AX6DD017, AX6DD018, AX6DD019, AX6DD020, AX6DD021, AX6DD022, AX6DD023, AX6DD024, AX6DD025, AX6DD026, AX6DD027, AX6DD028, AX6DD029, AX6DD030, and AX6DD031.

The main sources of radiation are the reactor vessel and the primary loop components, consisting of the steam generators, pressurizer, reactor coolant pumps, and associated piping. The reactor vessel is shielded by the concrete primary shield and by the concrete secondary shield which also surrounds other primary loop components. Air cooling is provided to prevent overheating, dehydration, and degradation of the shielding and structural properties of the primary shield.

The primary shield is a large mass of reinforced concrete surrounding the reactor vessel and extending upward from the containment floor to form a portion of the walls of the fuel transfer canal. The minimum concrete thickness of the primary shield is 8 ft. The primary shield meets the following objectives:

A. In conjunction with the secondary shield, to reduce the radiation level from sources within the reactor vessel and reactor coolant system (RCS) to allow limited access to the containment during normal, full-power operation.

B. After shutdown, to limit the radiation level from sources within the reactor vessel, to permit limited access to the reactor vessel, and to permit limited access to the RCS equipment.

C. To limit neutron activation of component and structural materials.

The secondary shield is a reinforced concrete structure surrounding the RCS equipment, including piping, pumps, and steam generators. This shield protects personnel from the direct gamma radiation resulting from reactor coolant activation products and fission products carried away from the core by the reactor coolant. In addition, the secondary shield supplements the primary shield by attenuating neutron and gamma radiation escaping from the primary shield. The secondary shield is sized to allow limited access to the containment during full-power operation. The minimum thickness of secondary shield walls is 3 ft.

Components of the letdown portion of the chemical and volume control system (CVCS) in the containment are located in shielded compartments that are normally Zone V, restricted access areas. Shielding is provided for each piece of equipment in the letdown system consistent with its postulated maximum activity (subsection 12.2.1) and with the access and zoning requirements of adjacent areas. This equipment includes the regenerative heat exchanger, the excess letdown heat exchanger, and the letdown lines.

After shutdown, the containment is accessible for limited periods of time and all access is controlled. Areas are surveyed to establish allowable working periods. Dose rates are expected to range from 0.5 to 1000 mrem/h, depending on the location inside the containment (excluding reactor cavity). These dose rates result from residual fission products and neutron activation products (components and corrosion products) in the RCS.

Spent fuel is the primary source of radiation during refueling. Because of the high activity of the fission products contained in the spent fuel elements, extensive shielding is provided for areas surrounding the refueling pool and the fuel transfer canal to ensure that radiation levels remain below zone levels specified for adjacent areas. Water provides the shielding over the spent fuel assemblies during fuel handling.

12.3.2.2.3 Auxiliary Building Shielding

During normal operations, the major components in the auxiliary building with potentially high radioactivity are those in the CVCS, steam generator blowdown, boron recycle, liquid radwaste, gaseous radwaste, and spent resin handling systems. Shielding is provided for each piece of equipment consistent with its postulated maximum activity (sections 11.1, 11.2, 11.3, and 12.2) and with the access and zoning requirements of adjacent areas. (See drawings 1X6DD001, 1X6DD002, 1X6DD003, 1X6DD004, 2X6DD001, 2X6DD002, 2X6DD003, AX6DD001, AX6DD002, AX6DD003, AX6DD004, AX6DD005, AX6DD006, AX6DD007, AX6DD008, AX6DD009, AX6DD010, AX6DD011, AX6DD012, AX6DD013, AX6DD014, AX6DD015, AX6DD016, AX6DD017, AX6DD018, AX6DD019, AX6DD020, AX6DD021, AX6DD022, AX6DD023, AX6DD024, AX6DD025, AX6DD026, AX6DD027, AX6DD028, AX6DD029, AX6DD030, and AX6DD031.)

Depending on the equipment in the compartments, the radiation zones vary from Zone II through V. Corridors are generally shielded to allow Zone II access, and operator areas for valve compartments are generally Zone II or III for access.

Removable sections of block shield walls and concrete plugs are utilized as necessary for equipment maintenance and spent filter cartridge replacement. Permanent or temporary shielding is used between equipment in compartments with more than one piece of equipment to permit access for maintenance. Where necessary, labyrinth entrances with provisions for adequate ingress and egress for equipment maintenance and inspection are provided and are designed to be consistent with the access and zoning requirements of adjacent areas.

Following reactor shutdown, the residual heat removal (RHR) system pumps and heat exchangers are in operation to remove heat from the RCS. The radiation levels in the vicinity of this equipment will temporarily reach Zone V levels due to corrosion and fission products in the reactor water. Shielding is provided to attenuate radiation from RHR equipment during shutdown cooling operations to levels consistent with the radiation zoning requirements of adjacent areas. The shielding around the RHR equipment is a minimum of 2.5 ft of concrete.

12.3.2.2.4 Fuel Handling Building Shielding Design

The concrete shield walls surrounding the spent fuel cask loading and storage area, and the shield walls surrounding the fuel transfer and storage areas, are sufficiently thick to limit radiation levels outside the shield walls in all accessible areas to Zone II. The building external walls are sufficient to shield external plant areas to Zone I.

All spent fuel removal and transfer operations are performed under borated water to provide radiation protection and maintain subcriticality. Nominal water depths above the active fuel during fuel handling are 10 ft in the reactor cavity and 10 ft in the fuel transfer canal and spent fuel pool. This limits the dose at the water surface to less than 2.5 mrem/h for an assembly in a vertical position. The minimum water depth in the spent fuel pool is a nominal 13 ft above the top of the active fuel in the storage racks; for this depth the dose rate at the water surface is less than 15 mrem/h. Normal water depth above the stored assemblies is about 26 ft, and for this depth the dose rate at the pool surface is significantly less than 2.5 mrem/h. The minimum 4 ft thick concrete walls of the fuel transfer canal and spent fuel pool walls supplement the water shielding and limit the maximum radiation dose levels in working areas to less than 2.5 mrem/h. In the event that fuel is damaged and fuel fragments or pellets are collected on filters or other devices, the shielding requirements for the movement of the filters govern rather than the requirements on the submergence of spent fuel.

The spent fuel cooling and purification (SFCP) system shielding (section 9.1) is based on the maximum activity discussed in subsection 12.2.1 and the access and zoning requirements of adjacent areas. Equipment in the SFCP system to be shielded includes the SFCP heat exchangers, pumps, and piping.

12.3.2.2.5 Turbine Building Shielding Design

Radiation shielding is not required for any process equipment located in the turbine building. Space has been provided so that shielding may be added around the condensate polishing demineralizers if they become radioactive.

12.3.2.2.6 Control Room Shielding Design

The design basis loss-of-coolant accident (LOCA) dictates the shielding requirements for the control room. Consideration is given to shielding provided by the containment structure. Shielding combined with other engineered safety features is provided to permit access and occupancy of the control room following a postulated LOCA, so that radiation doses are limited to 5 rem whole body from contributing modes of exposure for the duration of the accident, in accordance with 10 CFR 50, Appendix A, General Design Criterion 19.

The design basis LOCA is described in subsection 15.6.5. The contribution from direct radiation from airborne fission products inside the containment to personnel doses inside the control room following a postulated LOCA is discussed in section 15A.3. The shielding of the control room ensures compliance with 10 CFR 50, Appendix A, General Design Criterion 19.

The parameters used in the demonstration of control room habitability, in addition to those contained in Regulatory Guide 1.4, are listed in chapter 15. Control room ventilation system parameters are provided in section 6.4. Drawing AX6DD404 provides an isometric view of the control room shielding.

12.3.2.2.7 Miscellaneous Plant Areas and Plant Yard Areas

Sufficient shielding is provided for all plant buildings containing radiation sources so that radiation levels at the outside surfaces of the buildings are maintained below Zone I levels. Plant yard areas that are frequently occupied by plant personnel are fully accessible during normal operation and shutdown. These areas are surrounded by a security fence and closed off from areas accessible to the general public. Access to outside storage tanks that have a contact dose rate greater than 0.25 mrem/h is restricted by a concrete shield wall sufficiently high so that dose rates to personnel in plant yard areas are limited to less than 0.25 mrem/h. Access to the low-level waste storage facility is restricted by a fence and locked gate with shielding as necessary to assure that dose rates to personnel in plant yard areas surrounding the low-level waste storage facility are limited to less than 0.25 mrem/h.

12.3.2.2.8 Spent Fuel Transfer Tube Shielding

The spent fuel transfer tube (drawings 1X2D48J006, 1X2D48A050, and AX2D09A017) is shielded to within local radiation zone limits. This is primarily achieved through the use of permanent shielding. The only removable shielding consists of concrete or steel hatches which reduce radiation in accessible areas to within those levels prescribed in the normal operation radiation zone maps (drawings 1X6DD001, 1X6DD002, 1X6DD003, 1X6DD004, 2X6DD001, 2X6DD002, 2X6DD003, AX6DD001, AX6DD002, AX6DD003, AX6DD004, AX6DD005, AX6DD006, AX6DD007, AX6DD008, AX6DD009, AX6DD010, AX6DD011, AX6DD012, AX6DD013, AX6DD014, AX6DD015, AX6DD016, AX6DD017, AX6DD018, AX6DD019, AX6DD020, AX6DD021, AX6DD022, AX6DD023, AX6DD024, AX6DD025, AX6DD026, AX6DD027, AX6DD028, AX6DD029, AX6DD030, and AX6DD031.)

The removable hatch in the fuel handling building consists of two concrete blocks and provides access to the bellows inspection room outside containment. The hatch inside containment consists of seven laminated steel plates and provides access to the bellows inspection room inside containment. The opening of these hatches is administratively controlled; these hatches are in place during the fuel transfer operation. This will ensure that there will be no access to the spent fuel transfer tube during refueling. With the hatches in place, all accessible areas will be as indicated on the radiation zone maps.

The spent fuel transfer tube in the seismic gap between the containment wall and the internal containment structure and in the seismic gap between the containment wall and the fuel handling building is shielded by a combination of concrete, steel, and permanently installed lead-loaded silicone foam rubber to maintain radiation zone limits for normal operation. Therefore, there is no unshielded portion of the spent fuel transfer tube during the refueling operation.

12.3.2.3 Shielding Calculational Methods

The shielding thicknesses provided to ensure compliance with plant radiation zoning and to minimize plant personnel exposure are based on maximum equipment activities under the plant operating conditions described in chapter 11 and section 12.2.

The thickness of each shield wall surrounding radioactive equipment is determined by approximating as closely as possible the actual geometry and physical condition of the source or sources. The isotopic concentrations are converted to energy group sources using data from standard references.(1,5)

The geometric model assumed for shielding evaluation of tanks, heat exchangers, filters, ion exchangers, evaporators, and the containment is a finite cylindrical volume source. For shielding evaluation of piping, the geometric model is a finite shielded cylinder. In cases where radioactive materials are deposited on surfaces such as pipe, the latter is treated as an annular cylindrical surface source.

The methods and equations of the Reactor Shielding Design Manual (6) and the Engineering Compendium on Radiation Shielding (8) are used to calculate dose rates. Buildup is calculated using Taylor coefficients presented in ORNL-RSIC-10; (7).

In addition, two industry-accepted computer codes--ANISN(9) and QAD(10)--are used for shielding analysis. ANISN is a multigroup one-dimensional discrete ordinates transport code that solves the one-dimensional Boltzmann transport equation for neutrons and gamma rays in slab, sphere, or cylinder geometry. Using a finite-difference technique, ANISN allows general anisotropic scattering; i.e., an Lth order Legendre expansion of the scattering cross-sections. Monte Carlo techniques as described below may be used for more complicated geometries such as penetrations. QAD is a point-kernel general purpose code for estimating the penetration of gamma rays and neutrons that originate in a volume-distributed source. ANISN is used primarily for primary shield design, and QAD is used for configurations not conveniently modeled in one-dimensional geometrics.

For final design, a three-dimensional model is used to simulate radiation streaming from the reactor vessel surface to the containment using the MORSE Monte Carlo program. (11) The source terms used for the MORSE code are generated by the computer code DOT.(12) The source terms are divided into 13 neutron energy groups and 14 gamma energy groups.

The shielding thicknesses are selected to reduce the aggregate computed radiation level from all contributing sources below the upper limit of the radiation zone specified for each plant area.

Shielding requirements in each plant area are evaluated at the point of maximum radiation dose through any wall. In addition, for shielding design purposes the concrete density of 135 lb/ft^3 was assumed. Therefore, the actual anticipated radiation level in each plant area is less than this maximum dose and consequently less than the radiation zone upper limit.

Where shielded entryways to compartments containing high-radiation sources are necessary, labyrinths are designed using methods summarized in ORNL-RSIC-21.(13) The labyrinths are constructed so that the scattered dose rate, plus the transmitted dose rate through the shield wall from all contributing sources, is below the upper limit of the radiation zone specified for each plant area.

12.3.2.4 References

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13. Selph, W. E., "Neutron and Gamma Ray Albedos," ORNL-RSIC-21, Oak Ridge National Laboratory, February 1968.

12.3.3 VENTILATION

The plant heating, ventilating, and air-conditioning (HVAC) systems are designed to provide a suitable environment for personnel and equipment during normal operation, during anticipated events of moderate frequency, and during certain infrequent events. Parts of the plant HVAC systems perform safety-related functions.

12.3.3.1 Design Objectives

The plant HVAC systems for normal plant operation, anticipated events of moderate frequency, and certain infrequent events are designed to meet the requirements of 10 CFR 20, Standards for Protection Against Radiation, and 10 CFR 50, Licensing of Production and Utilization Facilities.

12.3.3.2 Design Criteria

Design criteria for the plant HVAC systems include the following.

A. During normal operation, anticipated events of moderate frequency, and certain infrequent events, the average and maximum airborne radioactivity levels to which plant personnel are exposed in radiation controlled areas of the plant are as low as reasonably achievable (ALARA) and within the limits specified in 10 CFR 20. The average and maximum airborne radioactivity levels outside radiation controlled areas of the plant during normal operation, events of moderate frequency, and certain infrequent events are ALARA and within the limits of Appendix B, Table II of 10 CFR 20.1 - 20.601.

B. During normal operations, anticipated events of moderate frequency, and certain infrequent events, the dose from concentrations of airborne radioactive material

in unrestricted areas beyond the site boundary is ALARA and within the limits specified in 10 CFR 20.1 - 20.601 and 10 CFR 50, Appendix I.

C. The plant siting dose guidelines of 10 CFR 100 will be satisfied following those hypothetical accidents described in chapter 15.

D. The dose to control room personnel shall not exceed the limits specified in General Design Criterion 19 of Appendix A to 10 CFR 50 following those hypothetical accidents described in chapter 15.

12.3.3.3 Design Guidelines

To accomplish the design objectives and to conform to the design criteria, the following design guidelines are employed wherever practicable.

12.3.3.3.1 Guidelines to Minimize Airborne Radioactivity

A. Access control and traffic patterns are considered in the basic plant layout to minimize the spread of contamination.

B. Equipment vents and drains are piped directly to a collection device connected to the collection system. This is to prevent any contaminated fluid from flowing across the floor to a floor drain.

C. All welded piping systems are employed on systems containing radioactive fluids to the maximum extent practicable. If welded piping systems are not employed, drip trays are provided at the points of potential leakage. Drains from drip trays are piped directly to the collection system.

D. Suitable coatings are applied to the concrete floors and walls of potentially contaminated areas to facilitate decontamination.

E. To minimize the amount of airborne radioactivity as a result of valve leakage, 2 1/2 in. and larger valves in radioactive systems are provided with a leakage control system consisting of double sets of packing with lantern rings and with leakoff connections that are piped to drain headers or a reduced packing configuration and the leakoff connection capped. Diaphragm or bellows seal valves are used on those systems where essentially no leakage can be tolerated.

F. Design of equipment incorporates features that minimize the spread of radioactivity during maintenance operations. These features include, but are not limited to, flush connections on pump casings for draining and flushing the pump prior to maintenance and flush connections on piping systems that could become highly radioactive.

12.3.3.3.2 Guidelines to Control Airborne Radioactivity

A. The airflow is directed from areas with lesser potential for contamination to areas with greater potential for contamination.

B. In building compartments with a potential for contamination, the exhaust is designed for greater volumetric flow than is supplied to that area. This minimizes the amount of uncontrolled exfiltration from the area.

C. Consideration is given to the possible disruption of normal airflow patterns by maintenance operations, and provisions are made in the design to prevent adverse airflow direction.

D. The air cleaning system's design, maintenance, and testing criteria are discussed in detail (in response to Regulatory Guides 1.52 and 1.140) in sections 1.9, 6.4, and 9.4 and in subsections 6.2.5 and 6.5.1. An illustrative example of an air cleaning system design is given in paragraph 12.3.3.5.

E. Air being discharged from potentially contaminated areas in the containment, fuel handling building, and the auxiliary building is passed through high efficiency particulate air filters and charcoal adsorbers to remove particulates and halogens. Means are provided to isolate the affected areas in the containment, fuel handling building, and auxiliary building upon indication of contamination. This minimizes the discharge of contaminants to the environment and in-plant exposures.

F. Means are provided to isolate the control room to minimize inleakage of contaminated air to protect personnel.

G. Suitable containment isolation valves are installed in accordance with General Design Criteria 54 and 56, including valve controls, to ensure that containment integrity is maintained. See additional discussion in subsection 6.2.4.

H. Redundant Seismic Category 1 systems and/or components are provided for portions of the ventilation system that serve areas required for safe shutdown of the reactor plant. Included herein are the plant control room and selected engineered safety feature equipment rooms.

I. Atmospheric tanks which contain radioactive materials are vented to the respective building ventilation system for filtration prior to release.

12.3.3.3.3 Guidelines to Minimize Personnel Exposure from HVAC Equipment

A. The guidelines of Regulatory Guide 8.8 have been utilized, as practicable, in the design of the plant ventilation systems.

B. Ventilation fans and filters are provided with adequate access space to permit servicing with minimum personnel radiation exposure. The HVAC system is designed to allow rapid replacement of components. Filter-adsorber unit conformance with the recommendations of Regulatory Guides 1.52 and 1.140 for access and service requirements is summarized in section 1.9.

C. Ventilation ducts are designed to minimize the buildup of radioactive contamination within the ducts.

D. Ventilating air is recirculated in clean areas only. Exhaust from potentially contaminated areas is filtered and then discharged.

E. Access and service of ventilation systems in potentially radioactive areas is provided by component location to minimize operator exposure during maintenance, inspection, and testing as follows.

1. The outside air supply units and building exhaust system components are enclosed in ventilation equipment rooms. These equipment rooms are accessible to the operators. Work space is provided around each unit for anticipated maintenance, testing, and inspection.

2. Local cooling equipment servicing the normal building requirements is located in areas of low contamination potential. (See drawings 1X6DD001, 1X6DD002, 1X6DD003, 1X6DD004, 2X6DD001, 2X6DD002, 2X6DD003, AX6DD001, AX6DD002, AX6DD003, AX6DD004, AX6DD005, AX6DD006, AX6DD007, AX6DD008, AX6DD009, AX6DD010, AX6DD011, AX6DD012, AX6DD013, AX6DD014, AX6DD015, AX6DD016, AX6DD017, AX6DD018, AX6DD019, AX6DD020, AX6DD021, AX6DD022, AX6DD023, AX6DD024, AX6DD025, AX6DD026, AX6DD027, AX6DD028, AX6DD029, AX6DD030, and AX6DD031.)

12.3.3.4 Design Description

The ventilation systems serving the following structures are considered to be potentially radioactive and are discussed in detail in subsections 6.2.2, 6.2.5, and 6.5.1 and in section 9.4.

- Containment building. (See subsections 6.2.2, 6.2.5, and 6.5.1.)
- Auxiliary building. (See subsection 9.4.3.)
- Fuel handling building. (See subsection 9.4.2.)

Although the control room is considered to be a nonradioactive area, radiation protection is provided to ensure habitability. (See section 6.4 and subsection 9.4.1.)

Other structures; e.g., pump intake structures, the administrative building, etc., contain no potential source of radioactivity and are not addressed in this chapter.

12.3.3.5 Air Cleaning Units

The guidance and recommendations of Regulatory Guides 1.52 and 1.140 concerning maintenance and inplace testing provisions for atmospheric cleanup systems, air filtration, and adsorption units have been used as a reference in the design of the various ventilation systems. The extent to which Regulatory Guide 1.52 and 1.140 have been followed is discussed in section 1.9. Drawing AX4DJ3101 shows the typical layout of an air cleaning unit.

Provisions specifically included to minimize personnel exposures and to facilitate maintenance or inplace testing operations are as follows.

A. The loading of the filters and adsorbers with radioactive material during normal plant operation is a slow process. Therefore, in addition to monitoring for pressure drop, the filters are checked for radioactivity on a scheduled maintenance basis with portable equipment. The filter elements are replaced before the radioactivity level is of sufficient magnitude to create a personnel hazard. No shielding is provided since it is not required for the level of radioactivity developed during normal operation. In case of excessive radioactivity caused by a postulated accident, the whole filter is replaced before

normal personnel access is resumed. It will not be necessary for workers to handle filter units immediately after a design basis accident, so exposures can be minimized by allowing the short-lived isotopes to decay before changing the filter.

B. Active components of the atmospheric cleanup systems are designed to permit ready removal.

C. Access to active components is direct from working platforms to simplify element handling. Ample space is provided on the platforms for accommodating safe personnel movement during replacement of components, including the use of necessary material handling facilities and inplace testing operations.

D. HEPA filter banks which are more than three filter units high, where each filter is 2 ft by 2 ft, have a platform to facilitate access to the upper filters.

E. The clear space for doors is a minimum of 20 in. by 50 in.

F. The filters are designed with replaceable units that are clamped in place against compression seals. The filter housing is designed, tested, and proven to be airtight with bulkhead type doors that are closed against compression seals.

12.3.4 AREA RADIATION AND AIRBORNE RADIOACTIVITY MONITORING INSTRUMENTATION

The radiation monitoring system consists of the following:

A. Area radiation monitoring system (ARMS).

B. Process and effluent radiation monitoring system PERMS).

C. Sampling system.

D. Post-accident monitoring systems (PAMS) radiation monitors.

The PERMS, sampling systems, and PAMS radiation monitors are described in section 11.5.

12.3.4.1 Area Radiation Monitoring

The ARMS is provided to supplement the personnel and area radiation survey provisions of the plant health physics program described in section 12.5 and to ensure compliance with the personnel radiation protection guidelines of 10 CFR 20, 10 CFR 50, 10 CFR 70, and Regulatory Guides 1.97, 8.2, 8.8, and 8.12.

The design of the fuel pool racks precludes criticality under all postulated normal and accident conditions. Therefore, criticality monitors, as stated in 10 CFR 70.24 and Regulatory Guide 8.12, are not needed.

The area radiation monitors (ARM) installed in the alternate radwaste building (ARB) monitor radiation levels in and around the ARB. The design objectives of these ARMs are described in paragraph 12.3.4.1.10.

The ARM, which are installed in the radwaste processing facility, monitor radiation levels in the radwaste processing facility process and dressout areas during liquid radwaste processing. The design objectives of the ARM are described in paragraph 12.3.4.1.11.

12.3.4.1.1 Design Objectives

The design objectives of the ARMS during normal operating plant conditions and anticipated operational occurrences are:

A. To furnish records of radiation levels in specific areas of the plant.

B. To warn of uncontrolled or inadvertent movement of radioactive material in the plant.

C. To provide local and remote indication of ambient gamma radiation and local and remote alarms at key points where substantial change in radiation levels might be of immediate importance to personnel frequenting the area.

D. To annunciate and warn of possible equipment malfunctions and leaks in specific areas of the plant.

E. To furnish information for making radiation surveys.

By meeting the above objectives, the area radiation monitoring system aids health physics personnel in keeping radiation exposures as low as reasonably achievable (ALARA).

The design objectives of the ARMS during postulated accidents are:

A. To provide the capability to alarm and initiate a containment ventilation isolation signal in the event of a loss-of-coolant accident (LOCA), fuel handling accident inside containment, or abnormally high radiation inside the containment (monitors RE-0002, RE-0003). In modes 1 through 4, two channels of radiation monitors are required to be operable to ensure radiation monitoring instrumentation necessary to initiate containment ventilation isolation. In mode 6 during core alterations or movement of irradiated fuel assemblies in containment, automatic or system level manual CVI capability is not required and the required channels provide input to control room alarms to ensure prompt operator action to manually close the containment purge and exhaust valves. (See subsection 6.2.4.)

B. To provide long-term post-accident monitoring of conditions at strategic locations. (See subsection 11.5.5.)

12.3.4.1.2 Criteria for Location of Area Monitors

The locations of the area monitors are shown in the flow of personnel drawings (drawings 1X6DD200, 1X6DD201, 1X6DD202, 1X6DD203, 2X6DD200, 2X6DD201, 2X6DD202, AX6DD200, AX6DD201, AX6DD202, AX6DD203, AX6DD204, AX6DD205, AX6DD206, AX6DD207, AX6DD208, AX6DD209, AX6DD210, AX6DD211, AX6DD212, AX6DD213, AX6DD214, AX6DD215, AX6DD216, AX6DD217, AX6DD218, AX6DD219, AX6DD220, AX6DD221, AX6DD222, AX6DD223, AX6DD224, AX6DD225, and AX6DD226).

Considerations for area monitor locations are based on the following:

A. Frequency and length of personnel occupancy of a specific area.

B. Potential for personnel to unknowingly receive high radiation doses.

C. Potential for equipment malfunction.

D. Access areas where, during normal plant operation (including refueling), radiation exposures could exceed the radiation limits due to system failure or personnel error.

E. Access areas where new and spent fuel is received and stored.

F. Containment area for indicating the level of radioactivity and detecting the presence of fission products due to a reactor coolant pressure boundary (RCPB) leak, or fuel handling accident.

G. Normally or potentially radioactive areas.

12.3.4.1.3 General System Description

The area radiation monitors are located at selected locations throughout the plant to detect, indicate, and store information through their associated data processing module on the radiation levels and, if necessary, annunciate abnormal radiation conditions. The ARMS monitors are an integral part of the PERMS, which is described in detail in section 11.5.

The ARMS consists of individual, locally mounted area monitors. Each monitor is composed of the requisite number of channels, with a channel consisting of a radiation detector and check source, except for monitors RE-0005 and RE-0006 which have "keep-alive sources" instead of check sources, and the radwaste processing monitors ARE-16971, ARE-16972, and ARE-16973 which have “no pulse” detection. The detectors for all area monitors are either gamma-sensitive Geiger-Mueller counter tubes or ionization chambers. If exposed to radiation in excess of full-scale indication, the area monitors indicate that the full-scale reading has been exceeded and remain at the full-scale value. If the radiation field causing the overload condition is removed, the system returns to its normal operating condition unless the detector has failed. An administrative procedure (positioning the check source) is initiated to ensure that radiation monitoring equipment has not been damaged. All channels, except for the alternate radwaste building monitors and radwaste processing facility monitors, associated with a monitor are served by a local dedicated data processing module. All channels are indicated and annunciated in the control room and, with the exception of containment area monitors, indicated and alarmed near the detector location (normally at the data processing module) and the alternate radwaste building monitors which alarm in the alternate radwaste building control room. Monitors RE-0002, RE-0003, RE-0005, and RE-0006, which are safety-related, Class 1E, are also indicated and annunciated at the safety-related display console. The radwaste processing facility monitors are indicated and alarmed locally near the detector in the radwaste processing facility local control room, and in the Unit 1 main control room.

12.3.4.1.4 Data Processing Module and Display Console

A description of these components is given in section 11.5.

12.3.4.1.5 Local Annunciation

All area monitors, except those in containment, have local annunciation consisting of an audible alarm rated at 80 dB at 10 ft and a warning light at the local readout.

12.3.4.1.6 Power Supplies

Each channel is provided with an independent power supply, designed such that a failure in that channel does not affect any other channel, except for radwaste processing facility monitors which share a common microprocessor assembly. Monitors that are identified as safety related are redundant and are supplied power from the plant 120-V safety-related buses. Power to the channels that monitor only normal operations is supplied from the regulated 120-V instrumentation bus that is backed by the diesel generator.

12.3.4.1.7 Redundancy, Diversity, and Independence

Monitors designated safety related are part of the safety-related portion of the PERMS and are designed for redundancy, diversity, and independence in accordance with Institute of Electrical and Electronic Engineers (IEEE) 344-1975, IEEE 336-1971, IEEE 279-1971, IEEE 308-1974, IEEE 323-1974, and IEEE 384-1974. All monitors which are Seismic Category 1 are also manufactured and rated to the above standards.

12.3.4.1.8 Area Monitor Description

Table 12.3.4-1 gives the conditions of service for the area monitors. A brief description of each area monitor's function is given below.

A. Control Room Area Monitor RE-0001

   To continuously indicate the radiation levels in the control room. A high alarm signal warns control room personnel of a deteriorated radiological condition inside the control room.

B. Containment Low-Range Area Monitors RE-0002 and RE-0003

   To continuously indicate the radiation levels inside the containment building at the operating deck. During refueling operations a high radiation alarm indicates a fuel drop accident and isolates the containment ventilation system. During power operations, a high radiation alarm indicates a possible LOCA and isolates the containment ventilation system.

C. Fuel Handling Building Area Monitor RE-0008

   To continuously indicate the radiation levels inside the fuel handling building. A high radiation alarm signal warns the occupants of the fuel handling building of a deteriorated radiological condition.

D. Sampling Room Area Monitor ARE-0007B

   To continuously indicate the radiation levels in the sampling room. A high radiation alarm signal warns the occupants of the sampling room of a deteriorated radiological condition.

E. Seal Table Room Area Monitor RE-0011

   To continuously indicate the radiation levels in the seal table room and establish radiological habitability prior to entry. A high radiation alarm signal warns occupants of the seal table room of a deteriorated radiological condition.

F. Containment Access Hatch Area Monitor RE-0004

To continuously indicate the radiation levels in the containment access hatch and establish radiological habitability prior to entry.

G. Containment High Range Area Monitors RE-0005 and RE-0006

To indicate, along with RE-0002 and RE-0003, the radiation levels inside the containment building at the operating deck following a design basis accident.

H. Technical Support Center Area Monitor

Upon activating the Technical Support Center, a portable area radiation monitor is used to continuously indicate the radiological condition of the area. A high alarm signal provides warning to personnel of a deteriorating radiological condition. The monitor is placed in a manned area which is closest to the principal filtration units adjacent to the TSC. The portable unit provides an alarm point of 5 millirem per hour and dose rate indication up to 100 millirem per hour.

I. Radwaste Processing Facility Area Monitor ARE-16971 (North Wall Near the High Integrity Container (HIC) Storage Area)

To continuously indicate the radiation levels at the north wall of the radwaste processing facility process area near the subterranean HIC storage area. A high alarm alerts personnel in the area of a deteriorating radiological condition.

J. Radwaste Processing Facility Area Monitor ARE-16972 (East Wall Near Demineriliz er Subterranean Vault Area)

To continuously indicate the radiation levels at the east wall of the radwaste processing facility process area near the subterranean demineralizer vault area. A high alarm alerts personnel in the area of a deteriorating radiological condition.

K. Radwaste Processing Facility Area Monitor ARE-16973 (Dressout Area)

To continuously indicate the radiation levels in the dressout area of the radwaste processing facility. A high alarm alerts personnel in the area of a deteriorating radiological condition.

12.3.4.1.9 Range and Alarm Setpoints

The range and control function of the PERMS area monitors are given in table 12.3.4-2.

Alarm setpoints are controlled by plant procedures and the Offsite Dose Calculation Manual (ODCM) where appropriate.

Radiation zones for VEGP are described in table 12.3.1-1.

With the exception of area monitors RE-0002, RE-0003, RE-0004, RE-0005, RE-0006, and RE-0011, all of the area monitors are located in radiation Zones I or II.

The control room monitor RE-0001 has a greater sensitivity than the other area monitors, since it is located in a Zone I radiation area; monitors RE-0002, RE-0003, RE-0005, and RE-0006 cover a wide range of radiation levels. During plant shutdown including refueling operations, the radiation level on and above the operating deck should be less than 2.5 mR/h. The high end of the range is dictated by the design basis accident, a LOCA.

Each area monitor has two alarm setpoints, intermediate and high. If a monitor has a control function; i.e., RE-0002 and RE-0003, the control function is triggered coincidentally with the high alarm setpoint. An intermediate alarm gives a visual indication in the control room and near the detector, except for containment area monitors which have no local alarm or indicator, that the radiation level has reached the intermediate setpoint. A high alarm gives both a visual and audible indication in the control room and near the detector, except for containment area monitors which have no local alarm indicators, that the high alarm setpoint has been reached. The radwaste processing facility monitors' alarm gives visual and audible indication near the detector in the radwaste processing facility local control room, and in the Unit 1 main control room.

For testing, each area monitor has a check source assembly (except for RE-0005 and RE-0006, which have keep-alive sources) which is operated from the safety-related display console for the safety-related monitors and from the communications display computer and PERMS display computer for the nonsafety-related monitors, and uses a sealed source. Inservice inspection, calibration, and maintenance of the ARMS monitors is discussed in paragraph 11.5.2.5.

12.3.4.1.10 Design Objectives for the Radwaste Processing Facility Area Radiation Monitors

The design objectives of the radwaste processing facility ARM during normal liquid radwaste processing and anticipated operational occurrences are:

A. To warn of uncontrolled or inadvertent movement of radioactive material in the radwaste processing facility.

B. To provide local and remote (in the radwaste processing facility control room and Unit 1 main control room) indication (audible and visual) of ambient gamma radiation levels in the radwaste processing facility.

C. To annunciate and warn of possible equipment malfunctions and radioactive leaks in the radwaste processing facility.

D. To furnish information for making radiation surveys.

By meeting the above objectives, the ARM aid health physics personnel in keeping radiation exposure as low as reasonably achievable (ALARA).

12.3.4.2 Standard Review Plan Evaluation

The VEGP did not utilize American National Standards Institute/ American Nuclear Society (ANSI/ANS) HPSSC-6.8.1-1981. Design of the VEGP ARMS began prior to the issuance in 1981 of ANSI/ANS HPSSC-6.8.1, Location and Design Criteria for Area Radiation Monitoring Systems for Light-Water Nuclear Reactors, and therefore the standard was not specifically addressed in the initial design stages. However, the ARMS is in conformance with other applicable regulations and guides. (See paragraph 12.3.4.1.)

Criticality monitors as defined in the Standard Review Plan are not included in the VEGP design. Paragraph 12.3.4.1 states that the design of the fuel pool racks precludes criticality under all postulated normal and accident conditions and that therefore criticality monitors are not needed. Supporting information of this subject is given in subsections 9.1.1, New Fuel Storage, and 9.1.2, Spent Fuel Storage, and paragraph 4.3.2.6, Criticality of the Reactor During Refueling.

TABLE 12.3.1-1

RADIATION ZONES

Zone	Maximum Dose Rate (mrem/h)	Description[a]
I	≤ 0.25	Controlled access, unlimited occupancy.
II	≤ 2.5	Controlled access, limited occupancy.
III	≤ 15	Controlled access, limited occupancy.
IV	≤ 100	Controlled access, limited occupancy.
V	> 100	Restricted access, limited occupancy for very short periods. Access controlled as stated in the Technical Specifications.

a. Controlled access, unlimited occupancy areas: where entry and exit by plant employees and visitors are not under the direct supervision of the plant health physics staff. These areas can be occupied by plant personnel or visitors on an unlimited time basis with a minimum probability of health hazard from radiation exposure.

Controlled access, limited occupancy areas: Where higher radiation levels arid/or radioactive contamination, which have a greater probability of radiation health hazard to individuals, can be expected. Only individuals directly involved in the operation of the plant will, in general, be allowed to enter these areas. Entry and exit are authorized and supervised by the plant health physics staff.

Occupancy: The time spent by an individual in a particular area. Occupancy is to be determined on an area by area and individual by individual basis.

Restricted access, limited occupancy areas: Where extremely high radiation levels and/or radioactive contamination is expected. Only individuals directly involved in the operation of the plant will, in general, be allowed to enter these areas. These areas are normally inaccessible with locked doors and positive control of access. Entry and exit are under the supervision of the plant health physics staff.

TABLE 12.3.1-2

EQUIPMENT SPECIFICATION LIMITS FOR
COBALT IMPURITY LEVELS

Component	Material	Maximum Weight Percent Cobalt
Reactor internals (nonactive region)	SS	0.20
Reactor internals (active region)	SS	0.12
Reactor vessel clad	SS	0.20
Reactor coolant piping	SS	0.20
Reactor internal bolting material	SS	0.25
Reactor coolant pumps	SS	0.20
Pressurizer	SS	0.20
Steam generators	Inconel	0.10
Fuel (nonactive region)	SS	0.12
Fuel (active region)	SS	0.08
Fuel	Inconel	0.10
Fuel	Zircaloy/ ZIRLO™	0.002

TABLE 12.3.1-3

APPROXIMATE REACTOR COOLANT SYSTEM
WETTED SURFACE AREAS

Component	Material	Surface Area (ft^2)
Reactor internals	SS	4236
Reactor vessel clad	SS	2190
Reactor coolant piping	SS	2758
Reactor internal bolting material	SS	Negligible
Reactor coolant pumps	SS	Negligible
Steam generators	Inconel	1.90×10^5
Fuel (nonactive region)	SS	2000
Fuel (active region)	SS	3600
Fuel	Inconel	7.80×10^3
Fuel	Zircaloy/ ZIRLO™	7.78×10^4

TABLE 12.3.1-4

APPROXIMATE REACTOR COOLANT SYSTEM WETTED SURFACE AREAS OF STELLITE

Component	Surface Area (ft^2)
Reactor internals	3.2
Reactor coolant pump journals	17.2
Control rod drive mechanisms	10.8
RCS valves	2.6

TABLE 12.3.1-5

POST-ACCIDENT RADIATION LEVELS IN VITAL AREAS (NUREG-0737, 11.B.2 SHIELDING REVIEW)

		Projected Dose Rate (rem/hr)				
Building Room	Description	0 h	1 h	24 h	168 h	720 h
Auxiliary building						
D105	Switchgear train A	2.10E1	1.11E1	2.56E-1	5.82E-2	1.78E-2
C109	MCCIABD	3.80E2	1.00E2	2.45	5.24E-1	1.64E-1
B16	MCC1BBC Unit 1[a]	1.10	5.85E-1	1.35E-2	3.07E-3	9.40E-4
B122	MCC2BBC Unit 2	1.40E1	7.38	1.71E-1	3.89E-2	1.18E-2
116	1BBB motor control center train B	1.00E-1	5.27E-2	1.22E-3	2.77E-4	8.47E-5
118	1ABB motor control center train A	1.50E1	3.96	9.69E-2	2.07E-2	6.48E-3
207	Switchgear room	<4.4E-4	Nil	Nil	Nil	Nil
Control building						
B47	Train B channel 2	3.08E-3	1.62E-3	3.75E-5	8.53E-6	2.61E-6
B48	Train D channel 4	3.08E-2	1.62E-2	3.75E-4	8.53E-5	2.61E-5
B52	Train A channel 1	3.08E-2	1.62E-2	3.75E-4	8.53E-5	2.61E-5
B55	Train C channel 3	3.08E-3	1.62E-3	3.75E-5	8.53E-6	2.61E-6
B61	Switchgear train B	1.37E2	7.22E1	1.67	3.79E-1	1.16E-1
B76	Switchgear train A	3.47	1.83	4.23E-2	9.61E-3	2.94E-3
B79	Motor control center room	1.40E2	7.38E1	1.71	3.88E-1	1.19E-1
A43	Shutdown room train B	9.87	5.20	1.20E-1	2.73E-2	8.36E-3
A48	4.16-kV switchgear train A	6.00E-1	3.16E-1	7.32E-3	1.66E-3	5.08E-4
A50	4.16-kV switchgear train B	1.52E2	8.01E1	1.85	4.21E-1	1.29E-1
A54	13.8-kV switchgear	3.62E2	1.91E2	4.42	1.00	3.07E-1
A64	13.8-kV switchgear	4.56E2	2.40E2	5.56	1.26	3.86E-1
A75	Shutdown room train A	2.58E-2	1.36E-2	3.15E-4	7.15E-5	2.19E-5
A77	Motor control center room	2.42E1	1.28E1	2.95E-1	6.7OE-2	2.05E-2
Diesel generator building		1.09	5.75E-1	1.33E-2	3.02E-3	9.23E-4
Auxiliary feedwater building		2.65E-1	1.40E-1	3.23E-3	7.34E-4	2.24E-4
Control room, from containment[b]		3.8E-5	8.7E-5	5.6E-5	2.0E-5	6.1E-6
Technical support center, from containment and TSC filter units		2.44E-1	1.74E-1	1.27E-2	4.84E-3	4.68E-4

a. Doses to the equivalent Unit 2 room are similar unless noted otherwise.

b. Doses to the radiochemistry laboratory from direct radiation are similar to that seen in the control room.

TABLE 12.3.4-1 (SHEET 1 OF 2)

CONDITIONS OF SERVICE FOR AREA RADIATION MONITORS

Area Monitor	Detectors Per Unit	Operating Temperature (°F)	Pressure	Relative Humidity (%)	Radiation Zone	Safety Classification	Location/ Elevation (ft)
RE-0001 control room	1	65-85	-1/8 in. to +1/2 in. WG	50 (max)	I	NNS	Control room at 224.5
RE-0002, RE-0003 containment low range	2	60-120	-1.5 in. WG to +3 psig	17.7 to 50	V	SC-3/1E	Containment at 224.5
RE-0008 fuel handling building	1	40-104	-1/4 in. to 0 in. WG	20 to 95	II	NNS	Fuel handling building at 224.5
ARE-0007B sampling room	1 (shared)	65-85	0 psig	40 to 60	II	NNS	Sample room at 224.5
RE-0011 seal table instrumentation	1	60-120	-1.5 in. WG to +3 psig	17.7 to 50	IV	NNS	Seal table room at 197
RE-0004 containment access hatch	1	60-120	-1.5 in. WG to +3 psig	17.7 to 50	III	NNS	Inside containment personnel airlock at 227
RE-0005, RE-0006 containment high range[(a)]	2	60-120	-1.5 in. WG to +3 psig	17.7 to 50	V	SC-3/1E	Outside surface of shield wall at 224.5. 2-RE-0006 is located near the personnel airlock.
ARE-16971 Radwaste processing facility HIC storage (north wall)	1 (shared)	50-100	Slightly negative to 0 psig	100(max)	V	NNS	Radwaste processing facility north wall near subterranean HIC storage
ARE-16972 Radwaste processing facility demin vault (east wall)	1 (shared)	50-100	Slightly negative to 0 psig	100(max)	V	NNS	Radwaste processing facility east wall near subterranean demin vault
ARE-16973 Radwaste processing facility dressout area	1 (shared)	60-80	0 psig to slightly positive	20 to 95	II	NNS	Radwaste processing facility dressout area

TABLE 12.3.4-1 (SHEET 2 OF 2)

Area Monitor	Detectors Per Unit	Operating Temperature (°F)	Pressure	Relative Humidity (%)	Radiation Zone	Safety Classification	Location/ Elevation (ft)
ARE-16851 ARB west wall	1 (shared)	17-104	0 psig to-3 psig	100 (max)	I	NNS	Outside, west wall of the alternate radwaste building (ARB)
ARE-16852 ARB east wall	1 (shared)	17-104	0 psig to-3 psig	100 (max)	I	NNS	Outside surface of the shield wall on the east side of the ARB
ARE-16853 ARB interior	1 (shared)	50-120	Slightly negative to 0 psig	100 (max)	IV	NNS	South wall of the demineralizer vault in the ARB
ARE-16854 ARB dressout area	1 (shared)	60-80	0 psig to slightly positive	100 (max)	II	NNS	Dressout area for the ARB

a. These monitors are qualified for post-LOCA environment.

TABLE 12.3.4-2 (SHEET 1 OF 2)

RANGE AND CONTROL FUNCTIONS FOR AREA RADIATION MONITORS

Monitor	Range (mR/h)	Sensitivity (mR/h)	Control Function	Accuracy
RE-0001 control room	10^{-2} to 10^{3}	10^{-2}	No	± 20 percent of actual radiation field
RE-0002, RE-0003 containment low range	$4.5x10^{-1}$ to $5.4x10^{3}$ (both)	10^{-1}	Yes, isolates containment ventilation	± 20 percent of actual radiation field system
RE-0008 fuel handling building	10^{-1} to 10^{4}	10^{-1}	No	± 20 percent of actual radiation field
ARE-0007B sampling room	10^{-1} to 10^{4}	10^{-1}	No	± 20 percent of actual radiation field
RE-0011 seal table room	10^{-1} to 10^{4}	10^{-1}	No	± 20 percent of actual radiation field
RE-0004 containment access hatch	10^{-1} to 10^{4}	10^{-1}	No	± 20 percent of actual radiation field
RE-0005, RE-0006 (A and B) containment high range	10^{3} to 10^{11} (both)	10^{3}	No	± 20 percent of actual radiation field

TABLE 12.3.4-2 (SHEET 2 OF 2)

Monitor	Range (mR/h)	Sensitivity (mR/h)	Control Function	Accuracy
ARE-16971 Radwaste processing facility HIC storage (north wall)	10^{-3} to 10^{4}	10^{-3}	No	± 23 percent of actual radiation field
ARE-16972 Radwaste processing facility demin vault (east wall)	10^{-3} to 10^{4}	10^{-3}	No	± 23 percent of actual radiation field
ARE-16973 Radwaste processing facility dressout area	10^{-3} to 10^{4}	10^{-3}	No	± 23 percent of actual radiation field

12.4 DOSE ASSESSMENT

Radiation exposures in the plant are primarily due to direct radiation from components and equipment containing radioactive fluids. In addition, in some plant radiation areas there can be radiation exposure to personnel due to the presence of airborne radionuclides. Inplant radiation exposures during normal operation and anticipated operational occurrences are discussed in subsection 12.4.1.

A summary paragraph on radiation exposures due to direct radiation and airborne radioactivity at locations outside the plant structures is provided in subsection 12.4.2.

Radiation exposures to operating personnel will be within 10 CFR 20 limits. Radiation protection design features described in section 12.3 and the health physics program outlined in section 12.5 will ensure that the occupational radiation exposures (ORE) to operating personnel during operation and anticipated operational concurrences will be as low as reasonably achievable (ALARA).

Regulatory Guide 8.19 establishes a procedure for providing an estimate of annual dose to personnel. This procedure provides an assessment by specifying all actions and/or operations which could potentially lead to significant levels of exposure and estimating the frequency and anticipated dose levels for said actions. These data are then combined to formulate an annual plant dose.

While the method employed in the dose assessment of the VEGP provides the same information, it is assessed in a more generalized manner. For the initial dose estimates, rather than performing a breakdown of all operations concerned (routine maintenance operations, refueling, etc.), an estimate of the manhours required annually was performed for each type operation. These manhours were broken down by the percent of time spent in the five radiation zones, and a total occupancy time for each radiation zone was determined. With the expected average dose for each of the zones, an estimated annual exposure was obtained for each zone. Further discussion of this method is provided in subsection 12.4.1. After the plants are in operation, annual dose estimates will be based on previous VEGP history.

In addition, Regulatory Guide 8.19 establishes a means of verifying that the designs of specific areas and systems are consistent with the ALARA philosophy. However, rather than relying upon more limited information upon which the specific manpower and dose breakdown as recommended in Regulatory Guide 8.19 are based, VEGP utilizes the concept of the ALARA review. This review is a systematic review performed to ensure that the ALARA considerations of Regulatory Guide 8.8 are followed and provides the same design control as the method proposed in Regulatory Guide 8.19. For further discussion of the ALARA review, see paragraph 12.1.1.1.

12.4.1 EXPOSURES WITHIN THE PLANT STRUCTURES

12.4.1.1 Direct Radiation Dose Estimates

Annual man-rem doses from direct radiation during the performance of the routine functions, such as operation and surveillance, normal maintenance, radwaste handling, refueling, and inservice inspection, have been estimated, using the following bases:

A. Radiation exposure data from operating pressurized- water reactors (PWRs) are given in tables 12.4.1-1 through 12.4.1-11.[1][2][3]

B. Expected average dose rates in plant radiation areas are discussed in this section.

C. Expected occupancy times for various work function personnel in the different plant radiation areas are listed in table 12.4.1-12.

D. Anticipated manpower requirements for each unit are given in table 12.4.1-12.

E. Table 12.4.1-13 provides an estimate of the distribution of the annual man-rem according to routine functions.

The maximum and expected average dose rates in the plant radiation areas are given below:

Zone	Maximum Dose Rate (mrem/h)	Expected Average Dose (mrem/h)
I	0.25	0.06
II	2.5	1.0
III	15.0	6.0
IV	100.0	40.0
V	>100.0	100.0

The maximum dose rates are determined by shielding calculations based on conservative assumptions regarding sources, self-shielding, locations, etc. The expected dose rates are estimated by assuming a failed fuel percentage of 0.12, by assuming that stringent water chemistry control and improved design will minimize crud buildup and hence the expected dose rates in various radiation zones, and by the recognition that real maximum doses in a given zone are localized effects. The expected average doses given above are used in computing the doses for personnel involved in all operations, except inservice inspection and special maintenance.

The direct radiation dose estimates have been developed from exposure models for each of the major job categories within routine functions. Each exposure model has been developed by breaking the job into individual packages and identifying expected radiation fields, time spent in each radiation field, and the number of men required to carry out each package. Engineering judgment and feedback from operating plant experience have been used to define typical values for each parameter in the exposure model. As such, the resultant exposure estimates should be used as typical values, keeping in mind the variability of the input data from which the estimates were developed.

Initial estimated exposure to plant personnel from direct gamma radiation during the performance of routine functions is estimated to be approximately 418 man-rem/year/unit. Details of these initial man-rem estimates are given in table 12.4.1-13.

12.4.1.2 Airborne Radioactivity Dose Estimates

Due to leakages of radioactive fluids into the auxiliary, containment, radwaste, fuel handling, and turbine buildings, plant personnel are exposed to radionuclides released into the atmosphere of these buildings by the leaked fluids. These atmospheric contaminants contribute to the total body, thyroid, and lung doses.

The peak airborne concentrations for most areas in the plant are within the limits specified in 10 CFR 20. By use of appropriate respiratory equipment and/or limitation of occupancy time, personnel are allowed to enter areas where the airborne activity levels exceed 10 CFR 20 limits.

The expected annual doses to plant personnel from airborne radioactivity for each building in the plant are presented in table 12.4.1-14. The assumptions used to determine airborne radioactivity in each building, along with the airborne concentrations for all areas, are presented in subsection 12.2.2 and tables 12.2.2-1 and 12.2.2-2.

Doses resulting from airborne radioactivity are calculated by the methods discussed below using appropriate portions of 10 CFR 20.1 - 20.601; TID-14844[4]; recommendations of Regulatory Guides 1.4, 1.5, 1.21, 1.24, and 1.25; and the assumptions of ICRP-2[5]. Accident doses are discussed in chapter 15.

12.4.1.2.1 Method for Calculating the Total Body Gamma Dose

The total body gamma dose for isotopes other than tritium can be calculated using the following formulas:

$$TBD = \sum_i (C_i)\,(DCF_i)\,(t)\,(10^9)$$

where:

- TBD = total body dose (rem/year).
- C_i = equilibrium airborne concentration of each radioisotope ($\mu Ci/cm^3$).
- DCF_i = dose conversion factor for each isotope ($mrem\text{-}m^3/pCi\text{-}h$).
- t = total exposure time in hours over 1 year.
- 10^9 = conversion factor ($pCi\text{-}cm^3\text{-}rem/\mu Ci\text{-}m^3mrem$).

The dose conversion factors are tabulated in table 12.4.1-15.

12.4.1.2.2 Method for Calculating the Lung Dose

The lung dose for isotopes other than tritium can be calculated using the following formula:

$$D_{lung} = \sum_i \frac{(C_i)\ (MPD)_{lung}}{(MPC)_{a_{lung\,i}}} \left(\frac{t}{250}\right)$$

where:

- D_{lung} = lung dose (rem/year).
- C_i = equilibrium airborne concentration of radioisotope i ($\mu Ci/cm^3$).
- $(MPC)_{a_{lung\,i}}$ = maximum permissible concentration in air resulting in the MPD to the lung for radioisotope i ($\mu Ci/cm^3$).
- $(MPD)_{lung}$ = maximum permissible dose to lung (rem/year).
- t = total exposure time in days (8-h basis).

The maximum permissible concentration in air and maximum permissible dose to the lungs are tabulated in table 12.4.1-16.

12.4.1.2.3 Method for Calculating the Inhalation Thyroid Dose

The thyroid dose can be calculated using the following formula:

$$D_{thyroid} = \sum_i (C_i)(t)(BR)(DCF)_i$$

where:

$D_{thyroid}$	=	inhalation thyroid dose (rem/year).
C_i	=	equilibrium concentration of each radio-isotope ($\mu Ci/cm^3$).
t	=	exposure time in days over 1 year.
BR	=	breathing rate (cm^3/day).
$(DCF)_i$	=	dose conversion factors for each isotope (rem/μCi).

The breathing rate and dose conversion factors are presented in table 12.4.1-16.

12.4.1.2.4 Method for Calculating the Tritium Dose

The dose for tritium can be calculated using the following formula:

$$D_{tritium} = (4.12 \times 10^3)(C)(t)$$

where:

$D_{tritium}$	=	dose caused by absorption of tritium by lungs and skin (rem/year).
C	=	equilibrium airborne concentration of tritium ($\mu Ci/cm^3$).
t	=	exposure time in days over 1 year (8-h basis).

The factor 4.12×10^3 is derived from the breathing rate, tritium uptake coefficient, and other factors given in table 12.4.1-16. The model assumes that the dose from absorption through the skin is equal to the dose absorbed by the lungs.

12.4.1.2.5 In-Plant Radiation Monitoring Program

A program shall be maintained, the in-plant radiation monitoring program, which ensures the capability to accurately determine the airborne iodine concentration in vital areas under accident conditions. This program shall include the following:

A. Training of personnel.

B. Procedures for monitoring, and

C. Provisions for maintenance of sampling and analysis equipment.

12.4.1.3 Standard Review Plan Evaluation

Dose assessments should be made in accordance with the procedures stated in Regulatory Guide 8.19.

An evaluation of the VEGP dose assessment methodology is provided in the introductory paragraphs of section 12.4.

12.4.1.4 References

1. U.S. Nuclear Regulatory Commission, "Occupational Radiation Exposure at Light Water Cooled Power Reactors," 1969-1975, NUREG-0109, 1976.
2. U.S. Nuclear Regulatory Commission, "Occupational Radiation Exposure at Light Water Cooled Power Reactors," 1969-1974, NUREG-75/032, 1975.
3. U.S. Nuclear Regulatory Commission, "Occupational Radiation Exposure at Commercial Nuclear Power Reactors," 1981, NUREG-0713, vol 3, 1982.
4. DiNunno, J. J., et al, "Calculations of Distance Factors for Power and Test Reactor Sites," TID-14844, March 1962.
5. "Report of Committee II on Permissible Dose for Internal Radiation," ICRP Publication 2, Pergamon Press, New York, 1959.
6. U.S. Nuclear Regulatory Commission letter, "Request for Exemption from Portion of General Design Criteria 4 of Appendix A to 10 CFR 50 (Units 1 and 2)," (H. R. Denton to D. O. Foster), dated February 5, 1985.
7. Georgia Power Company letter, "Alternative Pipe Break Design Considerations," (D. O. Foster to H. R. Denton), dated April 2, 1984.

12.4.2 RADIATION EXPOSURE OUTSIDE THE PLANT STRUCTURES

12.4.2.1 Direct Radiation Dose Estimates

The direct radiation from the containment, the low-level waste storage facility, and the auxiliary and turbine buildings is negligible compared to that from the radwaste processing facility and outside storage tanks. The principal sources of radioactivity not stored in the plant structures are the radwaste processing facility, the reactor makeup water storage tanks, and the refueling water storage tanks. The tanks are surrounded by concrete walls that shield the yard area to Zone I limits (0.25 mrem/h). The radwaste processing facility utilizes interior and exterior shield walls to maintain yard levels to Zone I. The annual dose at the nearest site boundary, based on 8760-h occupancy, is less than 1 mrem.

12.4.2.2 Exposures Due to Airborne Radioactivity

Doses at the site boundary due to released activity are given in subsection 11.3.3.

12.4.3 EXPOSURES TO CONSTRUCTION WORKERS

12.4.3.1 Direct Radiation Dose Estimates

For the purpose of estimating the direct radiation exposure to construction workers in Unit 2 resulting from the operation of Unit 1, it is assumed:

A. The radiation field at the interface between Units 1 and 2 is 0.25 mrem/h, ie., the interface between the two units is Zone 1. Because there are no contained

sources in the Unit 2 construction areas, this radiation level is reduced for most of the construction area due to distance from the interface and shielding provided by structures.

B. Construction manpower levels are as given in table 12.4.3-1 for the time period between the scheduled fuel loading of Unit 1 (September 1986) and the completion of Unit 2 construction (March 1988).

1. The construction workers are onsite 60 h/week for 50 weeks/year.

2. The construction workers spend 15 percent of their time in close proximity to the interface with Unit 1. The exposure rate for this time is 0.25 mrem/h (the full radiation field at the interface).

3. The construction workers spend 85 percent of their time in areas where the radiation field is reduced by distance and/or shielding. A dose reduction factor of 5 is assumed, resulting in an exposure rate of 0.05 mrem/h.

Based on these assumptions, the anticipated annual exposures to Unit 2 construction workers are given in table 12.4.3-2.

The average annual exposure to an individual is 0.26 rem/year; however, as can be seen from table 12.4.3-1, the majority of the workers will be on the site for less than a year. Assuming that no new workers are brought onto the site, thus maximizing individual exposure estimates, results in an anticipated individual cumulative exposure of 0.36 rem over 18 months.

12.4.3.2 Exposures Due to Airborne Radioactivity

The construction worker doses (table 12.4.3-1) due to airborne radioactivity were calculated using the individual dose option in the GASPAR code, in conjunction with the near field dispersion and deposition data for each release point (table 12.4.3-3) and the projected craft requirements for future periods at Unit 2 construction activities.

12.4.4 DECOMMISSIONING DOSE ESTIMATES

The radiation protection design features (section 12.3) established for station operation will aid in maintaining occupational radiation exposure as low as reasonably achievable during decommissioning. The shielding design allows for efficient mothballing and entombment. Decommissioning by removal of all contaminated and activated equipment will be aided by remote handling of equipment, equipment layout (subsection 12.3.1), and administrative planning which includes the health physics program (section 12.5).

Specifications and limitations on cobalt content in equipment components will serve to limit radiation doses from crud buildup during both operation and subsequent decommissioning. A summary of the features in Westinghouse pressurized water reactors (PWRs) that reduce occupational exposure are given in WCAP-8872, Design, Inspection, Operation, and Maintenance Aspects of the Westinghouse NSSS to Maintain Occupational Radiation Exposures As Low As Reasonably Achievable, April 1977.

A National Environmental Studies Project(1) study analyzed the decommissioning alternatives for light-water reactors and high-temperature gas-cooled reactors. The majority of the estimated PWR occupational radiation exposure due to removal/dismantling comes from decontaminating the primary and radwaste systems. Estimated occupational radiation

exposures from the study are given in table 12.4.4-1. The dominant radioactive isotopes are listed in table 12.4.4-2.

12.4.4.1 Reference

1. Atomic Industrial Forum, Inc., "An Engineering Evaluation of Nuclear Power Reactor Decommissioning Alternatives," AIF/NESP-009, 1976.

TABLE 12.4.1-1

AVERAGE NUMBER OF PERSONNEL PER PWR UNIT FOR THE PERIOD 1969-1981[3]

Year	Number of Units	Average Number of Personnel/Unit
1969	3	151
1970	3	447
1971	4	226
1972	5	377
1973	12	787
1974	20	485
1975	26	419
1976	30	586
1977	34	614
1978	39	659
1979	42	924
1980	42	1101
1981	44	1076
1969-1981	Overall average	761

NOTE:

During the years 1969-1972, all plants reported collective doses, but a few did not submit the number of personnel that received measurable doses.

TABLE 12.4.1-2

AVERAGE OCCUPATIONAL RADIATION EXPOSURE (MAN-REM DOSE)
PER PWR UNIT FOR THE PERIOD 1969-1981[3]

Year	Number of Units	Average MREM Dose/Unit
1969	4	165
1970	4	684
1971	6	307
1972	8	464
1973	12	783
1974	20	331
1975	26	318
1976	30	460
1977	34	396
1978	39	429
1979	42	516
1980	42	578
1981	44	652
1969-1981	Overall average	488

NOTE:

During the years 1969-1972, all plants reported collective doses, but a few did not submit the number of personnel that received measurable doses.

TABLE 12.4.1-3 (SHEET 1 OF 2)

AVERAGE OCCUPATIONAL RADIATION EXPOSURE (MAN-REM DOSE) BASED ON PWR PLANT AGE[3]

Year of Operation (Range of Months)	No. of Units	Man-rem per Unit
1	35	193
2	42	348
3	41	445
4	37	434
5	34	457
6	28	461
7	25	529
8	17	779
9	10	782
10	6	875
11	5	441
12	2	650
13	2	1870
14	2	2130

Initial Operation Date	Units Considered	
Jan 1968	Haddam Neck	- X X X X X X X X X X X X X
Jan 1968	San Onofre 1	- X X X X X X X X X X X X X
July 1970	Ginna	X X X X X X X X X X X
Dec 1970	Point Beach 1	X X X X X X X X X X X
March 1971	Robinson 2	- X X X X X X X X X X
Dec 1971	Palisades	X X X X X X X X X X
Oct 1972	Point Beach 2	X X X X X X X X X
Dec 1972	Maine Yankee	X X X X X X X X X
Dec 1972	Surry 1	X X X X X X X X X
Dec 1972	Turkey Point 3	X X X X X X X X X
May 1973	Surry 2	X X X X X X X X
July 1973	Oconee 1	X X X X X X X X
Sept 1973	Turkey Point 4	X X X X X X X X
Sept 1973	Fort Calhoun	X X X X X X X X
Aug 1973	Indian Point 2	X X X X X X X X
Dec 1973	Zion 1	X X X X X X X X
Dec 1973	Prairie Island 1	X X X X X X X X
June 1974	Kewaunee	X X X X X X X
Sept 1974	Three Mile Island 1	X X X X X X X
Sept 1974	Zion 2	X X X X X X X
Sept 1974	Oconee 2	X X X X X X X
Dec 1974	Arkansas 1	X X X X X X X
Dec 1974	Oconee 3	X X X X X X X
Dec 1974	Prairie Island 2	X X X X X X X

TABLE 12.4.1-3 (SHEET 2 OF 2)

Year of Operation (Range of Months)	No. of Units	Man-rem per Unit
1	35	193
2	42	348
3	41	445
4	37	434
5	34	457
6	28	461
7	25	529
8	17	779
9	10	782
10	6	875
11	5	441
12	2	650
13	2	1870
14	2	2130

Initial Operation Date	Units Considered	
April 1975	Rancho Seco	- X X X X X
May 1975	Calvert Cliffs 1	X X X X X
Aug 1975	Cook 1	X X X X X
Dec 1975	Millstone Point 2	X X X X X
May 1976	Trojan	X X X X
Aug 1976	Indian Point 3	X X X X
Oct 1976	Beaver Valley	X X X X
Dec 1976	St. Lucie	X X X X
March 1977	Crystal River 3	- X X X
April 1977	Calvert Cliffs 2	- X X X
June 1977	Salem	X X X
Nov 1977	Davis-Besse	X X X
Dec 1977	Farley	X X X
June 1978	North Anna 1	X X
June 1978	North Anna 2	X X
July 1978	Cook 2	X X
Dec 1978	Three Mile Island 2	X X
March 1980	Arkansas 2	- X

NOTE:

Only PWRs operating at power levels ≥ 450 MWe have been considered, with the exception of San Onofre 1 (430 MWe).

TABLE 12.4.1-4

DISTRIBUTION OF THE NUMBER OF PERSONNEL
(>100 MREM/YEAR)
ACCORDING TO WORK FUNCTION

Work Function	Percentage per NUREG-75/032 (for the Year 1974)[2]	Percentage per NUREG-0109 (for the Year 1975)[1]	Percentage per NUREG-0713 (for the Year 1981)[3]
Reactor operations	19.2	9.1	8.7
Routine maintenance	34.5	59.5	35.0
Inservice inspection	1.4	4.1	5.2
Special maintenance	28.7	16.4	39.3
Waste processing	2.1	7.6	6.9
Refueling	14.1	3.3	4.9
	100.0	100.0	100.0

TABLE 12.4.1-5

DISTRIBUTION OF PERSONNEL (>100 MREM/YEAR)
ACCORDING TO EMPLOYEE CATEGORY

Work Function	Percentage per NUREG-75/032 (for the Year 1974)[2]	Percentage per NUREG-0109 (for the Year 1975)[1]	Percentage per NUREG-0713 (for the Year 1981)[3]
Plant employees	47.4	34.7	22.7
Utility employees	18.1	6.1	9.2
Contract workers	34.5	59.2	68.0
	100.0	100.0	100.0

TABLE 12.4.1-6

PERCENTAGES OF PERSONNEL DOSE
BY WORK FUNCTION[3]

	Percentage of Dose						
Work Function	1975	1976	1977	1978	1979	1980	1981
Reactor operations and surveillance	10.8	10.2	10.5	13.3	12.2	9.5	8.7
Routine maintenance	52.6	31.0	28.1	31.5	29.2	35.5	35.0
Inservice inspection	3.0	6.0	6.4	7.7	9.0	5.5	5.2
Special maintenance	19.0	40.0	42.5	35.9	39.4	40.6	39.3
Waste processing	6.9	5.0	5.8	5.0	3.6	3.0	6.9
Refueling	7.7	7.9	6.7	6.6	6.6	6.1	4.9

TABLE 12.4.1-7

ANNUAL OCCUPATIONAL EXPOSURES FOR VARIOUS PWR VENDOR'S UNITS

Approximate Year of Operation	Westinghouse		CE		B and W	
	No. of Reactors	Man-Rem/ Year/Unit	No. of Reactors	Man-Rem/ Year/Unit	No. of Reactors	Man-Rem/ Year/Unit
1	23	193	6	110	6	277
2	27	357	7	460	8	218
3	26	442	7	567	8	346
4	23	464	7	348	7	420
5	21	457	7	508	6	394
6	18	536	5	296	5	329
7	17	580	3	535	5	350
8	13	851	3	591	1	404
9	8	872	2	424	-	-
10	5	869	1	902	-	-
11	5	441	-	-	-	-
12	2	650	-	-	-	-
13	2	1870	-	-	-	-
14	2	2130	-	-	-	-

NOTES:

Only PWRs operating at power levels 450 MWe have been considered, with the exception of San Onofre 1 (430 MWe).

The power plants for this table are the same as those appearing on table 12.4.1-3.

TABLE 12.4.1-8 (SHEET 1 OF 2)

AVERAGE ANNUAL OCCUPATIONAL EXPOSURE AND POWER
FOR INDIVIDUAL PWR[3]

Plant Name	Total Years of Operation (Approximate)	Full Power MWe (net)	Annual Average Percentage of Full Power	Annual Average Man-Rem	Comments
Haddam Neck	14	555	92	601	First year data not available. Highest annual exposures: 1979 - 1161 man-rem; 1980 - 1353 man-rem; 1981 - 1036 man-rem.
San Onofre	14	436	69	700	First year data not available. Highest annual exposures: 1980 - 2387 man-rem; 1981 - 3223 man-rem.
Ginna	11	470	73	626	Percentage of full power (about 52 percent, 51 percent) was low in 1974 and 1976. Highest annual exposures: 1972 - 1032 man-rem; 1974 - 1225 man-rem.
Robinson 2	11	665	75	863	1971 - only 53 percent of full power was produced. Highest annual exposure: 1980 – 1852 man-rem.
Palisades	10	635	51	588	1974 - only produced 1 percent of full power. Highest annual exposure: 1973 - 1133 man-rem.
Maine Yankee	9	810	69	294	Percentages of full power produced were low in 1973 (50 percent) and 1974 (53 percent).

TABLE 12.4.1-8 (SHEET 2 OF 2)

Plant Name	Total Years of Operation (Approximate)	Full Power MWe (net)	Annual Average Percentage of Full Power	Annual Average Man-Rem	Comments
Fort Calhoun	8	478	64	330	1975 - only 53 percent of full power was produced.
Kewaunee	7	512	84	146	
Rancho Seco	7	873	60	285	1976 - only 31 percent of full power was produced.
Millstone Point 2	6	864	66	612	1978 - highest annual exposure: 1621 man-rem.
Trojan	5	1080	60	356	1978 - only 19 percent of full power was produced.
Beaver Valley	5	810	37	238	1980 - only 5 percent of full power was produced.
St. Lucie	5	777	79	478	
Crystal River 3	5	782	53	462	1978 - only 40 percent of full power was produced.
Salem	4	1079	51	352	1979 - only 23 percent of full power was produced.
Davis-Besse	4	874	43	725	1980 - only 29 percent of full power was produced.
Farley	4	804	56	424	1979 - only 26 percent of full power was produced.

NOTES:

Only single unit PWRs have been considered.

Only plants that have been in operation for equal to or greater than 4 years were considered.

TABLE 12.4.1-9

AVERAGE INDIVIDUAL EXPOSURE BASED ON PWR PLANT AGE[3]

Year of Operation (approx.)	No. of Units	Average No. of Personnel/ Unit	Average Exposure per (man-rem/ year/unit)	Average Exposure per Individual (rem/year)
1	35	666	193	0.29
2	42	849	348	0.41
3	41	967	445	0.46
4	37	986	434	0.44
5	34	993	457	0.46
6	28	1124	461	0.41
7	25	1323	529	0.40
8	17	1443	779	0.54
9	10	1422	782	0.55
10	6	1268	875	0.69
11	5	832	441	0.53
12	2	1066	650	0.61
13	2	2640	1870	0.76
14	2	2393	2130	0.89

NOTE:

Units considered are the same as those shown on table 12.4.1-3.

TABLE 12.4.1-10 (SHEET 1 OF 2)

CUMULATIVE AVERAGE OF ANNUAL EXPOSURE BY YEARS OF OPERATION

	Years of Operation													
Plant	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Haddam Neck	-	106	689	342	325	697	201	703	449	614	117	1161	1353	1036
San Onofre	-	42	155	50	256	353	71	292	880	847	401	139	2387	3223
Ginna	430	1032	224	1225	538	636	401	450	592	708	655			
Point Beach 1	164	580	294	1475	229.5	185	214.5	160	322	299	298			
Robinson 2	-	215	695	672	1142	715	455	963	1188	1852	783			
Palisades	78	1133	627	306	696	100	764	854	424	902				
Point Beach 2	294	1475	229.5	185	214.5	160	322	299	298					
Maine Yankee	117	420	319	85	245	420	154	462	424					
Surry 1	152	442	824.5	1582.5	1153.5	918.5	1792	1918	2122					
Turkey Point 3	78	227	438	592	518	516	840	825.5	1125.5					
Surry 2	442	824.5	1582	1153.5	918.5	1792	1918	2122						
Oconee 1	517	165.67	342	422.67	464.3	333.67	351.67	403.7						
Indian Point 2	455	352.5	975	267.5	501.5	639.5	485.5	1185.5						
Fort Calhoun	71	294	313	297	410	126	688	458						
Turkey Point 4	227	438	592	518	516	840	825.5	1125.5						
Zion 1	56	63.5	286	501.5	508.5	637	460	860						
Prairie Island 1	18	61.5	223.5	150	110.5	90	176.5	164.5						
Kewaunee	28	270	139	154	127	165	141							
Three Mile Island 1	73	286	359	504	696	197	188							
Oconee 2	165.67	342	442.3	464.33	333.7	351.7	403.7							
Zion 2	63.5	285.5	501.5	508.5	637	460	860							
Arkansas 1	21	289	256	189	369	342	551							
Oconee 3	165.67	342	423	464.33	333.7	351.7	403.7							
Prairie Island 2	61.5	223.5	150	110.5	90	176.5	164.5							

TABLE 12.4.1-10 (SHEET 2 OF 2)

CUMULATIVE AVERAGE OF ANNUAL EXPOSURE BY YEARS OF OPERATION

	Years of Operation													
Plant	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Rancho Seco	-	58	390	323	126	412	402							
Calvert Cliffs 1	74	547	250	402.5	338.5	303.5								
Cook 1	116	299	336	359	246.5	327.5								
Millstone Point 2	168	242	1621	472	636	531								
Trojan	174	319	267	421	609									
Indian Point 2	535	1003	636	308	364									
Beaver Valley	87	190	132	553	229									
St. Lucie	152	337	438	532	929									
Crystal River 3	-	321	495	625	408									
Calvert Cliffs 2	-	250	402.5	338.5	303.5									
Salem	122	584	449	254										
Davis-Besse	48	30	154	58										
Farley 1	108	643	435	511										
North Anna 1	224.5	109	340											
North Anna 2	224.6	109	340											
Cook 2	359	246.5	327.5											
Three Mile Island 2	696	197	188											
Arkansas 2	-	551												
Total	6765	14616	18245	16058	15538	13369	13233	13245	7824	5245	2205	1300	3740	4259
No. of Units	35	42	41	37	34	20	25	17	10	6	5	2	2	2
Average	193	348	445	434	457	461	529	779	782	875	441	650	1870	2130

NOTE:

Plants considered are the same as those shown in table 12.4.1-3.

TABLE 12.4.1-11

OCCUPATIONAL RADIATION EXPOSURE ESTIMATE FOR INSERVICE INSPECTION AND SPECIAL MAINTENANCE (FOR ONE UNIT)[d]

Inservice Inspection	Average Expected Dose Rate (rem/h)	Total Time (man-h)	Frequency	Annual Dose Over 10-Year Period (man-rem)
Inservice inspection of reactor vessel and reactor coolant piping	0.100	520	1 per 10 years[a]	5.2
Insulation removal from reactor vessel nozzle and reactor coolant piping	0.100	200	1 per 10 years[a]	2.0
Snubber inspection of reactor coolant piping	0.005	2000	1 per 10 years[a]	1.0
Steam generator inservices inspection (4 steam generators)	0.125	470	1 per 10 years[a]	5.9
Steam generator eddy current inspection	0.015	600	(b)	1.5
Special Maintenance[c]				
Steam generator tube plugging	0.24	52	1 per 2 years	6.1
Steam generator tube plug welding	0.21	52	1 per 10 years	1.1
Sludge lancing	0.06	160	1 per 2 years	4.9
Control rod drive mechanism repair	0.50	5	1 per 5 years	0.5

a. Inservice inspection is performed over a 10-year period on a schedule defined in the Inservice Inspection Plan.

b. Per Regulatory Guide 1.83, eddy current inspection for steam generator tubes would occur at year 1, year 2, year 5, and year 8. Two steam generators are inspected at year 1 and one steam generator is inspected in subsequent years.

c. These data reflect dose estimate for projected special maintenance and repair tasks and do not include dose estimates for unique tasks that may be performed on a limited basis such as unforeseen major repair tasks or unusual inspection efforts.

d. Elimination of pipe whip restraints in the reactor coolant system main loop piping (GDC-4 exemption, Reference 6) results in a reduction of the occupational radiation exposure estimate as discussed in the safety balance shown in reference 7.

TABLE 12.4.1-12 (SHEET 1 OF 2)

PLANT STAFF CLASSIFICATION - WORKING TIME AND ZONE DISTRIBUTION
(VEGP UNITS 1 AND 2)[a]

Classification of Personnel	No.	Category[b]	Working h/Year	Percentage of Time Spent in Zone				
				I	II	III	IV	V
Plant management	11	0	2000	97	1.5	1.0	0.5	-
Administration	30	0	2000	94	1.5	1.0	0.5	-
Administration	34	M	2000	95	2.5	1.6	0.9	-
Engineering	24	0	2000	94	1.5	1.0	0.5	-
Engineering	40	M	2000	95	2.5	1.6	0.9	-
Engineering	11	R	2000	90	5.0	4.2	0.7	0.1
Operations	60	O	2200	97	1.5	1.0	0.5	-
Operations	100	M	2200	95	2.5	1.6	0.9	-
Operations	32	R	2200	90	5.0	4.2	0.7	0.1
Operations	17	RH	2200	65	25.0	9.3	0.6	0.1
Maintenance	10	0	2200	97	1.5	1.0	0.5	-
Maintenance	151	M	2200	95	2.5	1.6	0.9	-
Maintenance	45	R	2200	90	5.0	4.2	0.7	0.1
Maintenance	4	RH	2200	65	25.0	9.3	0.6	0.1
Radiochemistry	23	0	2200	97	1.5	1.0	0.5	-
Radiochemistry	35	M	2200	95	2.5	1.6	0.9	-

TABLE 12.4.1-12 (SHEET 2 OF 2)

Classification of Personnel	No.	Category[b]	Working h/Year	Percentage of Time Spent in Zone I	II	III	IV	V
Radiochemistry	15	R	2200	90	5.0	4.2	0.7	0.1
Radiochemisty	5	RH	2200	65	25.0	9.3	0.6	0.1
Training	15	0	2000	97	1.5	1.0	0.5	-

a. Does not include the doses due to inservice inspection and special maintenance which are respectively equal to 31.2 man-rem/year and 25.2 man-rem/year for Units 1 and 2 as shown in table 12.4.1-11.

b. Abbreviations:
- O - operation
- M - maintenance
- R - refueling
- RH - radwaste handling

TABLE 12.4.1-13

ESTIMATED ANNUAL GAMMA DOSE TO PLANT PERSONNEL[a]
(EXTRACTED NUMBER OF MANHOURS OF OCCUPANCY PER YEAR)[b]

Radiation Zone	Expected Average Dose Rate in Zone (mrem/h)	Operation	Maintenance[c]	Radwaste Handling	Refueling(c)	Total Manhours	Estimated Annual Exposure (man-rem)
I	0.06	375,002	738,340	37,180	201,960	1,352,482	81
II	1.0	5799	19,430	14,300	11,220	50,479	51
III	6.0	3866	12,435	5320	9425	31,046	186
IV	40.0	1933	6995	343	1571	10,842	434
V	100.0	-	-	57	224	281	28
	TOTALS	386,600	777,280	57,200	224,400	1,445,400	780

a. Detailed breakdown by job classification presented in table 12.4.1-12.

b. Does not include the doses due to inservice inspection and special maintenance which are respectively equal to 31.2 man-rem/year and 25.2 man-rem/year for Units 1 and 2 as shown in table 12.4.1-11.

c. Elimination of pipe whip restraints in the reactor coolant system main loop piping (GDC-4 exemption, Reference 6) results in a reduction of the occupational radiation exposure estimate as discussed in the safety balance shown in reference 7.

TABLE 12.4.1-14

DOSES TO PLANT PERSONNEL CAUSED BY AIRBORNE RADIOACTIVITY

Location	Assumed Occupancy (h/year)	Total Body Gamma Dose (man-rem/year)	Lung Dose (man-rem/year)	Inhalation Thyroid Dose (man-rem/year)	Airborne Tritium Dose (man-rem/year)
Auxiliary building corridor	2000 (40 h/week-50 weeks/year)	2.79E-2	3.93E-3	2.80E-1	2.55E-3
Auxiliary building letdown heat exchanger valve gallery	50	2.54E-2	3.64E-3	2.60E-1	2.37E-3
Turbine building	2000 (40 h/week-50 weeks/year)	5.52E-7	6.90E-6	4.96E-4	4.68E-4
Fuel building H-3 only	168 (56 h/week-3 weeks/year)	NA	NA	NA	2.16E-1
Containment (full power)	86	3.59E-1	6.20E-2	6.55E-1	6.67E-2
Containment (refuel) H-3 only	168 (56 h/week-3 weeks/year)	NA	NA	NA	2.16E-1

TABLE 12.4.1-15

TOTAL BODY DOSE FACTORS

Isotope	Total Body Dose Factor[(a)] (mrem/h per Pci/m^3)
I - 131	3.1×10^{-7}
I - 132	2.0×10^{-6}
I - 133	4.4×10^{-7}
I - 134	2.0×10^{-6}
I - 135	1.5×10^{-6}
Kr - 83m	8.6×10^{-12}
Kr - 85m	1.3×10^{-7}
Kr - 85	1.8×10^{-9}
Kr - 87	6.8×10^{-7}
Kr - 88	1.7×10^{-9}
Kr - 89	1.9×10^{-6}
Kr - 90	1.8×10^{-6}
Xe - 131m	1.0×10^{-8}
Xe - 133m	2.9×10^{-8}
Xe - 133	3.4×10^{-8}
Xe - 135m	3.6×10^{-7}
Xe - 135	2.1×10^{-7}
Xe - 137	1.6×10^{-7}
Xe - 138	1.0×10^{-6}
Ar - 41	1.0×10^{-6}

a. Obtained from WASH-1258, table D-3 for iodines, and from Regulatory Guide 1.109, table B-1 for noble gases.

TABLE 12.4.1-16 (SHEET 1 OF 2)

ASSUMPTIONS USED TO DETERMINE DOSES TO PLANT PERSONNEL CAUSED BY AIRBORNE RADIOACTIVITY

Isotope	$(MPC)_a$ with the Lung as the Critical Organ[(a)] ($\mu Ci/cm^3$)	Dose to the Thyroid per μCi Inhaled (rem/μCi)[(b)]
I-131	3×10^{-7}	1.48×10^{-0}
I-132	7×10^{-6}	5.35×10^{-2}
I-133	1×10^{-6}	4.00×10^{-1}
I-134	2×10^{-5}	2.50×10^{-2}
I-135	3×10^{-4}	1.24×10^{-1}
Rb-86	7×10^{-8}	-
Co-60	9×10^{-9}	-
Co-58	5×10^{-8}	-
Fe-59	5×10^{-8}	-
Mn-54	4×10^{-8}	-
Sr-89	4×10^{-8}	-
Sr-90	5×10^{-9}	-

$(MPD)_{lung}$ = 15.0 rem/year[(c)]

Breathing rate = $(3.47 \times 10^{-4}\ m^3/s)\ (10^6\ cm^3/m^3)(3600\ s/h)$

= $1.25 \times 10^6\ cm^3/h$

Tritium Dose

Tritium uptake coefficient	= 1
Tritium biological decay constant	= $2.41 \times 10^3\ (h^{-1})$
Tritium body mass	= 4.3×10^4 g
Tritium effective energy Mev/disintegration	= 0.01

TABLE 12.4.1-16 (SHEET 2 OF 2)

Total tritium dose[(d)](rem/year) =

$$\frac{(2)(BR)(C)(t)(U)(E)(8h/day)}{(D)(M)} \times$$

$$\frac{(3600\,s/h)(1.6\times10^{-6}\ ergs/Mev)(3.7\times10^{4}\ disintegrations/s/\mu Ci)}{(100\ ergs/g/rad)(1\,rad/rem)}$$

$$= (2)\,(t)\,(C)\,(2.57 \times 10^{2})\,(8\ h/day)$$

$$= (4.12 \times 10^{3})\,(t)\,(C)$$

where:

C = air concentration ($\mu Ci/cm^3$).

BR = breathing rate (cm^3/day).

D = tritium biological decay constant (h^{-1}).

E = tritium effective energy (Mev/disintegration).

M = tritium body mass (g).

t = number of 8-h days per year that individual is exposed to tritium environment (day/year).

U = tritium uptake coefficient.

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a. "Report of Committee II on Permissible Dose for Internal Radiation," ICRP Publication 2, p 61.

b. "Calculation of Distance Factors for Power and Test Reactor Sites," TID-14844, p 25.

c. "Basic Radiation Protection Criteria," NCRP Report No. 39, p 92.

d. The factor of 2 used in this equation arises from the fact that the dose due to absorption through the skin is equal to the dose due to the absorption through the lung as stated in ICRP-10, Appendix C.

TABLE 12.4.3-1

CONSTRUCTION WORKER DOSES

Time Period	Craft Requirement	Direct Airborne (man-rem)	Total Body (man-rem)[a]	Skin (man-rem)
3/87[b] - 6/87	565	3.4 E+01	4.8 E-01	4.9 E-01
6/87 - 9/87	328	2.0 E+01	2.8 E-01	2.9 E-01
9/87 - 12/87	162	1.4 E-01	1.4 E-01	1.4 E-01
12/87 - 3/88	110	6.6 E+00	9.4 E-02	9.6 E-02
3/88 - 9/88[c]	82	4.9 E+00	1.4 E-01	1.4 E-01
	Total	7.5 E+01	1.13 E+00	1.16 E+00

a. Total body dose rate is the same for various organs, e.g., gastro-intestinal tract, bone, liver, kidney, thyroid, and lung.

b. Date of Unit 1 commercial operation.

c. Date of Unit 2 commercial operation. The number of construction workers is assumed to be constant between 3/88 and 9/88.

TABLE 12.4.3-2

INDIVIDUAL DOSE

Release Point[a]	Direct Dose (mrem/year)	Total Body Dose[b] (mrem/year)	Skin Dose (mrem/year)
Turbine building vent	-	5.17 E-03	6.28 E-03
Plant vent	-	1.35 E+00	1.43 E+00
Radwaste building vent	-	2.04 E+00	2.04 E+00
Direct	2.6 E-01	-	-
Total	2.6 E-01	3.40 E+00	3.48 E+00

a. The GASPAR code calculates individual doses for various pathways. These doses are totals for the plume, ground, and inhalation pathways.

b. Total body dose is the same for various organs, e.g., gastro-intestinal tract, bone, liver, kidney, thyroid, and lung.

TABLE 12.4.3-3

NEAR FIELD DISPERSION (X/Q) AND DEPOSITION (D/Q) VALUES

Wind Direction (from)	Turbine Building Vent X/Q (s/m^3)	Turbine Building Vent D/Q (m^{-2})[a]	Plant Vent X/Q (s/m^3)	Plant Vent D/Q (m^{-2})
N	3.82 E-06	3.82 E-08	-	-
NNE	1.08 E-05	1.08 E-07	4.48 E-06	4.48 E-08
NE	2.06 E-05	2.06 E-07	7.73 E-06	7.73 E-08
ENE	1.65 E-05	1.65 E-07	8.78 E-06	8.78 E-08
E	4.48 E-06	4.48 E-08	1.01 E-05	1.01 E-07
ESE	-	-	-	-
SE	-	-	-	-
Total[b]	5.62 E-05	5.62 E-07	3.11 E-05	3.11 E-07

a. D/Q derived by multiplying X/Q by a conservatively assumed deposition velocity of 0.01 m/s.

b. To estimate the worst case X/Q and D/Q, the individual values are totaled. In effect, the worker is assumed to be moving with the wind direction changes or standing at the location of maximum relative concentration for each direction for 24 h/day.

TABLE 12.4.4-1

ESTIMATED OCCUPATIONAL RADIATION EXPOSURE DURING DECOMMISSIONING

Alternative	Man-Rem
A	150
B	130
C	630
A+C[(a)]	150+310
B+C[(a)]	130+310

A - Mothballing

B - Entombment

C - Prompt removal/dismantling

C[(a)] - 102-year delay period before removal/dismantling

a. This information was assembled from reference 1.

TABLE 12.4.4-2

DOMINANT RADIOACTIVE ISOTOPES AT DECOMMISSIONING

	After 2 Years of Decay	After 102 Years of Decay
Vessel and internals	Fe^{55} Co^{60} Ni^{63}	Ni^{63} Ni^{59} Nb^{94}
Other systems	Co^{60}	Sr^{90} Cs^{137}

12.5 HEALTH PHYSICS PROGRAM

12.5.1 ORGANIZATION

12.5.1.1 Program Objectives

The health physics program is organized with the objectives of:

A. Providing administrative control of the activities of plant personnel to ensure that personnel exposure to radiation and radioactive materials is within the guidelines of 10 CFR 20.

B. Providing administrative control of work practices to ensure that the as low as reasonably achievable (ALARA) techniques of paragraph 12.1.3.2 are applied as appropriate.

C. Providing administrative control of effluent releases from the plant to ensure that the releases are maintained ALARA and within the limits of the Offsite Dose Calculation Manual.

D. Providing administrative control of waste shipments from the plant to ensure that all applicable requirements for the shipment and receipt of the material at the burial ground are met.

E. Providing administrative control of records of surveys, radiation monitoring, effluent releases, and waste shipments, to ensure that the requirements of Subpart L of 10 CFR 20 are met.

F. Providing technical support for the radiological training program.

A more detailed discussion of the procedures to accomplish these objectives is presented in subsection 12.5.3.

12.5.1.2 Staff Organization

Responsibility for the operation of the health physics program at the plant is delegated to the health physics manager by the plant manager, who has the overall radiological safety responsibility. Responsibilities of the health physics staff relative to radiation protection are given below.

The plant manager will ensure that all station personnel support the health physics program. The plant manager will participate, via review and approval, in the selection of goals and objectives for the health physics program. Through the plant manager's review and controlling functions, support will be given to the efforts of the health physics manager in achieving the objectives stated in paragraph 12.5.1.1.

The health physics manager has functional control of and is responsible for establishing the health physics program. The health physics manager has the responsibility for ensuring that the ALARA policy is implemented and serves as the radiation protection manager referred to in Nuclear Regulatory Commission Regulatory Guides 8.8 and 8.10. The health physics manager's qualifications are listed in subsection 13.1.3. Additional health physics personnel

titles include health physics support supervisor, nuclear specialists, health physics foremen, health physics technicians, and plant health physicist.

12.5.1.3 Program Organization

The applicable portion of the VEGP health physics program was officially initiated when fuel was received for Unit 1 and will be in effect continuously thereafter until the units are decommissioned. This program consists of rules, management and worker philosophies, practices, instructions, and procedures that are used to accomplish the objectives stated above in a practical and safe manner. The program is consistent with the recommendations of Regulatory Guide 8.2.

The health physics program is described in detail in the health physics procedures described in subsection 12.5.3.

The organization and staff will ensure that the radiation surveys required to ensure that the radiological conditions for each job are well defined. The use of the radiation work permit (RWP) will ensure that health physics reviews all jobs for ALARA prior to starting work. The RWP will ensure that the requirements of paragraph 3.a of Regulatory Guide 8.8 are met before work is permitted to start.

For those jobs with significant radiological hazards, a health physics technician may be assigned to provide continuous surveillance for each crew. This continuous coverage will be reserved for significant hazards only, to ensure that exposures to the crew are maintained ALARA. Direct reading dosimeters will be used for all work requiring an RWP. Communications via the plant paging system will be available. These actions will ensure that the requirements of paragraph 3.b of Regulatory Guide 8.8 are met.

Dosage data from completed work will be maintained in plant records for use in planning future work. This will ensure compliance with paragraph 3.c of Regulatory Guide 8.8.

12.5.2 EQUIPMENT, INSTRUMENTATION, AND FACILITIES

12.5.2.1 Facilities

The health physics and radiochemistry facilities are located at el 220 ft in the control building. (See drawing AX1D11A04.) The health physics control point is located at el 220 ft in the northwest corner of the control building.

The health physics control point is located along a principal access corridor. Job planning, radiation work permit (RWP) coordination, and issuance of direct reading dosimeters may be conducted at the control point. Windows at the control point allow observation of the area by health physics personnel to facilitate positive entry control. Portable radiation survey instrumentation, air monitoring survey equipment, self-reading dosimeters, respirators, and other miscellaneous health physics supplies are stored at the health physics control point. The control point is equipped with filing cabinets and desks to provide work space for health physics personnel and storage space for some records.

Anticontamination clothing will be available at local contamination area access points as required. Change facilities will be provided for those areas where and when personnel traffic warrants. Such facilities will consist of change benches, clothing storage, frisker stations, and other health physics support equipment storage, as appropriate for the job in progress.

The equipment and instrumentation described in chapter 12 may not necessarily be the exact instrumentation installed and used at VEGP at the time the plant is operational. State-of-the-art radiation protection equipment will be used if possible.

Health physics sample counting equipment is located in the control building counting room at el 220 ft 0 in. Equipment used for routine counting of smears and air samples such as pancake probe or end window Geiger-Mueller counters, alpha and beta detectors, and/or gasflow proportional counters will be located in this area. Special samples requiring gamma isotopic analysis and/or low level counting will be performed in the counting room.

Personnel decontamination facilities in the west wing of the control building consist of standup shower facilities. The decontamination area may also be used for decontamination of small equipment. Large or highly contaminated equipment may be decontaminated in special areas established by the health physics staff. Decontamination equipment consists of ultrasonic cleaners, sand blasting cabinets, turbulators, stainless steel wash basins, and/or similar equipment. The ventilation systems for the control point are described in subsection 9.4.1. Drainage from the decontamination facilities is held up as liquid radwaste for appropriate treatment.

The radiochemistry and low level laboratories are located in the control building at el 220 ft 0 in., adjacent to the counting room. This area will be used for gamma isotopic analysis and other health physics/radiochemistry special sample counting and analysis.

12.5.2.1.1 Access and Exit of Radiation Controlled Areas

Access to the main radiation controlled area encompassing the auxiliary building, fuel handling building, and containment will normally be through the health physics control point located at the northwest corner of the control building at el 220 ft 0 in. This is normally the only approved ingress to the main radiation controlled area. Exit from the main radiation controlled area will normally be through the same door. Geiger-Mueller tube friskers and portal monitors will be placed at this area for personnel contamination monitoring. Personnel normally pass through a portal monitor prior to leaving the access control area.

12.5.2.2 Health Physics Instrumentation

12.5.2.2.1 Laboratory Instrumentation

Laboratory instrumentation located in the counting room and health physics workroom allows plant personnel to ascertain the radioactive material present in survey samples. Typical samples would be contamination survey smears, airborne survey filters, liquid samples, and charcoal cartridges, but tritium surveys and other samples may be processed also. The laboratory instrumentation is listed in table 12.5.2-1. Each laboratory counting system is checked and calibrated at regular intervals with standard radioactive sources traceable to a National Institute of Standards and Technology (NIST) source, in accordance with plant instructions. Counting efficiency, background count rates, and high voltage settings are checked by plant personnel, in accordance with plant instructions. Instrumentation will also be calibrated after repair, prior to use.

12.5.2.2.2 Whole Body Counting Instrumentation

The whole body counter(s) will be located in an appropriate low background radiation area. The whole body counting facility(s) will be equipped with a high-throughput vertical linear geometry whole body counter or equivalent capable of detecting fractional body burdens of gamma emitting radionuclides. The counter consists of a shadow shield representing a 4-π geometry, multiple detectors, a computer, a multichannel analyzer, data reduction software, and associated electronics. The counter will be capable of detecting radionuclides in the lungs, gastrointestinal tract, and thyroid.

12.5.2.2.3 Portable Survey Instrumentation

Portable survey instrumentation is located at the health physics control point and in-plant control points. This instrumentation will allow plant personnel to perform radiation, contamination, and neutron surveys, as needed, as well as collect samples for airborne analysis.

Each portable survey instrument will be calibrated at least annually, when in use, or after undergoing repair work, prior to use. Calibrations are normally performed using the instrument calibrator (discussed in paragraph 12.5.2.2.5) by plant personnel. Instruments will be source checked to verify proper operation, in accordance with plant instructions. Sufficient quantities of each type of instrument are available to permit calibration, maintenance, and repair, without causing a shortage in operational instrumentation. Portable radiological survey instrumentation is listed in table 12.5.2-2.

12.5.2.2.4 Personnel Monitoring Instruments

Personnel monitoring is provided by use of survey instrumentation (as applicable), air samplers, optically stimulated luminescent dosimeters (OSLDs), and/or direct reading dosimeters.

OSLD badges contain at least two luminescent chips with suitable filters to allow determination of anticipated radiation types.

Direct reading dosimeters will be worn by personnel in the radiologically controlled area as specified by plant instructions. These dosimeters can be used to provide official records of personnel exposure.

If a potential for neutron exposure exists, personnel OSLD badges will be read for exposure to neutron radiation, or a neutron dose rate instrument and personnel stay time will be used to compute neutron exposure in accordance with Regulatory Guide 8.14, section C.1.b.

Since film badges will not be used for personnel monitoring, the performance criterion in Regulatory Guide 8.3 is not applicable.

When audible alarm dosimeters are used as personnel monitoring devices, the guidance for selection and use will generally be in accordance with Regulatory Guide 8.28. Audible alarm dosimeters will be checked electronically before each use. Health physics will determine when dosimeter use is mandatory. If a dosimeter is dropped, proper operation will be verified visually prior to further use.

Extremity dosimetry will be provided as necessary.

The pocket ion chambers used as personnel monitoring equipment will be tested for calibration and leak rate in accordance with requirements of Regulatory Guide 8.4, sections C.1 and C.2.

Personnel survey instrumentation will consist of Geiger-Mueller count rate meters (contamination friskers), portal monitors, and hand and foot counters. These instruments will be calibrated at least annually, as applicable, when in use or after undergoing repair prior to use. Test procedures for Geiger-Mueller counters will be conducted in accordance with those test conditions of Regulatory Guide 8.6 which are specified by the manufacturer. The calibration laboratory is located in the control building at el 240 ft 0 in. Personnel monitoring instrumentation is listed in table 12.5.2-3.

12.5.2.2.5 Health Physics Equipment

Portable air samplers are used to survey airborne radioactive material concentrations. Air samplers are calibrated for flow at least annually. Surveys may be performed for radioactive particulate and radioiodine airborne concentrations.

Portable continuous air monitors are used to monitor airborne concentrations at specific work locations. Local indication and trend information for certain instruments are provided. Alarm setpoints are variable; visual and audible alarms are provided.

Respiratory protective equipment is available at the control point. Self-contained breathing apparatus for emergency use is available at the health physics control point. Equipment will be maintained and used in accordance with 10 CFR 20, Subpart H and Regulatory Guide 8.15.

An instrument calibrator will be used for calibrating gamma dose rate instrumentation. This will be a self-contained, heavily shielded, multiple dose rate calibrator. Neutron, beta, and alpha radiation sources will also be available for source checks and instrument calibration. Calibration sources are traceable to an NIST source.

Protective clothing will be supplied for personnel working in radiologically controlled areas. The clothing required for a specific job will be identified by health physics personnel.

An adequate inventory of protective clothing will be maintained on hand at the access control area or other control points, as necessary to support plant operations and maintenance activities. This clothing will include laboratory coats, coveralls, hoods, caps, plastic oversuits, gloves (plastic, rubber, cloth), shoe covers, boots, and rubbers.

Additional contamination control supplies will be available. These include vacuum cleaners, mops, absorbent paper, plastic sheets and bags, barricades, ropes, signs, and labels. Dedicated and marked disposal containers for discarded and used supplies and clothing are also provided.

A listing of health physics equipment is given in table 12.5.2-4.

12.5.2.2.6 Other Health Physics Instrumentation

The area radiation monitoring system is installed in areas where it is desirable to have constant dose rate information.

Monitors indicate dose rate in the control room and provide appropriate alarms upon reaching a preset dose rate. Fixed continuous airborne radioactivity monitors are also provided at strategic locations, where personnel exposure to airborne radionuclides is likely. More information on these fixed instruments is given in sections 12.3 and 11.5.

A portable monitor system, as required in NUREG-0737, item III.D.3.3, is available. The system uses an iodine silver zeolite or charcoal sample cartridge and a single-channel analyzer. The use of this portable monitor has been incorporated in the emergency procedures. The portable

monitor is part of the in-plant radiation monitoring program (paragraph 12.4.1.2.5) to accurately determine the airborne iodine concentration in areas where plant personnel may be present during an accident.

Sampling cartridges could be removed to a low background area (the counting room in the control building) for further analysis. These cartridge samples can be purged of any entrapped noble gases, when necessary, prior to being analyzed.

12.5.3 PROCEDURES

Procedures will be developed to cover all necessary areas of plant operations and maintenance. Section 13.5 provides information on the procedures to be developed. The procedures describe certain methods that will be used to ensure that occupational radiation exposures will be as low as reasonably achievable (ALARA). The Company's commitment to regulatory guides will be incorporated into procedures as appropriate.

Procedures for personnel radiation protection shall be prepared consistent with the requirements of 10 CFR Part 20 and shall be approved, maintained, and adhered to for all operations involving personnel radiation exposure. Strict adherence to the VEGP radiation protection procedures will ensure that personnel radiation exposures are both ALARA and within the limits of 10 CFR 20. Policy and operational considerations for radiation protection are set forth in subsections 12.1.1 and 12.1.3. Radiation protection procedures will be subject to the requirements of the quality assurance program for the operational phase of plant life described in the SNC Quality Assurance Topical Report (QATR).

12.5.3.1 Radiation and Contamination Surveys

Health physics personnel normally perform routine radiation and contamination surveys, the techniques of which are delineated in plant procedures. Surveys are performed on a frequency that varies with the potential radiological hazards associated with the given area. Survey frequency will be specified in plant procedures. These surveys consist of radiation measurements and/or contamination surveys, as appropriate for the specific area. Air samples are also routinely taken in portions of the controlled area. Survey information is factored into exposure stay time determinations and radiation work permit (RWP) specifications. (See paragraph 12.5.3.7.) An RWP may specify the need for additional surveys for specific operations and/or maintenance activities. Radiation level surveys will be performed for alpha, gamma, beta, and/or neutron exposure rates, as specified in plant procedures. Contamination surveys are normally performed to establish gross beta-gamma contamination level but may be processed for specific radionuclides (via gamma spectroscopy). Air samples are normally taken to establish airborne concentrations of particulates, noble gases, and/or radioiodine. Availability of current survey information will aid in keeping exposures ALARA.

12.5.3.2 Methods to Maintain Exposures ALARA

Methods to maintain exposures ALARA in accordance with Regulatory Guides 8.8 and 8.10 are not only included in radiation protection procedures but are also contained in applicable operating and maintenance procedures. Some examples of the types of methods that will be used to maintain exposures ALARA are discussed below for the following operational categories.

12.5.3.2.1 Refueling

After the reactor coolant system is depressurized, it is degassed and sampled to verify that the gaseous activity is low, prior to removing the reactor head. After flooding the refueling pool above the reactor, purification of the refueling pool water is continued to maintain exposures from activity in the water ALARA. Movement of irradiated fuel assemblies will be accomplished with the assembly maintained underwater. By following these procedures, the normal radiation level on the refueling bridge is expected to be less than 5 mrem/h. The RWP system is used to maintain positive radiological control over work in progress.

12.5.3.2.2 Inservice Inspection

Prior to entry into radiation areas to perform inspections, personnel may study, as appropriate: blueprints, drawings, photographs, videotapes, previous inspection reports, previous radiation and contamination surveys, or previous RWPs appropriate to the particular job. This will acquaint personnel with the job location, the work to be done, and radiation and contamination levels previously experienced. Surveys are performed to the extent required to determine present contamination and/or radiation levels. From this data, previous data, and past experience of personnel, an RWP (paragraph 12.5.3.7) is issued. Equipment is checked and/or calibrated to verify it is operating properly prior to entry into the radiation area. Temporary shielding will be used, where practicable, to reduce radiation exposure.

12.5.3.2.3 Radwaste Handling

The handling of radwaste has been minimized by plant design. The radwaste system is shielded and incorporates remotely operated liquid radwaste systems. The systems are designed to minimize operator exposure in all waste processing and handling operations. The radwaste system is described in chapter 11.

12.5.3.2.4 Spent Fuel Handling and Spent Fuel Cask Loading

Spent fuel handling and spent fuel cask loading is performed underwater, using the fuel handling cranes and/or manual extension tools. This normally requires a small crew working in the fuel handling area and usually involves little exposure. The RWP system will be used to maintain positive radiological control over this task.

Some of the methods used to maintain exposure ALARA are:

A. Maintain at least 8 ft of water above the fuel assembly to minimize direct radiation.

B. Purify fuel pool water to minimize exposure due to water activity.

C. Cool the spent fuel pool water.

D. Provide continuous air sampling while moving fuel to evaluate airborne activity.

E. Have emergency procedures immediately available.

After the cask is loaded, it is decontaminated using pressurized water to minimize loose contamination and the amount of hand cleaning of the cask.

12.5.3.2.5 Normal Operation

The plant was designed so that significant radiation sources are minimized, shielded, and/or placed in cubicles. Much of the instrumentation required for normal operation reads out remotely in the control room or in other low radiation areas. Instrumentation that cannot be placed remotely or that is read infrequently is situated, where possible, so that it can be read from the entrance to the cubicle or from a low radiation area within the cubicle. Operators are instructed to stay outside cubicles in which radiation levels are high as much as possible, and they are appraised of the areas inside cubicles where the radiation level is usually the lowest. If an operator plans to enter a high radiation area cubicle, he notifies health physics and specifies the cubicle. Upon leaving the health physics control point, he enters exposure data and time spent in the area on the RWP.

Area radiation monitoring equipment, part of the process effluent radiological monitoring system, is available and provides indication of radiation levels and local alarms, except for containment area radiation monitors which do not have local alarms and indicators. The ventilation system is designed to minimize spread of airborne contamination. The RWP system is used to maintain positive radiological control over radiation work. (See paragraph 12.5.3.7.)

12.5.3.2.6 Routine Maintenance

Routine maintenance falls into the categories of preventive maintenance (planned and scheduled maintenance such as lubrication, adjustments, and tests) and corrective maintenance (unscheduled maintenance such as valve packing, pump seal replacement, and stopping leaks). Procedures are written for the usual preventive maintenance jobs and for some recurring corrective maintenance jobs. These procedures specify the precautions to be taken. The procedures list the required lubricants, special tools and equipment, and the acceptance standards. This serves to minimize the time spent in the radiation area.

In addition, the usual preventive maintenance procedure normally states whether an RWP is required. When the RWP is issued, the radiation and/or contamination levels are listed, shielding is specified if appropriate, and additional specific instructions are given to personnel. For corrective maintenance jobs in radiation areas, for which a general procedure is used, a similar approach is used.

Extension tools are used when practical. Detailed surveys are performed and the RWP is issued (if required) with specific instructions. The individuals performing the work may be required to read procedure manuals or may be shown pictures or sketches to aid in understanding what is to be accomplished, how it is to be accomplished as safely and quickly as possible, and what the acceptance criteria are. Additional requirements may be imposed to reduce exposures at the discretion of health physics personnel.

After the job is completed, debriefings may be utilized to obtain input from personnel actually performing the work in addition to supervisory and support personnel. This will assist in revising procedures for ALARA considerations.

12.5.3.2.7 Sampling

Most sampling of radioactive systems is performed inside the hoods in the sampling station, which protect personnel from airborne activity. Protective clothing and gloves are required when sampling radioactive systems to prevent contamination of personnel.

A survey instrument may be used to check radiation levels. The liquid sample container is normally washed with clean water and dried before being brought into the radiochemistry laboratory for analysis. The dose from sample bottles is minimized by grasping the bottle at the top, by using tongs, or by using a sample carrier.

12.5.3.2.8 Calibration

Calibration of most ranges of the portable gamma detection instruments is performed inside a shielded calibrator, thus almost eliminating exposure from calibration of portable instruments. Portable sources used to calibrate fixed instruments (such as the area radiation monitoring system) are transported in shielded containers.

Where possible, fixed instruments requiring routine calibration are situated so that the necessary test signals can be inserted from a low radiation area with the instruments in place.

12.5.3.2.9 Housekeeping

Plant procedures specify that the plant be maintained in a clean condition to avoid, to the extent practicable, conditions which lead to possible contamination. Plant personnel will be made aware of the importance of a clean plant to radiation protection. Housekeeping requirements are performed in accordance with NQA-1-1994, as described in the SNC QATR.

12.5.3.3 Controlling Access and Stay Time

The area within the primary security boundary is broken down into nonradiation (clean) areas and radiation controlled areas (for radiation protection purposes). Radiation controlled areas are further categorized as radiation areas, high radiation areas, very high radiation areas, airborne radioactivity areas, contamination areas, and radioactive materials areas, to comply with 10 CFR 20 and plant procedures and instructions.

Personnel entering the radiation controlled areas of the plant must be trained in radiation protection and emergency procedures as specified in paragraph 12.5.3.8 or must be escorted. Entrance to the radiation controlled area is normally through the access control area (previously discussed in subsection 12.5.2) where an RWP will be issued, if required, for the areas to be entered. (See paragraph 12.5.3.7.)

Radiation, high radiation, and very high radiation areas are segregated and identified in accordance with 10 CFR 20. The entrance to high radiation areas is controlled in accordance with Technical Specifications, section 5.7. Measures taken to control access to very high radiation areas will meet the intent of Regulatory Guide 8.38. Control over entries into radiation controlled areas is provided by using RWPs. (See paragraph 12.5.3.7.)

During major outages, such as refueling, the RWP is prepared to designate which individual job tasks and associated doses are to be included under specific RWPs. Thus, exposure information relating to specific jobs can be accumulated for preplanning sessions in the future. This technique is discussed in subsection 12.1.3.

12.5.3.4 Contamination Control

Contamination limits for personnel, equipment, and areas are listed in the plant procedures. Surveys are performed routinely, as discussed in paragraph 12.5.3.1, to determine contamination levels. Additional surveys may be performed after maintenance work or after an operation that may have increased contamination levels. Any area found contaminated is roped off or otherwise delineated with a physical barrier, posted with appropriate signs, and decontaminated as soon as practicable. In areas where the radiation level is high or where it is considered impractical to decontaminate the area to general radiation controlled area limits, a stepoff pad is used to prevent the spread of contamination.

Tools and equipment used in contaminated areas are monitored and/or bagged (or wrapped in polyethylene sheeting) prior to being removed from the work area to prevent the spread of contamination. All tools and equipment being removed from the radiation controlled area are surveyed for contamination by health physics personnel (or other qualified personnel as specified in plant radiation protection procedures) to ensure that they meet clean area limits. If the tools or equipment do not meet the limits, they are decontaminated to the extent practicable. Decontamination facilities are discussed in subsection 12.5.2.

Control of personnel contamination (external and internal) may be provided by using protective clothing and respiratory equipment. Each individual who exits a contaminated or potentially contaminated area is responsible for surveying himself when he crosses a local control point equipped with either friskers or personnel contamination monitors. Personnel contamination monitors will be used at the control point. If contamination is found, the individual is decontaminated, using the facilities described in subsection 12.5.2, under the direction of health physics or other qualified personnel.

Special coatings are applied to walls and floors of areas containing radioactive fluids, which will aid in decontaminating these areas if it should become necessary.

12.5.3.5 Respiratory Protection

If personnel entry is required into areas where the sources of airborne radioactivity cannot be removed or controlled, occupancy may be restricted and/or respiratory protection equipment may be provided to maintain the total effective dose equivalent ALARA. When airborne radioactivity is detected in excess of the limits in 10 CFR 20, the area is isolated and posted as an airborne radioactivity area and access is controlled. Entry into these areas requires the issuance of an RWP. The RWP system (paragraph 12.5.3.7) provides radiation exposure control by controlling and recording conditions under which work in airborne radioactivity areas is performed. Air sampling techniques are used to evaluate the need for respiratory protective equipment. If needed, selection of the appropriate type of respiratory equipment is then determined. The respiratory protection program is organized to conform to 10 CFR 20, Subpart H and Regulatory Guide 8.15.

The major portion of the respiratory equipment is available in the access control area of the control building. Supplementary emergency respiratory equipment is available in the control room and emergency kits. The respiratory equipment that is available includes full face masks and self-contained breathing equipment. Table 12.5.2-4 lists respiratory equipment.

12.5.3.6 Personnel Dosimetry

All personnel entering a radiation controlled area are required to wear, as a minimum, a direct reading dosimeter. Personnel entering containment are required to wear, as a minimum, a direct reading dosimeter capable of alarming excessive dose and dose rate. Setpoints for dose and dose rate are determined based on the work to be performed. Visitors are restricted from access into high radiation areas and very high radiation areas. Temporary radiation workers and visitors are escorted by a radiation worker and are limited to the requirements of 10 CFR 20.2104 and 10 CFR 20.1502.

Only those individuals who have completed training in radiation protection and emergency procedures will be authorized to enter radiation controlled areas unescorted. When visitors and other persons who have not completed this training enter a radiation controlled area, they will obtain an escort trained in these procedures prior to entering the area.

Direct reading dosimeters will be used to provide the daily measurement of personnel exposure. When OSLDs are required, they will be processed on a routine basis by a dosimetry processor holding a current personnel dosimetry accreditation from the National Voluntary Laboratory Accreditation Program (NVLAP). OSLD badges may require special processing during outages and refueling or when an individual's exposure status is in doubt. Individual dosimeter and OSLD records will be maintained by Health Physics personnel and document control.

Personnel dosimetry record keeping will be in accordance with 10 CFR 20, Subpart L and will meet the intent of Regulatory Guide 8.7.

Whole body counts and/or bioassays will be performed on an individual basis as necessary. These special individual measurements will be initiated when the results of monitoring in the workplace indicate that significant intakes may have occurred or when workers have been associated with known incidents possibly involving significant intakes of radioactivity. Air monitoring and surface contamination tests in the workplace and tests of skin contamination, nose blows, and nasal smears will be used for determining whether special measurements will be required. Investigation levels are established in plant procedures.

The development of plant procedures for bioassay will include the development of detailed criteria for the performance of bioassay and methods of data analysis and interpretation. The criteria and methods used for developing and implementing the bioassay program will be based on the recommendations of Regulatory Guides 8.20, and 8.26, and will meet the intent of Regulatory Guide 8.9.

Control of exposure to declared pregnant female workers will be in accordance with 10 CFR 20.1208. The success of such a limitation of exposure will largely depend upon the female worker's understanding the reasons for such limitations. See paragraph 12.5.3.8 for further discussion. At the time of declaration, their exposure will be limited to 0.5 rem for the duration of their pregnancy.

Exposure data of all personnel will be collected and recorded on form NRC-5, Occupational Exposure Record for a Monitoring Period, or the equivalent. Occupational exposures incurred by individuals prior to working at VEGP will meet the requirements of 10 CFR 20.2104 for determination of prior exposure. Prior dose information may be summarized on form NRC-4, accumulative occupational exposure history, or the equivalent. These records will be maintained at the plant and will be preserved indefinitely or until the Nuclear Regulatory Commission authorizes their disposal pursuant to Subpart L of 10 CFR 20.

Results of unusual or excessive exposures will be reported to the Nuclear Regulatory Commission as required by 10 CFR 20 Subpart M.

12.5.3.7 Radiation Work Permits

The RWP is a document issued by health physics personnel to ensure proper control of work performed in radiologically controlled areas.

The RWPs will be issued by health physics personnel prior to allowing work to be performed in radiologically controlled areas. The health physics foreman or a designated alternate has direct management responsibility for the issuance of RWPs. No RWPs will be issued without approval from the health physics foreman or a designated alternate.

The RWPs identify personnel involved in the job and state the job to be performed, allowable stay times (if appropriate), protective clothing and equipment required, monitoring requirements, and any special instructions or cautions pertinent to radiation hazards. The permit lists the radiological hazards present in the work area, including area dose rate and the presence of hot spots, airborne contamination, loose surface contamination, and other hazards as appropriate. These permits provide the administrative control to ensure that all work is performed in a radiologically safe manner.

Most RWPs will be issued for a specific job. Typically the RWP will be valid only for the duration of the job or some shorter interval specified by health physics personnel. General RWPs will be issued to cover the performance of routine surveillance work in radiation controlled areas. General RWPs will not be issued for work in high radiation areas, very high radiation areas, areas with significant contamination potential, or areas requiring a survey by health physics personnel prior to entry.

Dosimeter readings and entry and exit times will be recorded for each worker on the RWP.

12.5.3.8 Radiation Protection Training

Each member of the permanent operating organization, whose duties entail entering radiologically controlled areas or directing the activities of others who enter radiation controlled areas, will be instructed in the fundamentals of radiation safety. A description of the radiation safety course is given in section 13.2. All permanent plant personnel will be required to attend a retraining program in radiation protection every year. Personnel whose duties do not require entry into radiation controlled areas will be made aware of the reasons for keeping out of such areas. The training program will be developed based on the recommendations of Regulatory Guides 8.27 and 8.29. Instructions on risks associated with occupational radiation exposure will be implemented in accordance with the recommendations of Regulatory Guide 8.8.

Considerable time and effort will be devoted to ensure that employees understand radiation and radiation safety as it applies to their work. Supervisors are responsible for ensuring that their employees follow proper radiation protection procedures and instructions. The amount and type of training will depend on the radiological hazards associated with the work they perform. Orientation lectures on radiation and radiation protection will be given to all new employees. Training will continue with detailed discussions of the specific radiological hazards associated with work assignments. In the course of their work, employees will receive additional training in radiation protection practices from supervisors, senior coworkers, and health physics personnel. The necessity for each employee to recognize radiological conditions and minimize his exposure will be stressed in training sessions.

The training of female radiation workers will include a special discussion concerning the hazards of radiation exposure to unborn children.

12.5.3.8.1 Respiratory Protection Training Program

Individuals and their supervisors requiring access to areas where respiratory protection will be utilized will complete the respiratory protection training program. The instructor will be a qualified individual with a thorough knowledge and considerable experience regarding the application and use of respiratory protective equipment and the hazards associated with radioactive airborne contaminants.

Training will include lectures, demonstrations, discussions of pertinent station procedures, and actual wearing of respirators to become familiar with the various devices utilized at VEGP. The program will include as a minimum: discussion of the airborne contaminants against which the wearer is to be protected, including their physical properties, derived air concentrations (DACs), annual limits on intake (ALIs), physiological effect, toxicity, and means of detection; discussion of the construction, operating principles, and limitations of the respirator and the reasons the respirator is the proper type for the particular purpose; discussion of the reasons for using the respirators and an explanation of why more positive control is not immediately feasible, including recognition that every reasonable effort is being made to reduce or eliminate the need for respirators; instruction in procedures for ensuring that the respirator is in proper working condition; instruction in fitting the respirator properly and checking adequacy of fit; instruction in the proper use of the respirator; discussion of the application of various cartridges and canisters available for air-purifying respirators; instruction in emergency action to be taken in the event of malfunction of the respiratory protective devices; review of radiation and contamination hazards, including the use of other protective equipment that may be used with respirators; classroom and/or field training to recognize and cope with emergency situations; and other special training as needed for special use.

Individuals will be required to don the device(s) that may be used, perform appropriate pressure test for leak detection, and be exposed to a challenging atmosphere. If a quantitative test device is available, it will be utilized to quantitatively measure and record leakage. If leakage exceeds the device's rated protection factor and retests confirm this, the individual will not be approved to use the device. If quantitative testing is not practicable or unavailable, qualitative tests such as irritant smoke or isoamyl acetate may be used as a challenging atmosphere. Detection of odor will be considered a fitting failure. After successful completion of training and fitting programs, appropriate records will be maintained to ensure individuals are issued only the approved type and model of protective device(s). These records will reflect expiration dates. Individuals will receive retraining every year and reconfirmation of respirator fit every 5 years. Related records will be maintained by the training supervisor or designated alternate.

12.5.3.9 Radioactive Materials Safety Program

Various types and quantities of radioactive sources are employed to calibrate the process and effluent radiation monitors, the area radiation monitors, and portable and laboratory radiation detectors. Check sources that are integral to the area, process, and effluent monitors consist of small quantities of byproduct material and do not require special handling and storage procedures for radiation protection purposes. The same consideration applies to solid and liquid radionuclide sources of exempt quantities or concentrations which are used to calibrate or check the portable and laboratory radiation measurement instruments. Recognized methods for the safe handling of radioactive materials, such as those recommended by the National Council of Radiation Protection and Measurements, will be incorporated in procedures to ensure proper handling of radioactive material. External doses will be minimized by a combination of time, distance, and shielding considerations. Internal doses will be minimized by the measurement and control of loose contamination.

Sealed radionuclide sources will be subject to material controls for radiological protection. Those controls will include:

A. Monitoring of all shipments containing radioactive materials in accordance with 10 CFR 20.1906.

B. Monitoring of each sealed source for removable surface contamination (leakage testing) in accordance with the Technical Requirements Manual.

C. Labeling of each source or source holder with the radiation symbol, stating the activity, isotope, and source identification number.

D. Storage in a locked area of each source that is not installed in an instrument or other piece of equipment.

E. Inventory of sources once every 6 months.

F. Maintenance of records on the results of inventories, leakage tests, use, location, condition, principal user, and the receipt and final disposition dates for all sources.

Additional details of the materials safety program are provided in the plant health physics procedures.

Radioactive sources that are subject to the radioactive material controls described herein will be used or handled only by or under the direction of health physics personnel or appropriately qualified individuals. Each individual using these sources will be familiar with the radiological restrictions and limitations placed on their use. These limitations protect both the user and the source. The experience, qualifications, and training programs for personnel who are responsible for handling and monitoring radioactive materials are described in paragraph 12.5.3.8 and section 13.2.

TABLE 12.5.2-1

FIXED LABORATORY INSTRUMENTATION

Instrument	Quantity	Radiation Detected	Accuracy	Range	Remarks
Counting laboratory					
Gamma spectrometer	2	Gamma	16% relative to 3 in. x 3 in. NaI	0-2 MeV	Computer based pulse height analyzer system with Germanium detector
Geiger-Mueller counter scaler	1	Beta, gamma	10% Co-60	0-10 counts/min	To be used for counting air samples and smears
Scintillation counter scaler	1	Beta, alpha	Determined during calibration	0-10 counts/min	To be used for counting air samples and smears

TABLE 12.5.2-2

PORTABLE HEALTH PHYSICS SURVEY INSTRUMENTS

Instrument	Quantity	Radiation Detected	Accuracy	Range	Remarks
Ion chamber survey meter	15	Beta, gamma	±10% of full scale	0-50 R/h	"RO-2A" type dose rate instrument
Geiger-Mueller survey meter	12	Beta, gamma	±10% of full scale	0.1 -1000 R/h	Extended probe type; dose rate instrument
Geiger-Mueller survey meter	15	Beta, gamma	±10% of full scale	0-50,000 counts/min	Count rate instrument
Geiger-Mueller survey meter	15	Beta, gamma	±10% of full scale	0-50 mR/h	Survey instrument
Geiger-Mueller survey meter (frisker)	40	Beta, gamma	±10% of full scale	0-50,000 counts/min	Count rate instrument with adjustment alarm for personnel surveys
Neutron survey meter	2	Neutrons	±15% directional response +8% linearity	0-5000 mR/h	BF tube within cadmium loaded polyethylene sphere; neutron dose rate instrument (rem counter)
Alpha survey meter	2	Alpha particles	±10% of full scale	0-500,000 counts/min	Alpha scintillation(ZnS) or gas proportional count rate instrument
Portable high volume air sampler	10	Not applicable	Not applicable	0-5 $ft^{(3)}$/min flowrate	Used for grab samples in work areas
Ion chamber survey meter	2	Beta, gamma	±20% of reading	0-10,000 R/h	Ion chamber dose rate instrument

TABLE 12.5.2-3

PERSONNEL MONITORING EQUIPMENT

Instrument	Quantity	Radiation Detected	Accuracy	Range	Remarks
Personnel dosimeters	130	Gamma	±10%	0-9999 mR	Electronic direct reading
Portal monitor	3	Beta, gamma		0-1000 counts/min	Multiple detector gas flow proportional
Hand and foot counter	1	Beta, gamma		0-10 counts/min	4 channels
Portal monitor	3	Gamma		0-1000 counts/min	Multiple detector gamma scintillation

TABLE 12.5.2-4

HEALTH PHYSICS EQUIPMENT

Instrument	Quantity	Radiation Detected	Accuracy	Range	Remarks
Continuous air monitor	1	Beta, gamma	±10% of reading	0-50,000 counts/min	Adjustable alarm
Instrument calibrator	1	Gamma	-	0.002-500 R/h	Shielded self-contained calibrator
Air purifying respirator	40	-	-	-	Full face – negative pressure
Self-contained breathing apparatus	5	-	-	-	Supplements emergency equipment
Atmosphere supplying respirators	15	-	-	-	1- or 2-piece, ventilated hood, constant airflow